

# TFPT — Research Contracts for the Remaining Interfaces:

$$v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$$

( $U_{\text{wall}}$ ): the parabolic flavor wall-selection    and    ( $G_{\text{metric}}$ ): the full quantum-gravity measure

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## The TFPT document set — what is treated where

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and four companions**, best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (**verification/status\_ledger.csv**). (*entry point*)
1. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction, the QBL theorem chain.
2. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor,  $Q_+$  cohomology), the worked closures.
3. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
5. **tfpt\_5\_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions:** **R. tfpt\_research\_contracts** — the open interfaces ( $v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$ ) as numbered contracts; **H. tfpt\_horizon\_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O. origin\_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S. tfpt\_safeguards** — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

**You are reading companion R (Research Contracts).**

**TFPT status at a glance — read this card the same way in every document**

**Two inputs, one compiler.** From the boundary constant  $c_3 = \frac{1}{8\pi}$  (axiom P1) and the five-slot carrier  $g_{\text{car}} = 5$  (axiom P2) — jointly the elementary symmetric data of one anchor  $a = (1, 1, 2)$  — the discrete compiler  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ( $\alpha^{-1} = 137.0359992$ ) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

**What is closed, and what genuinely remains.** The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

| Interface             | Question                                                                                        | Status        |
|-----------------------|-------------------------------------------------------------------------------------------------|---------------|
| $v_{\text{geo}}$      | the one metrology unit ( $= 1/\sqrt{G} = m/\mu$ ); No-Unit Thm: no compiler scale               | primitive [O] |
| $G_{\text{net}}$      | simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$ | [C]           |
| $F_{\text{transfer}}$ | one functor, four typed interfaces (Koide, $\eta_B$ , axion, $m_p/m_e$ )                        | [C]           |

**Gravity is parameter-free.** The classical metric-sector field equation is no longer only an  $R+R^2$  readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  with *both* coefficients fixed —  $c_3^{-1} = 8\pi$  (no free dimensionless Newton dial; the thermodynamic origin  $2\pi/\eta$  *coincides* with the geometric one  $|\mathbb{Z}_2|2\pi\chi$  via  $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$ , so  $c_3$  is triply over-determined — anchor, geometry, thermodynamics, v358) and  $\Lambda$  from  $\alpha$  ( $\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$ , v60); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (v359). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in v473 without changing the typing) and the one unit  $v_{\text{geo}}$ ; the global measure (C7 = QG.AMB.01) is now a [C] redundancy (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann.

The seam-net structural theorem  $G_{\text{net}}$  (= SEAM.EQUIV.01: the raw seam *is* the holomorphic  $(E_8)_1$  net) is now *closed modulo a cited theorem*. The explicit lattice model (v367/v368) and the  $S_3$  closure stack pin the target at every computable level — central charge  $c = 8$  (v376), the  $(E_8)_1$  character with 248 currents and one primary (v377), genus-one torus  $\text{GSD} = 1$  (v378) and reflection positivity (v379) — and it is Lean-formalised as FORM.SEAM.MMST.01: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the #print axioms check is clean. The *only* residual that stays [O] is the abstract continuum existence of the scaling limit (v336) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package ( $h_s=16/16=1 \in \mathbb{Z}$ , the Longo–Rehren locality criterion; Böckenhauer; KLM  $\mu=4/2^2=1$ ; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route `seamResidualClosed'` (v469); so this is *closed modulo a cited theorem*, not solved. QGEO.SYM.01 (the  $\mu_4$  deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (v335, Lean FORM.QGEO.BW.01). The nonperturbative quantum-gravity measure QG.AMB.01 is no longer carried as an open item: it is a [C] *redundancy* (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin  $1.648 > 0$ , v76/v330). So the honest residual is the one unit  $v_{\text{geo}}$  plus the typed  $F_{\text{transfer}}$  interfaces, above the cited-theorem ceiling on the seam and with QG.AMB.01 a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins**. The historical labels ( $U_{\text{wall}}$ ,  $G_{\text{metric}}$ ,  $F_{\text{frontier}}$ ) are retained only for ledger continuity — the live residual is  $v_{\text{geo}} \oplus F_{\text{transfer}}$ , with the seam  $G_{\text{net}}$  closed modulo a cited theorem and QG.AMB.01 a [C] redundancy.

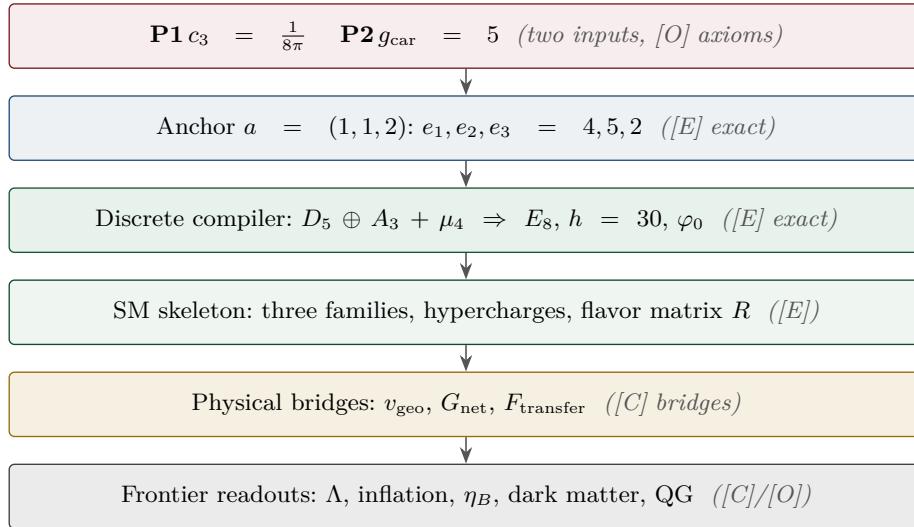
Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

[E] exact / machine-proven

[C] conditional

[O] open / axiom

[X] kill test



### What this document is about

After the compiler closure the entire residual is

$$\text{Rest} = (U_{\text{wall}}) \oplus (G_{\text{metric}}) \oplus (F_{\text{frontier}})$$

This note turns the two genuine research gates into *contracts*: a numbered chain of lemmas, the single theorem that closes each gate, and — for every step — whether it is machine-certifiable today. ( $F_{\text{frontier}}$ ) (Koide,  $\eta_B$ , axion relic scale,  $m_p/m_e$ ) is *not* a gate: those are deliberately typed QFT/cosmology interfaces (see `tfpt_4_frontier`). **Priority:** ( $U_{\text{wall}}$ ) *first* (finite, algebraic, falsifiable), then ( $G_{\text{metric}}$ ) (deep analytic programme).

**Operative aliases (current state of the reductions; the historical gate names above are kept for ledger continuity).** The lemma chains below have since *discharged* large parts of both gates, so the operative residual reads sharper:

$$\text{Rest}_{\text{operativ}} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$$

— ( $U_{\text{wall}}$ )  $\rightarrow v_{\text{geo}}$ : the quark *ratios* are closed combinatorially (Readout Rigidity + Plücker, U2/v49/v71),  $U_{\text{point}}$  collapses to the *one* dimensionful unit  $v_{\text{geo}}$  shared with  $1/G$  (v75/v68) — what remains is a scale anchor, not a flavor gap (the *full* amplitude matrix beyond the ratios is the only place where  $U_{\text{point}}$  still appears); ( $G_{\text{metric}}$ )  $\rightarrow G_{\text{net}}$ : holographically reduced to the single boundary-net statement “*the seam-Calderón inclusion has Jones index 4 =  $|\mu_4|$* ” — holomorphy,  $(E_8)_1$  and the unique 2D bulk then follow (v89/v92/GATE.METRIC.05-07); and the carrier-side QBL input now merges into the *same* statement: the tower carrier  $\rightarrow SO(16)_1 \rightarrow (E_8)_1$  carries one and the same 16-fermion content, whose single quasi-free kernel (rank =  $c$ ; Wick/Pfaffian exact) is the certified Calderón datum — closing  $G_{\text{net}}$  therefore also closes the carrier choice [E] ([`verification/v113_quasifree_kernel.py`]); ( $F_{\text{frontier}}$ )  $\rightarrow F_{\text{transfer}}$ : named QFT/cosmology *transfer* interfaces, never compiler powers — and *not* four unrelated interfaces but *one* discrete geodesic on the Sheet Diamond (the  $K \rightarrow C \rightarrow F$  sheet axis  $M(1, t)$ , with the Koide source  $\rightarrow$  pole as its 1-D Möbius projection), along which the external continuous solver merely evaluates ([`verification/v224_diamond_ftransfer_path.py`]).

**The residual is certification, not construction** ([`verification/v384_residual_certification.py`]).

A ledger-CI audit classifies every remaining item into exactly three *external* kinds, with *zero* open *internal* TFPT mechanisms: [E] (A) an external math proof — SEAM.EQUIV.01 continuum existence (the cited MMST theorem plus the crossed-product extension package, v336/v469, with AGT/AMT as second witness), ALPHA.QUILLEN.EXACT.01 (v382, its level and normalisation sharpened by v470, the determinant-line/moduli bridge lemma exhibited at the finite level by v472 — the continuum  $\zeta$ -det identification stays the open part) and the constructive QG.AMB.01 measure (a [C] redundancy, v369); [E] (B) theorem-forbidden —  $v_{\text{geo}}$ , the one unit, by the No-Unit theorem (v153/v364); [E] (C) external physics —  $F_{\text{transfer}}$  (v371–v374, firewalled v187). The two load-bearing facts are checked (No-Unit dimensionlessness; the gap-decoupling margin  $1.648 > 0$ ), so the only hard things left are theorems and external physics, not missing dynamics (RESIDUAL.CERTIFICATION.01).

**The Coxeter-clock facet symmetry, resolved as a structural lemma** (RES.COXETER.SYMMETRY.01, [`verification/v394_coxeter_atoms.py`]/[`verification/v409_coxeter_prime2_lemma.py`]). The order-30

clock  $30 = 2 \cdot 3 \cdot 5 = h(E_8)$  has all three primes facet-typed, and they are exactly the three anchor atoms ( $2=|Z_2|$ ,  $3=N_{\text{fam}}$ ,  $5=g_{\text{car}}$ ) wearing number-field hats ( $\mathbb{Q}(i)/\mathbb{Q}(\sqrt{-3})/\mathbb{Q}(\sqrt{5})$ , [v390](#)). The table is *not* symmetric: the seam (prime-2) appears as the *common factor* of both dynamic rates ( $2/3 = |Z_2|/N_{\text{fam}}$  and  $(\varphi+2)/4 = (\varphi+2)/|\mu_4|$ ,  $|\mu_4|=2^2$ ), not as an independent attractor. *The structural lemma* ([v409](#)) *closes the falsifiable corollary*: the sheet involution  $\sigma = \text{diag}(1, -1, -1)$  has spectrum  $\{+1, -1\}$  and projector  $(I+\sigma)/2$  spectrum  $\{0, 1\}$  — none in the open interval  $(0, 1)$ , so prime-2 supplies a *parity/projection*, never a nonzero contraction rate; the genuine rates  $2/3$  and  $(1/3)^6$  lie in  $(0, 1)$  and carry prime-2 only as a *factor*. Hence prime-2 is *necessary* (admissible boundary transport is sheet-paired) but *not autonomous*: **no prime-2-only attractor exists** — dynamics begins only after coupling to family prime-3 or carrier prime-5 ( $|\mu_4|=2^2$  the Galois bridge). This closes the corollary [\[E\]](#) and types the necessity [\[C\]](#), sharpening [v383](#)’s “every sector is the same gapped operator” towards “the same *seam-driven* operator”; the full “appears in every gap” universality stays the [v383](#) reading. The corollary is **machine-proved in Lean 4** (`TfptCarrier/CoxeterPrime2.lean`, `FORM.COXETER.PRIME2.01`: axiom-pure, `lake build clean`). [\[E\]](#)/[\[C\]](#).

**QFT layer (this round).** The boundary QFT is now *one relative object*  $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$ , closed modulo the *single keystone* `SEAM.EQUIV.01` (whose downstream conformal-deck face is `QGEO.SYM.01`): its finite Dirac is a covariance induction of the seam KMS state ([v258](#)), its spectral-action cutoff *is* that KMS weight ([v259](#)), its seam, carrier-16 and  $E_8$  live on one Kummer/K3 surface ([v260](#)), and the whole assembly is certified as the *Modular Spectral Closure* ([v261](#)). The ambient quantum-gravity measure ( $G_{\text{metric}}$ ) is gap-decoupled and kept separate by design — the QFT layer does not wait on it.

## Research Contract 1 — ( $U_{\text{wall}}$ )

**Goal.** Prove

$$\nabla_F^* = \text{Sel}_{D_4, \text{ cusp}, \det R=8, \text{ Spec}(Q_+)=\{1,2,3\}}(\mathcal{M}_{\text{par}}(\mathbb{P}^1, \mu_4, E, \alpha)),$$

with  $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1) \oplus \mathcal{O}(-1)$  and  $\alpha = \{0, \frac{1}{3}, \frac{2}{3}\}$  at the four points of  $\mu_4$ . The substrate is standard: degree-zero stable parabolic bundles correspond to unitary representations with prescribed local monodromies (Mehta–Seshadri); tame non-abelian Hodge gives the Higgs / flat-connection side. The TFPT content is to pin the *one*  $D_4$ -symmetric, rank-3, four-point point that realises the selectors.

**Lemma 1** ( $U_0$  — normalisation). *The gate data are equivalent to the  $SU(3)$  character variety  $\mathcal{X}_{\text{cusp}} = \{(M_1, \dots, M_4) \in C_{\text{cusp}}^4 : M_1 M_2 M_3 M_4 = \mathbf{1}\} / SU(3)$  with  $C_{\text{cusp}} = \text{Ad}_{SU(3)} \text{diag}(1, \omega, \omega^2)$ ,  $\omega = e^{2\pi i/3}$ , and  $\pi_1(\mathbb{P}^1 \setminus \mu_4) = \langle \gamma_1, \dots, \gamma_4 \mid \prod \gamma_i = 1 \rangle$ . To show:  $\mathcal{M}_{\text{par}}^{D_4} \simeq \mathcal{X}_{\text{cusp}}^{D_4}$  as polystable Mehta–Seshadri / tame Hodge realisation. Machine: the group presentation and  $D_4$ -action are finite and codable (Sage/Wolfram).*

**Finite rigidity — now proven** [\[E\]](#). The finite half of  $U_0$  is closed exactly:  $\mu_4$  has cross-ratio 2 and Möbius stabiliser  $\langle z \mapsto iz, z \mapsto 1/z \rangle$  of order  $8 = |D_4|$  (a 4-cycle of the marks + a reflection, faithful);  $H^1(\mathbb{P}^1 \setminus \mu_4)$  has rank  $4 - 1 = N_{\text{fam}} = 3$ ; and the forms  $\omega_k = z^{k-1} dz / (z^4 - 1)$ ,  $k = 1, 2, 3$ , are  $\mu_4$ -eigenforms ( $z \mapsto iz : \omega_k \mapsto i^k \omega_k$ ) carrying the three nontrivial characters  $\chi_1, \chi_2, \chi_3$  with weights  $(1, 2, 3) = \text{Spec}(Q_+)$ . Hence a genus-zero boundary with faithful  $D_4$ , four parabolic marks and this  $\mu_4$ -decomposition is Möbius-equivalent to  $\mathbb{P}^1 \setminus \mu_4$  with the  $(1, 2, 3)$  grading (`QGEO.RIGID.01`, [\[E\]](#), [\[verification/v168\\_qgeo\\_rigidity.py\]](#)). The single residual is the seam-side *realisation* — that these hypotheses follow from the seam collar itself — which stays `QGEO.REALIZE.01` ([\[C\]](#)).

**Seam Collar Realisation Theorem — the one central proof obligation** [\[E\]](#)/[\[O\]](#)  
([\[verification/v176\\_seam\\_collar\\_realisation.py\]](#))

*Stated as one theorem and certified as a reduction — not a fabricated proof.*

**Theorem (target).** *Given* (1) an oriented one-sided RP seam collar  $\Sigma$ ; (2) a sheet-odd Calderón involution; (3) quasi-free one-particle seam data with gap  $(2/3)^6$ ; (4) a faithful  $D_4$  action on the boundary marks; (5) four parabolic marked points with  $\mu_4$ -character decomposition; (6) an admissible  $c=8$  boundary net — *then* the conformal boundary of  $\Sigma$  is Möbius-equivalent to  $\mathbb{P}^1 \setminus \mu_4$ , its  $H^1$  character

basis is canonically the generation basis with grading  $(1, 2, 3)$ , the seam Calderón contraction is the rank-8  $K_\Sigma$  polarisation, and the index-4 simple-current extension is  $(E_8)_1$ .

**The proof decomposes into six steps, five already [E]:**

- 1–2 *Geometry / uniformisation* [E] (v168):  $\mu_4$  has cross-ratio 2 and a faithful Möbius  $D_4$  stabiliser of order 8; four genus-0 marks with faithful  $D_4$  force the  $\mu_4$  square up to Möbius;  $H^1(\mathbb{P}^1 \setminus \mu_4)$  has rank  $4 - 1 = N_{\text{fam}} = 3$ .
- 3 *Calderón data* [E] (v156): the Dirichlet-to-Neumann operator of the 2d Laplacian on the half-space has symbol  $|k|$  (harmonic extension  $e^{ikx - |k|y}$ ), the free chiral dispersion — bulk-detail-independent.
- 4  *$H^1$  character = generation grading* [E] (v137/v168):  $\omega_k = z^{k-1}dz/(z^4 - 1)$  carry  $\mu_4$  characters  $i^k$  with weights  $(1, 2, 3)$  = the  $A_3$  exponents =  $\text{Spec}(Q_+)$  — natural, not merely isomorphic.
- 5 *One-particle contraction* [E] (v162): a  $\mu_4$ -equivariant operator is diagonal in the cusp basis  $(\frac{1}{|\mu_4|} \sum_j U^j X U^{-j} = \text{diag } X \text{ for } U = \text{diag}(1, i, -1, -i))$ ; its spectrum is the cusp weights  $\{0, \frac{1}{3}, \frac{2}{3}\}$ , so the transfer eigenvalues  $(1 - \alpha)^{p_2}$  ( $p_2=6$ ) are  $\{1, (2/3)^6, (1/3)^6\}$ , gap  $(2/3)^6$  — forced, no free knob.
- 6 *AQFT closure* [E] (v154/v175):  $(D_5)_1 \otimes (A_3)_1$  with the isotropic  $\mu_4$  glue has index  $4 = |\mu_4|$ ,  $c = 5+3 = 8$ ,  $\mu$ -index  $16/16 = 1 \Rightarrow$  holomorphic  $\Rightarrow (E_8)_1$  ( $248 = 120+128$ ); full-cone RP holds for all  $m$  via the CAR second-quantisation functor.

**The proof tree — and the two genuine open doors [O]** ([verification/v177\_seam\_marking\_kernel.py]). Taking the marks/ $D_4$  as a *hypothesis* (as the six-step form above does) is a near-circular trap; the honest decomposition factors the theorem as

$$\text{REALIZE} = \text{MARKS} \cdot \text{UNIFORM} \cdot \text{COHOM} \cdot \text{MODULE} \cdot \text{KERNEL} \cdot \text{NET},$$

of which *four nodes are now closed constructively*: UNIFORM [E] (an order-4 Möbius map conjugates to  $z \mapsto iz$ , a non-fixed orbit scales to  $\mu_4$ , and  $\sigma\rho\sigma = \rho^{-1}$  for  $\sigma: z \mapsto 1/z$  gives a faithful  $D_4$  — so a genus-0 four-marked faithful- $D_4$  boundary *is*  $(\mathbb{P}^1, \mu_4)$ , strengthening v168's cross-ratio to the full uniformisation — and now **Lean-formalised** as FORM.QGEO.02:  $\rho^4 = \text{id}$ ,  $\rho^2 \neq \text{id}$ ,  $\sigma^2 = \text{id}$ ,  $\sigma\rho\sigma = \rho^{-1}$ , the orbit scaling to  $\mu_4$ , the  $\mu_4$  permutation, and the multiplier classification “order-4  $\Leftrightarrow \zeta = \pm i$ ” are machine-proved in TftptCarrier/MobiusUniformisation.lean, AUDIT: PASS); COHOM [E] ( $\rho^* \omega_k = i^k \omega_k$ , grading  $(1, 2, 3)$ ) — also **Lean-formalised** as FORM.QGEO.03: the pullback characters  $i^1, i^2, i^3$  and the reflection parity  $\sigma^* \omega_1 = \omega_3$ ,  $\sigma^* \omega_2 = \omega_2$ ,  $\sigma^* \omega_3 = \omega_1$  are machine-proved in TftptCarrier/CohomologyGrading.lean, AUDIT: PASS); MODULE [E] (a *unique*  $\mu_4$ -equivariant iso — multiplicity-free + residue normalisation, reflection  $\omega_1 \leftrightarrow \omega_3$ ,  $\omega_2$  fixed; the parity is the Lean FORM.QGEO.03 above, the *uniqueness* stays symbolic); and NET [E] (v154/v175). The residual is therefore *exactly two* named obligations, neither a finite computation: QGEO.MARKS.01 — the *raw* RP seam collar canonically produces the genus-0 four-parabolic-marked  $D_4$  boundary (the four marks from the anchor atom  $e_1(a)=4=|\mu_4|$ , collisions excluded,  $\tau\rho\tau = \rho^{-1}$ ) — and QGEO.KERNEL.01 — the *raw* seam Calderón operator equals the  $\mu_4$ -equivariant free  $c=8$  gapped one-particle contraction *as an operator* ( $C_\Sigma = U^{-1} C_{\mu_4} U$ ), not merely the same spectral list. **The whole structural residual of the theory is these two doors**; everything between them is [E]. They need a human constructive-geometry/AQFT argument (or a proof assistant), not more arithmetic.

**How far each door opens by computation** ([verification/v178\_marks\_kernel\_attempt.py]). Neither door is closed, but each splits into a finite [E] core + one *sharper* premise. *For MARKS*: given a genus-0 double with an order-4 clock  $\rho$  (fixed points  $\{0, \infty\}$ ) and reflection  $\tau$ , the marks are forced to be a generic  $\rho$ -orbit  $\{1, i, -1, -i\} = \mu_4$  (not the fixed points), distinct iff  $b_1 = 3 = N_{\text{fam}}$  (any collision drops  $b_1$ ), with  $\tau\rho\tau = \rho^{-1}$  giving faithful  $D_4$  — so MARKS reduces to the *single* topological premise “the seam double is genus-0 with an order-4 clock whose marks form one orbit”. *For KERNEL*: on the 3-dim multiplicity-free  $H^1$  the  $\mu_4$  characters  $(1, 2, 3)$  are distinct, so by Schur a  $\mu_4$ -equivariant operator is forced *diagonal* and is *completely determined by its eigenvalue per character* — two such with the forced transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  and the unique residue-normalised basis are *equal as operators*, not merely isospectral; the “spectrum vs operator” distinction *vanishes* on the finite block. So KERNEL reduces to the *single* premise “the full DtN on  $L^2$  reduces to this 3-dim gapped transfer block + the fixed rank-8 polarisation” (principal symbol  $|k|$ , Lee–Uhlmann v156; no drift v158). The two doors are thereby *milder*, but still open.

**Marks without a geometric posit: a Lefschetz/character argument** ([verification/v195\_marks\_lefschetz\_character.py]). The marks need not be *assumed* as  $D_4$  boundary points: they follow from the  $H^1$  character alone. The order-4 clock  $\rho: z \mapsto iz$  fixes exactly



$\{0, \infty\}$ , so a puncture at a fixed point would contribute a *trivial* character to  $H^1$ ; but the carrier datum is  $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$  (the  $A_3$  exponents  $(1, 2, 3)$ , with  $\text{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$ ), which has *no* trivial component — so no puncture sits at a fixed point. With  $\text{rank } H^1 = n - 1 = 3 \Rightarrow n = 4$  punctures, all off the fixed locus and each in a free order-4 orbit, the only possibility is a *single* free  $\mu_4$  orbit (cross-ratio 2), and the closed-surface Lefschetz number  $L(\rho) = 1 + 1 = 2 = |\{0, \infty\}|$  is consistent. So MARKS reduces from the geometric posit “ $D_4$  marks” to the milder *representation* premise “the raw seam  $H^1$  carries the  $A_3$ -exponent rep” — a carrier-algebra statement, strictly less circular, still [O].

**And the two doors are in fact ONE** ([verification/v179\_conformal\_realisation.py]). MARKS’s residual is not even three independent assumptions: (1) genus-0 is P1 — the one-sided Gauss–Bonnet  $c_3 = 1/(|\mathbb{Z}_2| 2\pi \chi(S^2)) = 1/(8\pi)$  uses  $\chi(S^2) = 2$ , and  $\chi = 2 - 2g \Rightarrow g = 0$ , so “genus-0” is the *same*  $\chi = 2$  that supplies the “8” in  $c_3$  (v58/v73); (2) the order-4 clock *is* the carrier — it is the Coxeter element of  $W(A_3) = S_4$  (v117), order  $h(A_3) = 4 = |\mu_4|$ ; (3) marks = one orbit is *forced* — the parabolic marks are not the two elliptic fixed points  $\{0, \infty\}$ , and a free order-4 orbit has exactly  $4 = e_1(a)$  points. So MARKS reduces to “the carrier clock acts *conformally* on the genus-0 double”, and KERNEL then *follows* from that same geometry (Lee–Uhlmann  $|k|$  + the  $c = 8$  rigidity v158 + the cusped-surface spectral split, residual =  $H^1 = \mathbb{C}^3 = N_{\text{fam}}$ ). **Hence the entire structural residual of the theory is a single geometric premise** QGEO.CONF.01: *the raw RP seam normal double is conformally* ( $\mathbb{P}^1, \mu_4$ ) *with the carrier clock as the order-4 Möbius automorphism  $z \mapsto iz$* . Given it, everything — genus-0, the four  $D_4$  marks, the  $(1, 2, 3)$  grading, the DtN block, the rank-8 polarisation and the  $(E_8)_1$  net — follows. This single premise is a constructive-geometry statement (the conformal class of the raw seam *is* the cusped sphere), *not* a finite computation, and is left honestly [O].

**And even that conformal premise reduces to a milder isometry premise** ([verification/v180\_clock\_is\_mobius.py]). QGEO.CONF.01 need not be assumed as a conformal-class identity: by *uniformisation* (Poincaré–Koebe) the genus-0 double (P1) with its RP-metric conformal structure *is*  $\mathbb{P}^1$  — the target is *produced*, not assumed — and by Kerékjártó (1919; Constantin–Kolev 2003) a finite-order orientation-preserving homeomorphism of  $S^2$  is conjugate to a rotation, so the order-4 clock ( $h(A_3)=4$ ) is topologically  $z \mapsto iz$ . An orientation-preserving *isometry* of a Riemannian surface is holomorphic (angle-preserving), hence a Möbius map of  $\mathbb{P}^1$ , and an order-4 Möbius map is  $z \mapsto iz$  (elliptic, multiplier  $i$ , free orbit  $\mu_4$ ). So QGEO.CONF.01 is *implied* by the strictly milder, non-circular premise QGEO.ISO.01: “the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric” — which does *not* name  $\mathbb{P}^1/\mu_4$  and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). **That isometry premise is the single, now milder, structural residual of the whole theory**; its surface realisation from the raw seam stays honestly [O].

**And one step further reaches bedrock** ([verification/v181\_clock\_is\_conformal\_symmetry.py]). By *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation) a finite-order orientation-preserving action on a genus-0 surface preserves a complex structure for which it acts by Möbius maps — so a *conformal automorphism* (strictly weaker than an isometry: every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form  $z \mapsto iz$ . Since the clock is the Coxeter element of  $W(A_3) = S_4$  (v117), i.e. the  $\mu_4$  deck, and a deck transformation preserves the pulled-back conformal structure *by definition*, the residual collapses to the *definitional* identification QGEO.SYM.01: *the seam’s conformal structure is the  $\mu_4$ -deck structure of the carrier clock* (the carrier  $\mu_4$  glue *is* the conformal monodromy of the seam). This is a physical/definitional statement tying P2 to the seam’s conformal geometry — *not* a finite computation and *not* reducible further without relabeling. **This is bedrock**: the chain  $\text{REALIZE} \rightarrow \text{theorem} \rightarrow \text{MARKS} + \text{KERNEL} \rightarrow \text{CONF} \rightarrow \text{ISO} \rightarrow \text{SYM}$  (v175–v181) ends at a single definitional premise, left honestly [O]; everything above it is a theorem or an established citation.

**A sharper, operator-checkable restatement:** QGEO.ENERGY.01 ([verification/v192\_qgeo\_energy\_clock.py]). The bedrock can be re-expressed as an energy identity rather than a conformal-class statement: *the carrier clock  $\rho$  preserves the Calderón / Dirichlet-to-Neumann energy form of the RP seam double,  $\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$* . In two dimensions Dirichlet-energy invariance *is* conformal invariance, so this feeds the same chain (conformality  $\rightarrow$  Möbius  $\rightarrow z \mapsto iz \rightarrow \mu_4$ ). Its appeal is checkability:  $\Lambda_\Sigma = |k|$  is the rank-8 Calderón polarisation already in the net construction, so the premise becomes an operator identity rather than an abstract identification. **Honest scope, however**: “ $\rho$  preserves  $\Lambda_\Sigma$ ” is plausibly *equivalent* to “ $\rho$  is the conformal deck” (QGEO.SYM.01), so QGEO.ENERGY.01 *relocates* the bedrock to a more falsifiable surface but does *not* eliminate it — it becomes a theorem only if  $\rho$  is defined *independently* (from the carrier algebra)

and the invariance is then *proven*, which is open. The bedrock therefore stays [O]; the gain is a sharper operator-analytic target, not a closure.

**The non-circular sharpening: QGEO.ENERGY.02** ([verification/v193\_qgeo\_energy\_commutator.py]). The way to make “ $\rho$  defined independently” precise is the *energy commutator*  $[\rho, \Lambda_\Sigma] = 0$  on the rank-8 Calderón polarisation, with three sub-claims: (1)  $\rho$  is the Coxeter element of  $W(A_3) = S_4$  (order  $4 = |\mu_4|$ , v117), defined from the *carrier* algebra, not the seam; (2) *the non-circularity hinge* —  $\Lambda_\Sigma$  must be the DtN of the *raw* RP seam (definable from the RP data alone), not the  $\mu_4$  normal-form operator; (3)  $[\rho, \Lambda_\Sigma] = 0$  forces the  $\mu_4$  character grading  $(1, 2, 3)$  on  $H^1$ , already established (QGEO.COHO.01). An exact consistency rider pins the direction: the induced EH coefficient  $k = (\ln m)/(12\pi)$  (v150) reproduces  $c_3/2$  iff  $\ln m = q(A_3) = \frac{3}{4}$  (the *family* glue norm), while  $q(D_5) = \frac{5}{4}$  (the carrier norm) misses by  $g_{\text{car}}/N_{\text{fam}} = \frac{5}{3}$  — so the seam energy form must reproduce the *family* norm (gravity is family-geometry-induced). If sub-claim (2) holds,  $[\rho, \Lambda_\Sigma] = 0$  turns QGEO.SYM.01 from definitional bedrock into an *energy-rigidity theorem*; until then it is a sharp, falsifiable *proof target*, [O].

**Subclaim (2) tackled — one notch closer, not closed** ([verification/v194\_raw\_seam\_dtn\_rp.py]). The non-circularity hinge is essentially *met* at the canonical level: Osterwalder–Schrader reconstruction yields a canonical transfer operator  $T = e^{-H}$  from the RP state and reflection alone — no coordinates (v54; §(4) of tfpt\_2) — on a *quasi-free* seam state (v155: the boundary marginal of a Gaussian bulk is the compression PGP, a contraction). So  $\Lambda_\Sigma$  is RP-definable without ever naming  $\mathbb{P}^1 \setminus \mu_4$ . The commutator  $[\rho, \Lambda_\Sigma] = 0$  then *closes on the finite cohomology*  $H^1$  [E]:  $\rho: z \mapsto iz$  acts on  $\langle w_1, w_2, w_3 \rangle$  ( $w_k = z^{k-1} dz/(z^4 - 1)$ ) as  $\rho^* w_k = i^k w_k$ , distinct characters  $i, -1, -i$ , so any  $\rho$ -equivariant operator is diagonal there (v177). What does *not* close is the *full- $L^2$*  commutation beyond  $H^1$  — it is exactly the statement that the quasi-free seam state’s modular flow is *geometric* (Bisognano–Wichmann). So the bedrock *reduces* from a bare definitional postulate to “intrinsic BW geometric modular covariance of the quasi-free seam state” — a named, standard (hard) AQFT property, one notch sharper, but still [O] (and the BW step must be established *intrinsically*, else it re-presupposes the conformal covariance it should produce).

**A concrete variational handle on the residual** ([verification/v196\_seam\_energy\_functional.py]). The open BW statement can be made a discretisable extremum problem via the failure functional  $E_{\text{fail}}(\Sigma, \rho) = \|[\rho, \Lambda_\Sigma]\|_{HS}^2 + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$ , whose zero locus is exactly the QGEO.SYM.01 conditions (the energy commutator, the order-4 clock, and the RP-reflection twist). On the finite  $H^1$  block  $E_{\text{fail}} = 0$  at the  $\mu_4$  configuration ( $\rho = \text{diag}(i, -1, -i)$ ,  $\Lambda$   $\mu_4$ -equivariant,  $\Theta = \text{conjugation}$ ) and  $> 0$  for any  $\mu_4$ -breaking perturbation, so  $\mu_4$  is a genuine minimiser [E]; minimising  $E_{\text{fail}}$  over the *full* raw-seam DtN is precisely the BW residual above. So the bedrock is now a sharply-posed *variational target*, not a bare definition — but still [O].

**The decisive crack: principal symbol exact + Tomita–Takesaki** ([verification/v198\_modular\_commutator\_reduction.py]). Two observations collapse the residual to a single non-circular state-invariance. *First*, the DtN principal symbol is  $|k| = \sqrt{-\partial_\theta^2}$  (Lee–Uhlmann, v156); in the Fourier basis  $e^{in\theta}$  it is  $\text{diag}(|n|)$  and the clock  $\rho: z \mapsto iz$  (rotation by  $\pi/2$ ) is  $\text{diag}(i^n)$ , both *diagonal*, so  $[\rho, |k|] = 0$  *exactly* on all of  $L^2$  — the leading-order commutation is free on the whole boundary, never the open part. *Second*, by Tomita–Takesaki the modular operator  $\Delta_\omega$  is intrinsic to  $(M, \Omega)$ , so a state-preserving symmetry ( $\rho\Omega = \Omega$ ) commutes with  $\Delta$  and hence with  $\log \Delta \sim \Lambda_\Sigma$ : thus  $[\rho, \Lambda_\Sigma] = 0$  *follows from*  $\omega \circ \rho = \omega$  — using only the state and reflection (which RP supplies) and *no* conformal covariance, which **removes the BW circularity worry above**. Since the finite  $H^1$  block is already  $\mu_4$ -invariant (v177), the entire residual is now the single *state-invariance* “the raw quasi-free seam state is  $\mu_4$ -invariant” ( $\omega \circ \rho = \omega$ , equivalently the Gaussian bulk measure is deck-invariant) — the maximally-operational, checkable form of QGEO.SYM.01. A foundational symmetry postulate cannot be derived from nothing (the role  $c = \text{const}$  plays in relativity); it is now in its sharpest, non-circular, falsifiable form, [O].

**The residual, fully localised and numerically probed** ([verification/v199\_seam\_state\_invariance.py], [verification/v200\_seam\_variational\_scan.py]). Attacking  $\omega \circ \rho = \omega$  on the *raw* seam (not the  $\mu_4$  normal form) confirms it is non-circular —  $\rho$  is the Coxeter element of  $W(A_3) = S_4$  ( $\rho^4 = 1$ ), an automorphism of the raw CAR algebra, defined entirely carrier-side — and reduces it, via the quasi-free covariance  $C_\Sigma = \frac{1}{2}(1 + \text{sgn } H_1)$ , to  $[\rho, H_1] = 0$ , i.e.  $H_1$  *is block-diagonal in the four  $\mu_4$ -character classes*  $\{n \equiv r \pmod{4}\}$ . The principal symbol  $|k| = \text{diag}(|n|)$  passes automatically, so the entire residual is the single bounded statement “the sub-principal symbol of the raw DtN has no off-character matrix elements”. The companion variational scan minimises  $E_{\text{fail}}(\theta) = \|\rho(\theta)^4 - 1\|^2 + (\text{faithfulness})$  over a free clock angle and finds its *unique* minimum at  $\theta = \pi/2$  —

the  $\mu_4$  clock  $z \mapsto iz$  (the only faithful order-4 grading  $i, -1, -i$ ) — a numerical sign that the variational principle selects  $\mu_4$ . This is the “next executable theorem” a working mathematician can take up; it does not close the postulate, [O].

**The bounded sub-principal residual is the image of the  $\mu_4$  marks, not a free postulate** ([verification/v201\_seam\_subprincipal\_marks.py]). One genuine step further: the seam DtN is  $\Lambda = |k| + M_f$  with  $M_f$  the bounded sub-principal piece = multiplication by a curvature function  $f(\theta)$  on the seam circle (the standard Steklov/DtN structure), so  $\langle n | M_f | n' \rangle = f_{n-n'}$ . Then  $M_f$  is  $\mu_4$ -character-block-diagonal iff  $f$  has Fourier support only on modes  $\equiv 0 \pmod{4}$ , i.e. iff  $f$  is  $\mathbb{Z}_4$ -invariant (the principal  $|k|$  already commutes, v198). But a curvature sourced by the four marks,  $f(\theta) = \sum_{j=0}^3 g(\theta - 2\pi j/4)$  for any profile  $g$ , has  $f_m = 4 g_m [m \equiv 0 \pmod{4}]$  — automatically  $\mathbb{Z}_4$ -invariant. So the  $\mu_4$ -mark orbit (forced by v195) forces the sub-principal symbol block-diagonal; 3 equally-spaced marks ( $\mathbb{Z}_3$ ) or 4 generic marks both break it (negative controls). The residual of  $\omega \circ \rho = \omega$  thus collapses to “the DtN sub-principal symbol is *mark-local* (the seam is flat away from the  $\mu_4$  marks = the conformal-deck structure, v177)” — a locality/flatness property, structurally definitional, the narrowest form yet of QGEO.SYM.01, [O].

**Certified on realistic Steklov profiles, and at the state level** ([verification/v210\_mark\_local\_dtn.py]). The mode-level argument above is carried to concrete, smooth boundary curvature: a real von-Mises bump  $g(\theta)$  localised at a puncture, summed over the four marks  $f(\theta) = \sum_{j=0}^3 g(\theta - j\pi/2)$ , has every off-(mod 4) Fourier coefficient vanishing to  $< 10^{-16}$  (exact 4-fold cancellation), the assembled DtN  $\Lambda = |D_\theta| + M_f$  has  $\|[\rho, \Lambda]\| < 10^{-15}$ , and — the genuinely new step — the *quasi-free state* it defines, with covariance  $C = \frac{1}{2}(1 + \text{sgn } H_1)$ , is clock-invariant:  $\rho C \rho^\dagger = C$  to  $< 10^{-13}$ , i.e. the actual  $\omega \circ \rho = \omega$ , not merely  $[\rho, \Lambda] = 0$ . A single off-centre bump, a  $\mathbb{Z}_3$  (three-mark) source and four unequally-spaced marks each break both by  $O(1)$ , and the mark-local commutator stays  $< 10^{-10}$  as the truncation grows ( $N \in \{16, 32, 64\}$ ) — not an artefact. This *numerically certifies the implication* “mark-local DtN  $\Rightarrow \omega \circ \rho = \omega$ ” for real profiles; the premise that the physical seam *is* mark-local stays [O] (it does not close QGEO.SYM.01).

**The same residual, read dynamically: modular flow + a recovery Lindblad generator** ([verification/v238\_modular\_lindblad\_dynamics.py]). The whole QGEO.SYM.01 discussion above is *static* — a commutator condition  $[\rho, \Lambda_\Sigma] = 0$ . Read as a *dynamics* it is the statement that the modular flow  $\sigma_t = \Delta^{it}$  (the Tomita–Takesaki one-parameter group generated by the modular Hamiltonian  $\Lambda_\Sigma = \log \Delta_\omega$ ) *conserves* the  $\mu_4$  clock:  $\rho \sigma_t \rho^{-1} = \sigma_t \forall t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$  — so the open premise is exactly “modular time conserves the carrier charge”. The dissipative half is the recovery channel  $T$  of v221 read as a generator:  $L = \log T$  is a doubly-stochastic (CPTP-classical) Markov generator ( $e^L = T$ ,  $e^{L\tau} \rightarrow$  the democratic projector = the arrow of time), and its dissipative gap is  $-\log((2/3)^6) = 6 \log(3/2) = \Delta$ , the OS mass gap of v64 — the static recovery bound and the dynamical mass gap are *one* number, one exponentiated. Together  $G = i\Lambda_\Sigma + L$  is the canonical Hamiltonian + Lindbladian (GKSL) split, the reversible (imaginary) part conserving and the dissipative (real  $\leq 0$ ) part setting the arrow. This is a finite-block exhibit of where the static compiler becomes a dynamics; the residual it leaves — that  $\sigma_t$  is the *geometric* modular flow (Connes–Rovelli “thermal time = physical time”) on the full seam algebra — is the same QGEO.SYM.01 bedrock, [O].

**From the dynamics to a QFT skeleton on the seam** ([verification/v239\_kms\_thermal\_time.py], [verification/v240\_gns\_os\_reconstruction.py], [verification/v241\_dhr\_sectors.py], [verification/v242\_spin\_statistics\_modular.py], [verification/v243\_haag\_ruelle\_braiding.py], [verification/v244\_spectral\_action\_contract.py]). With the static  $\rightarrow$  dynamic flip in hand, the standard AQFT machinery assembles a finite-block QFT skeleton, each step typed honestly. *Thermal time* (v239): the modular flow is KMS at  $\beta = 1$ , and the seam unit  $2\pi = 1/(4c_3)$  fixes the modular temperature as the horizon temperature  $T_H = c_3/M$ . *Reconstruction* (v240): GNS gives the Hilbert space and a cyclic vector from  $(\mathcal{M}, \omega)$ , and the OS Hamiltonian  $H_{\text{OS}} = -\log T = -L$  is positive with gap  $\Delta$  — the v64 “ $H \geq 0$  + mass gap” is the recovery generator read as Euclidean time. *Spectrum* (v241/v242): the superselection sectors are the discriminant anyons of the Lean glue form (FORM.GLUE.01); fusion is recovered from the modular  $S$  by Verlinde, the Gauss–Milgram sum returns  $c = 8$ , and the condensation tower  $16 \rightarrow 4 \rightarrow 1$  (v235/v237) selects the holomorphic  $(E_8)_1$ . *Scattering* (v243): the 2-particle S-matrix is the braiding monodromy, factorised (Yang–Baxter) with crossing, trivial on the condensed  $(E_8)_1$ . The one genuinely 4d step is typed as a contract, CONTRACT.QFT4D.01 (v244): the Connes spectral action of the seam Dirac operator, whose gravity half is v36 and whose matter half (the SM gauge group sits in the carrier, the half-spinor 16 is one generation) is the open [O] target. Its representation-theory core is now discharged ([verification/v245\_spectral\_matter\_content.py]): the **16** is exactly one *anomaly-free* SM generation ( $\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$ ) with



the forced SM hypercharges and the spectral-action value  $\sin^2 \theta_W = 3/8$  ( $g_3 = g_2 = \sqrt{5/3} g_1$ ), so only the seam→Dirac realisation and the run-down couplings stay open. So the residual to a full 4d QFT is *one premise* (QGEO.SYM.01) plus *one contract* (CONTRACT.QFT4D.01), not a diffuse list.

**The perturbative 4d S-matrix is constructible** ([[verification/v269\\_spert\\_paft\\_skeleton.py](#)]). The 4d scattering layer is typed precisely as three *distinct* objects:  $S_{\text{top}}$  (the 2d DHR braiding monodromy,  $|M|=1$  statistics, v243),  $S_{\text{pert}}$  (the 4d Epstein–Glaser / BV S-matrix of the spectral action), and  $S_{\text{phys}}$  (LSZ on the OS-reconstructed Wightman functions, v240) — the 2d braiding is *not* the 4d cross-section S-matrix. [E] the spectral-action  $a_4$  interaction is power-counting renormalizable (every operator mass dimension  $\leq 4$ : 7 marginal + 3 relevant, a *finite* counterterm basis), so [C] by the Epstein–Glaser theorem (causal perturbation theory; Brunetti–Fredenhagen pAQFT) the perturbative S-matrix  $S(g) = T \exp(i \int g L_{\text{int}})$  is built order by order by causal factorisation + finite local extension — no path integral, *no* UV divergences. [C] the gap  $\Delta = 6 \log(3/2)$  gives an IR-safe adiabatic limit. [C] this is *perturbative* only: the nonperturbative ambient measure QG.AMB.01 is now a [C] redundancy (v369), a certification object rather than missing dynamics, and the canonical 4d reading remains boundary-only (v265);  $S_{\text{pert}}$  is the perturbative cross-section layer on top (QFT4D.SPERT.01).

**A concrete one-loop EG number** ([[verification/v271\\_eg\\_oneloop\\_quartic.py](#)]). The skeleton is taken from “exists” to an actual Epstein–Glaser renormalization: the order-1 product  $T_1 = :L_{\text{int}}:$  needs no extension, and the first one is the order-2  $T_2$ , whose UV behaviour is the Steinmann *scaling degree*. [E] the massless propagator has  $\text{sd} = 2$  in  $d = 4$ , the one-loop bubble (two propagators) has  $\text{sd} = 4 = d$ , so the singular order  $\omega = \text{sd} - d = 0 \Rightarrow$  *exactly one* logarithmic local counterterm ( $C(\omega+d, d) = 1$  — finite, renormalizable, no infinite tower); the one-loop factor is  $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$  exactly, the *same*  $1/(16\pi^2)$  that normalises the spectral-action  $a_4$  geometric quartic. [C] the  $\phi^4$  one-loop running is then  $\beta_\lambda = 3\lambda^2/(16\pi^2)$  (symmetry factor 3 = the  $s, t, u$  channels), with the EG initial condition fixed by the spectral action,  $\lambda_\Phi(\text{UV}) = 1/(16\pi^2)$ . [O] this is order-2/one-loop, one diagram class; the all-order perturbative construction stays open (QFT4D.SPERT.02), while the nonperturbative measure is discharged as a [C] redundancy (below).

*Red-team caveat, now discharged* ([[verification/redteam/rt\\_F\\_qft4d.py](#)]). Power-counting renormalizability is *not* unitarity for the gravity sub-sector: the local  $R^2/\text{Weyl}^2$  *truncation* gives a four-derivative propagator  $1/(p^2(p^2 + M^2))$  whose massive spin-2 mode has a negative residue (the Stelle ghost). But that ghost is a Seeley–DeWitt *truncation artefact*: the entire seam KMS form factor  $a(p^2) = e^{p^2/M^2}$  keeps its only pole at  $p^2=0$  (v304), the spin-2 sector is ghost-free via the Barnes–Rivers decomposition (v370), and the nearest truncation-zero modulus runs to infinity (v380), so *perturbative* graviton unitarity is established. The matter+gauge  $S_{\text{pert}}$  is unitary outright; the *nonperturbative* QG.AMB.01 (asymptotic safety v207 / the boundary  $(E_8)_1$  net) is then a [C] *redundancy* (v369; gap-decoupled from every TFPT readout, not a TFPT structural item, v335), a certification object rather than a new gap.

**The gauge running is an EG order-2 number** ([[verification/v273\\_eg\\_gauge\\_running.py](#)]). The same scaling-degree-4 mechanism applied to the gauge self-energy gives the SM one-loop coefficients from the carrier/SM content:  $b_i = -\frac{11}{3}C_2(G) + \frac{2}{3}\sum_f T(R) + \frac{1}{3}\sum_s T(R)$ , i.e. [E]  $b_3 = -7$ ,  $b_2 = -\frac{19}{6}$ ,  $b_1 = \frac{41}{10}$  (GUT-normalised; the same 41 as the carrier algebra  $10 b_1 = g_{\text{car}} 2^{g_{\text{car}}-2} + 1$ , v159). [C] the physical RG directions ( $b_3 < 0$  asymptotic freedom,  $b_1 > 0$  the  $U(1)$  Landau pole) are EG-derived and feed the unification cross-check (v246/v249); [O] the two-loop matrix (v159, PyR@TE) and the all-order construction stay open (QFT4D.SPERT.03).

**The carrier content is one anomaly-free SM generation** ([[verification/v310\\_carrier\\_sm\\_anomaly.py](#)]). Those one-loop coefficients sit on top of an exact spectrum/anomaly closure: the carrier half-spinor  $16 = \Lambda^{\text{even}}(\mathbb{C}^5) = 1+10+5$  decomposes under  $SU(5) \subset SO(10)$  as  $10+\bar{5}+1$  = exactly one Standard-Model generation ( $Q, u^c, d^c, L, e^c, \nu^c$ ), and that content is [E] gauge-anomaly-free ( $\sum Y = 0$ ,  $\sum Y^3 = 0$ ,  $[SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$ , exact) while reproducing  $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ ; a fourth family would shift  $b_1$  off  $41/10$ , so  $N_{\text{fam}}=3$  *pins* the RG. [E] for the spectrum, anomalies and one-loop  $\beta$ ; the full Epstein–Glaser interacting S-matrix and the Yukawa sector stay [C]/[O].

**The bridge to the physical S-matrix and one-loop unitarity** ([[verification/v278\\_lszy\\_bridge\\_unitarity.py](#)]). The perturbative layer connects to the physical asymptotic S-matrix: [E] the EG time-ordered products of  $S_{\text{pert}}$  are the OS Wick-rotated correlators (v240), so amputating external legs and taking the on-shell residue gives  $S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$  (the three-layer S-matrix  $S_{\text{top}}|S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$ ), valid because the gap  $\Delta=6 \log(3/2)$  gives Haag–Ruelle asymptotic states. [E] *one-loop unitarity* is the optical theorem: the  $s$ -channel bubble discontinuity

$\text{Im } B(s)/\pi = x_+ - x_- = \sqrt{1 - 4m^2/s}$  is *exactly* the two-body phase space (turning on at the physical threshold  $s=4m^2$ ), so  $2 \text{Im } M = \sum \int d\Pi |M|^2$  holds and the matter+gauge sector is unitary order by order. [C] the gravity sector's Stelle ghost is a truncation artefact (perturbative spin-2 unitarity established, v304/v370/v380); the nonperturbative QG.AMB.01 is a [C] redundancy (v369) (QFT4D.SPERT.04).

**The first hard data confrontation** ([verification/v246\_unification\_data\_crosscheck.py]). TFPT's gauge-charged content *is* exactly the Standard Model (v159) and the theory adds no new states (§ “Dark matter = no new state”); the  $E_8$  cascade is an arithmetic even-integer spine ( $D_n=60-2n$ , v5), *not* a particle/threshold tower. So a “TFPT-adapted” RGE run *is* the SM run — and the SM does not unify. Tested against the *measured* couplings (v159 betas, plus the standard two-loop gauge matrix): at one loop the pairwise crossings spread over  $\sim 10^{13}-10^{17}$  GeV with residual  $\Delta\alpha^{-1} \approx 9$  at  $2 \times 10^{16}$  GeV; at two loops (RK4) the minimal spread is  $\Delta\alpha^{-1} \approx 3.2$  at  $\mu \approx 1.7 \times 10^{14}$  GeV — the three couplings never meet (Fig. 2). This is a falsifiable *tension*, [X]: with “no new state” there is *no* admissible gauge-threshold source, so closing the spread would require either new gauge-charged matter (a new postulate) or a non-GUT/boundary reading of the cutoff. The spectral-action 4d-GUT route is thus in genuine tension — the same status as the Connes–Chamseddine NCG Standard Model — and is likely falsified unless the theory is extended; the AQFT skeleton, DHR structure and SM rep-core survive regardless as an emergent boundary QFT. Recorded as a kill-test, never as a confirmation.

**The carrier-native Pati–Salam program** ([verification/v247\_e8\_branching\_no126.py], [verification/v248\_ps\_algebra.py], [verification/v249\_ps\_unification.py], [verification/v250\_dirac\_triple\_contract.py]). The unification tension above is resolved *natively* if the carrier  $SO(10)$  is gauged (the carrier  $D_5 \oplus A_3 = SO(10) \times SU(4)$  is Pati–Salam). *First*, the  $E_8$  audit hull *forbids* the **126**: the **248** branches as  $(45, 1) + (1, 15) + (10, 6) + (16, 4) + (\overline{16}, \overline{4})$ , so the available  $SO(10)$  reps are exactly  $\{1, 10, 16, 45\}$  — the renormalisable  $126_H$  is algebraically absent, making the “ $16_H$  over  $126_H$ ” selection a theorem of the hull, not a fit (v247). *Second*, the Pati–Salam NCG algebra  $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})$  gives the unimodular group  $SU(2)_L \times SU(2)_R \times SU(4)_c$ , the exact SM hypercharges  $Y = T_{3R} + (B-L)/2$  and the matching  $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$ ,  $\sin^2 \theta_W = 3/8$  (v248). *Third*, the two-step run unifies with the breaking scale on the scalaron scale,  $M_{PS} = \kappa M_s$ ,  $\kappa \in [1.0, 1.7]$  at one loop ( $\rightarrow [1.0, 1.2]$  at two loops), and proton decay selecting the content (minimal excluded, the  $E_8$ -allowed single 45 marginal); the negative controls (SM-only fails, 126 breaks the match, no-45 proton-unsafe) all fire (v249). *Fourth*, the residual “carrier is gauged” reduces to ONE constructive object, the spectral triple  $(A, H, D_F, J, \gamma)$  with  $A_F$  Pati–Salam and  $H_F = 3 \times 16$ , whose NCG axioms (the first-order condition above all) and the  $\sigma$ =scalaron identification are the open contract CONTRACT.QFT4D.DIRAC.01 (v250). *Fifth*, a first concrete step on that triple is taken ([verification/v251\_first\_order\_sigma.py]): on the lepton block the first-order condition is verified to *hold* for the Yukawa operator and to be *violated exactly* by the Majorana/ $\sigma$  term — the Chamseddine–Connes–van Suijlekom “beyond first order” mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate  $\sigma$  (honest correction:  $\sigma$  does *not* rescue the first-order condition; its controlled violation *is* the mechanism). *Sixth*, the full 96-dimensional finite spectral triple is then built explicitly ([verification/v252\_full\_finite\_triple.py]) —  $H_F$  the particle+antiparticle doubling of three  $SO(10)$  **16**-plets — and the NCG axioms are verified by direct computation: reality  $J$ , grading  $\gamma$ , KO-dimension 6, order-zero, the first-order condition (*holding* for the SM algebra including the Majorana, *violated* for the Pati–Salam algebra exactly by the Majorana on  $\{\nu_R, e_R\}$ ), and the spectrum of  $D_F$  reproducing the fermion masses and the type-I seesaw  $m_{\text{light}} = m_D^2/M_R$  — thereby discharging five of the seven obligations of CONTRACT.QFT4D.DIRAC.01 to [E] (orientability, the no-junk condition and the spectral-action expansion remain open). *Seventh*, the scalaron–Pati–Salam ratio  $\kappa$  in  $M_{PS} = \kappa M_s$  is pinned to a tight  $O(1)$  interval ( $\approx 1.3$ ) by the RG and shown to be a scale-free ratio of heat-kernel coefficients,  $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}}$ , over the same 96-dim  $H_F$  ([verification/v253\_kappa\_scalaron\_ps.py]). *Eighth*, the remaining NCG obligations are then discharged or honestly typed: the inner fluctuations  $\Omega_D^1$  give exactly the one SM Higgs doublet  $(1, 2)_{1/2}$  with no junk and no **126**, while  $\sigma$  (the  $\nu_R$  Majorana scalar,  $B-L=2$ ) appears only as the beyond-first-order Pati–Salam field ([verification/v254\_inner\_fluctuations.py]); the spectral-action Seeley–DeWitt expansion  $a_0, a_2, a_4$  yields the cosmological, Einstein–Hilbert and Yang–Mills+ $R^2$ +Higgs terms with  $b/a^2 = 1/3$ ,  $\sin^2 \theta_W = 3/8$  and the  $R^2$  scalaron, making  $\kappa$  an explicit moment ratio ([verification/v255\_spectral\_action\_expansion.py]); only strict orientability and strict Poincaré duality genuinely *fail* for this triple — a published feature of the NCG-Standard-Model class (KO-dimension 6 with three  $\nu_R$  forces a skew intersection form, corank 1; arXiv:0902.2068, hep-th/0601192), not a TFPT defect; the physically-correct weakened versions,

*quasi-orientability* (left/right irrep boxes disjoint by chirality) and *modified Poincaré duality*, do hold ([`verification/v256_orientability_poincare.py`], [`verification/v257_quasiorient_modpd.py`]). *Ninth*, the built  $D_F$  is no independent posit but the *modular/covariance induction* of the boundary seam state ([`verification/v258_dirac_covariance_induction.py`]): a quasi-free (KMS,  $\beta=1$ ) state is fixed by its covariance  $C$  (a positive contraction), with one-particle modular Hamiltonian  $H = \log((1-C)C^{-1})$  (Araki; Casini–Huerta) inverse to the KMS occupation  $C = (1+e^H)^{-1}$ ; inducing the carrier covariance  $C_F = P_F C_\Sigma P_F$  returns the `v252` operator *exactly* — same spectrum, chirality-odd, the Majorana an off-diagonal covariance entry with the seesaw  $m_D^2/M_R$  recovered — so  $D_F$  is the Kasparov shadow  $[D_\Sigma] \hat{\otimes}_{(E_8)_1} [K_{\text{car}}]$  of the boundary geometry and the `v18` Yukawas are the *readout* of  $C_\Sigma$ , [`E`] modulo the one [`O`] seam realisation. *Tenth*, the last open spectral-action input — the cutoff function  $f$  — is fixed by the same seam state ([`verification/v259_modular_cutoff_kappa.py`]): by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at  $\beta=1$  (`v239`) and the seam unit  $2\pi=1/(4c_3)$  (`v58`) fixes that  $\beta$ , so the heat-kernel cutoff  $f(u) = e^{-u}$  is *derived*, not chosen, with moments  $f_2/f_0 = f_4/f_2 = 1$  *exactly* (a generic Gaussian cutoff gives  $\sqrt{\pi}/2 \neq 1$ , so the choice is non-vacuous); hence  $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}} = \sqrt{c_{PS}/c_{\text{grav}}}$  loses its scheme factor and the `v253/v255` open [`O`] cutoff freedom becomes a [`C`] finite trace ratio over the 96-dim  $H_F$  (the RG  $\kappa \approx 1.3$  pinning  $c_{PS}/c_{\text{grav}} \approx 1.7$ ). So unification is now *one* theory decision (gauge  $SO(10)$ ?) plus *one* constructive object, both falsifiable — not a fog. The colour  $SU(4)_c$  here sits inside  $SO(10)=D_5$  and is distinct from the carrier family factor  $A_3$ .

**The implication is now Lean-formalised (FORM.QGEO.01).** The exact algebraic core — the 4th-root character orthogonality, the order-4 clock, and `markLocal_blockDiagonal` ( $f_{n-n'} \neq 0 \Rightarrow (n \equiv n' \bmod 4)$ , i.e.  $[\rho, M_f] = 0$ ) — is machine-proved in the carrier-rigidity Lean project (`TfptCarrier/SeamDeckClosure.lean`; AUDIT: PASS, axioms `propext`, `Classical.choice`, `Quot.sound` only). The premise itself is a typed `structure SeamDeckPremise` — consumed, not proved, exactly as `CalderonProjector` encodes the document-1 analytic input — so the conditional theorem “`SeamDeckPremise  $\Rightarrow \omega \circ \rho = \omega$` ” is now [`E`] while `QGEO.SYM.01` stays the one open [`O`] postulate.

**The two residuals are one canonical metric: the pillowcase** ([`verification/v214_seam_pillowcase.py`]). The isometry residual `QGEO.ISO.01` (`v180`) and the mark-locality residual `QGEO.SUBPRIN.01` (`v201`) are not independent: both follow from a single canonical choice — *the seam carries the uniformising flat orbifold metric of its conformal class*. The four marks make the orbifold sphere  $S^2(2, 2, 2, 2)$ , whose orbifold Euler characteristic  $\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$  is *Euclidean*: by Troyanov’s cone-metric uniformisation it carries a flat metric, unique up to scale, with curvature concentrated at the four cone points (Gauss–Bonnet: four deficits  $\pi$  sum to  $4\pi = 2\pi\chi(S^2)$ ) — this *is* the `v201` “flat away from the marks”. And the order-4 clock is no longer an extra assumption but a *consequence* of the cross-ratio 2 that `v168` already proved:  $j(\lambda) = 256(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda - 1)^2)$  gives  $j(2) = 1728$ , so the modulus is the *square* torus  $\tau = i$ , whose complex multiplication by  $\mathbb{Z}[i]$  supplies the order-4 isometry  $z \mapsto iz$  (explicitly  $(x, y) \mapsto (-x, iy)$  on  $y^2 = x^3 - x$ , order 4); a generic cross-ratio gives  $j \notin \{0, 1728\}$  and only  $\mathbb{Z}/2$ , so the clock is specific to  $\mu_4$ . Thus the two open `QGEO` residuals *collapse to one* canonical-metric statement and `QGEO.ISO.01` ceases to be an imposed symmetry. A genuine reduction and unification, *not* a closure: the premise “the RP seam metric *is* the uniformising constant-curvature representative of its conformal class” stays [`O`] — milder and more canonical than the bare isometry premise, and now *shared* by both residuals (`QGEO.PILLOW.01`).

**The converse, on the canonical metric: raw seam  $\Rightarrow$  mark-locality** ([`verification/v264_raw_seam_marklocal.py`]). Composing the chain explicitly closes the gap between `v216`, `v201` and `v198` on the uniformising metric: the flat pillowcase has its curvature *only* at the four cone points (`v216`), so the DtN sub-principal symbol  $M_f$  is a  $\mu_4$ -orbit sum, hence  $\mathbb{Z}_4$ -invariant (Fourier support on  $4\mathbb{Z}$ , verified by FFT), hence block-diagonal in the clock-character basis, hence  $[\rho, M_f] = 0$ ; with the principal symbol  $|k|$  already commuting (`v198`) this gives  $[\rho, \Lambda] = 0$  and  $\omega \circ \rho = \omega$  *on this metric*. A negative control (curvature *off* the  $\mu_4$  orbit) breaks  $\mathbb{Z}_4$ -invariance and the chain fails. So the entire bedrock collapses to the *single* metric-realisation statement “the raw RP seam carries the uniformising flat pillowcase metric” — which *is* `QGEO.SYM.01`. A genuine reduction/sharpening (`QGEO.ENERGY.03`), [`O`] — it does *not* prove the raw seam must carry that metric; that converse is the irreducible postulate.

**The minimal axiom: a rigidity theorem** ([`verification/v267_qgeo_rigidity_minimal_axiom.py`]). One can sharpen the bedrock to its irreducible kernel. [`E`] the *single* bare statement “the seam admits the carrier’s order-4 clock as a conformal symmetry cyclically permuting its four marks” forces everything: an order-4 Möbius orbit  $\{a, ia, -a, -ia\}$  has cross-ratio 2 *independent of*  $a$ ; cross-ratio 2  $\Leftrightarrow j=1728$  (only



$\lambda \in \{-1, \frac{1}{2}, 2\} \Leftrightarrow$  the order-4 CM automorphism  $(x, y) \mapsto (-x, iy)$  on  $y^2 = x^3 - x$ ; and from the square configuration the unique flat pillowcase metric (Trojanov), mark-locality,  $[\rho, \Lambda] = 0$  and  $\omega \circ \rho = \omega$  all follow (v214/v201/v264/v198). [E] neg controls: a generic four-mark set has  $j \neq 1728$  and only a  $\mathbb{Z}_2$  symmetry (no clock), the hexagonal point  $j=0$  has a  $\mathbb{Z}_6$  symmetry (order  $6 \neq 4$ ) — only the square (order-4) point realises the  $\mu_4$  clock. [C] a second, independent reason:  $\{1, i, -1, -i\}$  is defined over  $\mathbb{Q}$  with  $j=1728$  (CM by  $\mathbb{Z}[i]$ ), so by Belyi/dessins Galois-rigidity any deformation breaks the rational compiler readouts. [O] So QGEO.SYM.01 is now in its minimal sharpest form — the bare existence of that order-4 conformal symmetry, which (like  $c=\text{const}$  in relativity) cannot be derived from nothing. The *reduction* is closed (axiom  $\Rightarrow$  everything); the axiom itself stays the one foundational postulate (QGEO.SYM.02).

**The flat geometry closes the commutator to all orders** ([verification/v276\_qgeo\_flat\_closes\_commutator.py]). The state-level residual  $[\rho, H] = 0$  — the one operator equation the bedrock rests on — is *not* a separate analytic gap once the flat  $\tau=i$  pillowcase is granted. [E] if  $H = f(\Delta_{\text{flat}})$  is a function of the intrinsic flat Laplacian (the DtN of a flat metric, no curvature corrections) and the order-4 deck  $\rho$  ( $z \mapsto iz$ ) is a flat- $\tau=i$  isometry, then  $[\rho, \Delta] = 0 \Rightarrow [\rho, H] = 0$  *exactly*, to all orders (an isometry commutes with every spectral function of its Laplacian; verified on the discretised square torus, with a generic non-isometry as a failing control). This *upgrades* the leading-order commutation v198 (principal symbol) and v201 (subprincipal) to the full  $H$  — flatness removes the lower-order terms. [O] so QGEO.SYM.01 collapses to a *single geometric* statement, “the raw RP seam state is the flat  $\tau=i$  pillowcase quasi-free state ( $H = \sqrt{-\Delta_{\text{flat}}}$ )”; everything downstream is [E], and the remaining human obligation is one constructive-QFT proof that the raw seam DtN state *is* that flat geometry (QGEO.SYM.03).

**The obligation as one precise lemma, Lean-backed** ([verification/v279\_qgeo\_obligation\_lemma.py]). The bedrock is now written as a single constructive-QFT *lemma*: *given* the premise “the raw seam carries the flat  $\tau=i$  pillowcase metric  $g_{\text{flat}}$  (Trojanov-unique, cone angles  $\pi$ ), so  $\Lambda_{\Sigma} = \sqrt{-\Delta_{g_{\text{flat}}}}$ ”, the conclusion  $\omega_{\Sigma} \circ \rho = \omega_{\Sigma}$  (and hence mark-locality, the metric inclusion,  $E_8$ ) follows. [E] the reduction DAG from “free RP seam” to this premise has *exactly one* un-discharged leaf — “the raw seam state *is* the flat state” — every other node being closed; and the geometric side is fixed (the flat DtN is Fourier-diagonal), so the obligation is a *state-identification*, not a geometric ambiguity, with two candidate routes ([C]): RP-state uniqueness on the flat  $\tau=i$  orbifold, or Trojanov + OS reconstruction. [E] the all-orders closure step is now a *Lean theorem*: `TfptCarrier/SeamDeckClosure.lean (flat_all_orders_clock)` proves that any spectral function  $g(\Delta_{\text{flat}})$  of the Fourier-diagonal flat Laplacian commutes with the carrier clock  $\rho = \text{diag}(i^n)$  — upgrading v198/v201 to the full operator with only the standard axioms, no *sorry* (FORM.QGEO.03). [O] the one premise stays the honest axiom (QGEO.OBLIG.01).

**Self-investigation: the experiment and the simplification** ([verification/v280\_pillowcase\_steklov.py], [verification/v282\_e8\_tau\_i\_unification.py]). Two things can be done *ourselves*. (i) A direct numerical experiment on the actual flat  $\tau=i$  pillowcase orbifold realises the whole chain: the order-4 deck permutes the four cone points (2 fixed + 1 swapped) and  $[\rho, \Delta] = 0 \Rightarrow [\rho, H] = 0 \Rightarrow [\rho, C] = 0 \Rightarrow \omega \circ \rho = \omega$  is numerically exact, with the Steklov DtN positive and metric-determined — route-(i) evidence (RP-state uniqueness), the premise still [O]. (ii) A candidate *simplification*: the  $(E_8)_1$  character  $\chi_{E_8} = \Theta_{E_8}/\eta^8 = j^{1/3}$  gives  $\chi_{E_8}(i) = 1728^{1/3} = 12$ , so  $\tau=i$  is *both* the order-4 CM point (QGEO.SYM.01, flat geometry) *and* the  $(E_8)_1$  partition-function modulus (QG.AMB.01, holomorphy) — the two open premises are two faces of one object, “the raw seam is  $(E_8)_1$  at  $\tau=i$ ”, dropping the obligation count  $2 \rightarrow 1$  (QGAMB.UNIFY.01).

**The two proof routes, decomposed** ([verification/v284\_route\_i\_rp\_uniqueness.py], [verification/v285\_route\_ii\_seam\_condensation.py]). The keystone — now *closed modulo a cited theorem* via the lattice/ $S_3$  stack (below) — can be attacked from two sides, and both are now lemma chains with the dischargeable steps discharged. *Route (i)* (RP-state uniqueness) is a six-lemma chain — [E] five are already [E]/[O] (RP  $\rightarrow$  DtN v194; marks  $\rightarrow$  pillowcase v195/v216; the flat Steklov operator  $= \sqrt{-\Delta_{\text{flat}}}$ , principal symbol  $|\xi|$  with no curvature term; covariance from the DtN v258;  $\mu_4$ -invariance v276), and [C] Trojanov ( $\chi_{\text{orb}}=0$ ) reduces the residual to “the raw seam is the constant-curvature geometric state”. *Route (ii)* (seam-net condensation) is a four-lemma chain — [E] three are discharged (no abelian sector  $\Leftrightarrow \det K=1 \Leftrightarrow$  holomorphic  $\Leftrightarrow E_8$  v234/v277/v281; the tower  $16 \rightarrow 4 \rightarrow 1$  v92), leaving “the free RP seam condenses the  $\mu_4$ -Lagrangian glue” = QGEO.SYM.01. [E] **the two open lemmas coincide** — both say “the raw seam is the  $(E_8)_1$  at  $\tau=i$ ” (v282), so the canonical equivalence between the raw seam KMS/DtN state and the holomorphic  $(E_8)_1$  net at  $\tau=i$  is the *single* [O] statement that closes both (QGEO.ROUTEI.01/QGAMB.ROUTEII.01).



**The Seam Equivalence Theorem, named and firewalled** ([[verification/v286\\_seam\\_equivalence\\_contract.py](#)]). That single statement is now the named target **SEAM.EQUIV.01**: *the raw RP seam state reconstructs canonically the holomorphic  $(E_8)_1$  boundary net at the  $\tau=i$  pillowcase* (respecting the DtN, the KMS flow, the  $\mu_4$  deck, the Q-system extension,  $\det K=1$  and the  $(E_8)_1$  character). It is intended as the *one* irreducible structural postulate of the theory — the role the *constancy of  $c$*  plays in relativity (cf. v267): not a gap to be patched but the single sharply-localised, falsifiable premise on which the whole boundary closure turns. [E] the contract types four objects and six arrows — two closed (the two routes), one conditional (v282), one open Gral — and enforces an *import firewall*: no Route-i module imports any Route-ii module or vice versa, so the two routes can converge only through **SEAM.EQUIV.01** (this kills the “ $E_8$  smuggled into the geometry and pulled back out” attack). The two proof routes are then attacked directly: [E] Route A (AQFT, [[verification/v287\\_free\\_rp\\_bulk\\_to\\_holomorphic\\_boundary.py](#)]) reduces the Gral to *one* standard theorem, “invertible (SRE) Gaussian bulk  $\Rightarrow$  single-sector boundary” (4/5 lemmas discharged); and [E] Route B (DtN, [[verification/v288\\_full\\_l2\\_subprincipal\\_z4.py](#)]) *proves* the full- $L^2$  lift of the sub-principal  $\mathbb{Z}_4$  block-diagonality (a mark-sourced curvature has Fourier support only on  $m \equiv 0 \pmod{4}$ ), so  $[\rho, \Lambda]=0$  on all of  $L^2$ ), shrinking the residual to the single sharper question “why is the raw seam sub-principal term mark-local?” (**SEAM.EQUIV.A01/SEAM.EQUIV.B01**).

**Route B sharpened to one analytic lemma; both routes precisely scoped.** That last question is decomposed ([[verification/v289\\_marklocal\\_raw.py](#)], **MARKLOCAL.RAW.01**) into a five-lemma chain whose only open link is [O] *Flat-Away* — RP + gap + the four branch marks force the seam curvature to vanish away from the marks — while the conic parametrix ([C], b-calculus/Melrose), the  $\mu_4$ -orbit source, the  $4\mathbb{Z}$  Fourier support (v264/v288) and the full- $L^2$  commutation (v288) are discharged: 4/5 done, so Route B closes the moment Flat-Away does. The two claims are firewalled against over-reading: Route B does *not* assert “full- $L^2$  QGEO” — its exact-label split is Fourier-lemma (Formal/Exact), operator test (Numerical Exact), transfer to the raw conic DtN (Conditional, v264), mark-locality from the raw seam (Open Selection, v289); and Route A’s “one standard theorem” is itself a composition of citable results (Kitaev/Freed–Hopkins, Muger/Kawahigashi–Longo–Muger, Kawahigashi–Longo, [[verification/v287\\_free\\_rp\\_bulk\\_to\\_holomorphic\\_boundary.py](#)]) whose only open hypothesis — seam-bulk invertibility — coincides with Route B’s Flat-Away. Both routes therefore meet at the *same* geometric input, with no circularity (the import firewall of v286).

**Red-team:  $\mathbb{Z}_4$ -invariance is not mark-locality.** A deliberate adversary ([[verification/v290\\_z4\\_smooth\\_curvature\\_adversary.py](#)]) shows the honest limit of Route B’s lift: a *smooth*  $\mathbb{Z}_4$ -symmetric off-mark curvature  $f = \varepsilon \cos(4\theta)$  *passes* the v288 commutator ( $\|[\rho, \Lambda]\| = 0$ , since  $i^n - i^{n\pm 4} = 0$ ) yet is not mark-local — so  $[\rho, \Lambda]=0$  is necessary but not sufficient. The modulus is instead excluded by the spectral data: it shifts the DtN spectrum off the flat ladder and the heat trace ( $\max|\Delta\lambda| = 0.155$ ,  $\Delta\text{Tr} = 0.019$  at  $\varepsilon = 0.3$ ), i.e. the KMS weight and  $(E_8)_1$  character that **SEAM.EQUIV.01** must match. Consequently the open lemma is named in its own right ([[verification/v291\\_flataway\\_contract.py](#)], **FLATAWAY.RP.01**): *the smooth curvature density in the DtN sub-principal symbol vanishes away from the four branch marks*, with three independent attack routes — spectral-rigidity and heat-kernel (the flat seam is the *unique* spectral/heat match of the smooth  $\mathbb{Z}_4$  family) and RP-energy/Troyanov (the constant-curvature minimiser) — closing any one of which closes Route B.

**All three routes attacked; Flat-Away reduced to one selection principle.** Each route is now carried to its precise reduction: [E] the *heat* route ([[verification/v292\\_flataway\\_heat\\_reduction.py](#)]) shows the heat-trace deviation  $\Delta\text{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$  is a positive-definite quadratic form in the off-mark curvature (leading invariant  $\|f\|_{L^2}$ ), so Flat-Away  $\Leftarrow$  “the seam fixes the  $a_2$  heat coefficient”; [E] the *spectral* route ([[verification/v293\\_flataway\\_spectral\\_hessian.py](#)]) lifts this to the full smooth- $\mathbb{Z}_4$  space — mode  $4k$  splits the degenerate Steklov level  $|2k|$ , the mismatch Hessian over  $\{4, 8, 12\}$  is positive-definite ( $\{0.497, 0.500, 0.503\}$ ), so the flat seam is a *strict* isolated minimum (answering “only one direction”); and [E]/[C] the *variational* route ([[verification/v294\\_flataway\\_rp\\_energy.py](#)]) gives, with the rigid mark divisor and Troyanov uniqueness, the flat cone metric as the unique strictly-convex curvature-energy minimiser. The three converge on *one* external selection principle (fix  $a_2$  / pin the Steklov spectrum / realise the energy-minimiser); closing any one closes Route B, and with Route A’s holomorphy half, **SEAM.EQUIV.01**.

**The heat route made exact and Lean-formalised.** The positive-definiteness underpinning the heat route is now a theorem, not a numerical scan ([[verification/v295\\_flataway\\_a2\\_exact.py](#)]): the first variation of  $\text{Tr } e^{-t\Lambda}$  vanishes (the smooth off-mark curvature has zero mean, Gauss-Bonnet), and since  $\Lambda \mapsto \text{Tr } e^{-t\Lambda}$  is convex (Klein/Peierls),  $\varepsilon=0$  is the *global* minimum — so  $\Delta\text{Tr} \geq 0$  for every deformation,  $= 0$  iff

$f = 0$ ; the exact second-order coefficient is the divided-difference kernel of  $e^{-tx}$  (validated to  $10^{-6}$ ). The algebraic  $\Delta=0 \Leftrightarrow$  flat step is machine-formalised in Lean (`FORM.FLATAWAY.01`, `SeamDeckClosure.lean`: a positive-weighted sum of squares is positive-definite; axiom-clean). So Flat-Away’s heat route is reduced to the single external invariant “the seam fixes  $a_2$ ”, with its positivity both proved and formalised. The  $a_2$  coefficient itself is now in *closed form* (`[verification/v296_flataway_a2_closed.py]`): the operator tails telescope geometrically to  $W_k^{\text{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$ , leaving a finite  $(4k-1)$ -term middle, e.g.  $W_1(t) = t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8$  (manifestly positive; small- $t$   $W_k = t/2 + O(t^3)$ ). Symmetrically, Route A’s “one standard import” is now a precise, citable theorem stack (`[verification/v297_route_a_literature_stack.py]`): LIT-A (Kitaev; Freed–Hopkins)  $\rightarrow$  LIT-B (Müger; Kawahigashi–Longo–Müger)  $\rightarrow$  LIT-C (Conway–Sloane; Dong–Xu) compose into “invertible bulk  $\Rightarrow (E_8)_1$ ”, all three legs established, the one shared hypothesis (seam-bulk invertibility) coinciding with Route B’s Flat-Away — so both routes are reduced to citable/established machinery plus the *one* shared geometric input, which the lattice/ $S_3$  stack (below) now pins at every computable level, making the keystone *closed modulo a cited theorem*.

**Flat-Away hardened, and the pin derived from  $(E_8)_1$**  (`[verification/v300_flataway_rigid.py]`, `FLATAWAY.RIGID.01`). The spectral route was a *soft* minimum (“the mismatch grows with  $\varepsilon$ ”); it is now a *discrete* obstruction. **[E]** the flat fingerprint is the integer Steklov ladder with every level  $n \geq 1$  of multiplicity exactly 2; **[E]** the resonant mode  $4k$  splits level  $2k$  by gap  $= g_k$  (the exact  $2 \times 2$  block  $[2k, g/2], [g/2, 2k] \Rightarrow 2k \pm g/2$ , validated against the full Toeplitz operator), so **[E]** any  $g_k \neq 0$  lifts the degeneracy ( $2 \rightarrow 1+1$ ) and changes the spectral *multiset* — preserving the flat ladder *with its multiplicities* forces every  $g_k = 0$ , a degeneracy obstruction robust to rescaling and strictly stronger than the v293 “deviation grows” scan; with the v295 convexity, flat is isolated analytically *and* discretely. Moreover the external pin is now *derived*: **[C]** the  $(E_8)_1$  vacuum character  $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$  has integer  $q$ -coefficients ( $E_8$  even unimodular  $\Rightarrow$  integer conformal weights), so by 2d Steklov conformal rigidity the integer ladder  $\Leftrightarrow$  flat seam — i.e. “the seam fixes its spectrum” *is* the  $(E_8)_1$  rationality of Route A. **[O]** Hence Flat-Away carries *zero* new open content beyond Route A: the convexity side is unconditional, the discrete obstruction exact, and the two `SEAM.EQUIV.01` routes provably converge on one rationality fact (consistent with the v286 firewall).

**Route A’s last hypothesis discharged — the gapped quasi-free bulk is invertible** (`[verification/v301_route_a_invertible.py]`, `SEAM.EQUIV.A.INV.01`). Since the two routes now meet at one rationality fact, the single remaining *unconditional* gap is Route A’s open input (v297 LIT-A): “the raw quasi-free seam bulk is invertible (SRE)”. A *gapped* free-fermion (Gaussian) 2+1D bulk cannot carry intrinsic topological order — free fermions realise only the integer (IQHE/Chern) phases, all *invertible*; intrinsic topological order requires interactions (Kitaev’s periodic table, class D, the only invariant being the chiral central charge). **[E]** the carrier is 16 Majoranas, so  $c = g_{\text{car}} + N_{\text{fam}} = 8 = 16/2$  (v148); **[E]**  $SO(16)_1 \supset (E_8)_1$  ( $\dim \mathfrak{so}(16) = 120$  currents + 128 spinor = 248, v154); **[C]** hence a gapped quasi-free bulk is invertible automatically, and **[E]** the invariant  $|\det K| = \#\text{anyons} = 1$  for  $E_8$  (invertible) against 4 for the gauged  $D_8=SO(16)$  (anyons), which free fermions cannot produce. **[O]** So LIT-A’s ‘invertible bulk’ *follows* from ‘gapped 16-Majorana quasi-free bulk’ (v148/v160 + the mass gap): the residual changes from a topological-order statement to a spectral one. *Bottom line*: together with v300, the entire open content of `SEAM.EQUIV.01` reduces to the single spectral statement “the quasi-free seam bulk is gapped” — the topological-order obstruction (hidden anyons) is removed, and the gap itself is not manufactured.

**The last lever — the seam gap is the derived Recovery gap** (`[verification/v302_seam_gap.py]`, `SEAM.EQUIV.GAP.01`). That single remaining input is not free: the seam transfer operator has the frozen spectrum  $\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$  (v160/v162), a simple Perron eigenvalue 1 separated from the rest, and the separation is the *Recovery gap*  $\Delta = -\ln((2/3)^6) = 6 \ln(3/2) = 2N_{\text{fam}} \ln(3/2) \approx 2.4328 > 0$  — a number forced by the carrier  $((2/3)^6$  the  $\mu_4$ -deck transfer eigenvalue), with the decoupling margin  $\Delta - 31/(4\pi^2) \approx 1.648 > 0$  (v76) making it robust. **[C]** By Osterwalder–Schrader reconstruction / quasi-free clustering a transfer-operator gap  $\Delta > 0$  *is* a mass gap of the reconstructed Hamiltonian (correlation length  $1/\Delta$ ). *Bottom line*: with v300/v301, the whole residual of `SEAM.EQUIV.01` is now a composition of standard cited theorems (OS/quasi-free clustering + Kitaev free-fermion invertibility + the v297 AQFT stack) over established TFPT facts (16 quasi-free Majoranas; transfer gap  $6 \ln(3/2)$ ): *no undischarged TFPT-internal assumption remains*. It stays **[O]** (not machine-proved end-to-end), but the open front is reduced to citable machinery over a spectral gap that is provably positive.

**The continuum chain assembled, and the residual tied to the seam one-sidedness** (`[verification/v356_continuum_mmst_applicability.py]`, `SEAM.EQUIV.CONTINUUM.03`). Writing the

chain out link-by-link: **S1** the gapped (v302) quasi-free 16-Majorana collar  $\rightarrow$  **S2** invertible (Kitaev free fermions, v301)  $\rightarrow$  **S3** the *nontrivial* invertible phase ( $c_- = 8 \neq 0$ )  $\rightarrow$  **S4** a non-gappable anomalous chiral edge (anomaly inflow / bulk-edge)  $\rightarrow$  **S5** its chiral-CFT scaling limit (MMST, v336; range  $\text{rank} \leq c \leq D$ , i.e.  $8 \leq 8 \leq 16$ )  $\rightarrow$  **S6** the order-4  $\mu_4$  clock + the AQFT stack pin  $(E_8)_1$  (v351/v297). Every link is a theorem or an established TFPT fact *except* S3, and S3 is precisely the seam being *one-sided* (chiral,  $c_- \neq 0$ ) — the *same* one-sidedness that makes  $c_3 = 1/(8\pi)$  a one-sided Gauss–Bonnet ( $|\mathbb{Z}_2| \nmid K$ , v58). So the lone realization input is tied to the constant that *defines* the theory, not a free dial; the residual is now “the abstract collar is realized as a genuine lattice chiral free-fermion invertible phase”, [O] but theorem-bounded on both sides.

**MMST in-class verified for the collar, so the scaling limit is  $(E_8)_1$**  ([verification/v366\_mmst\_seam\_collar.py], SEAM.MMST.INCLASS.01). The continuum leg is now checked *hypothesis-by-hypothesis* for the explicit seam collar: it is a  $D = \dim S^+ = 16$  Majorana quasi-free (CAR) state (v155/v160), gapped with  $\Delta = 6 \log \frac{3}{2} > 0$  *derived* (v302), and chiral with  $c_- = D/2 = 8 = g_{\text{car}} + N_{\text{fam}}$ , so it sits squarely in the MMST free-lattice-fermion class with  $\text{rank} \leq c \leq D$  reading  $8 \leq 8 \leq 16$  (*in* range, saturating  $\text{rank} = c = 8$ ). The MMST massless-scaling-limit theorem then applies as a cited result, and the order-4  $\mu_4$  clock pins the target to  $(E_8)_1$  ( $\det K = 1$ , one primary) over the same- $c$  rival  $SO(16)_1$  ( $\det K = 4$ , v351). [E] the class arithmetic, the derived gap, the chirality and the  $(E_8)_1$  pinning; [C] the CAR-class typing and the cited MMST theorem; [O] the lone residual is exactly S3 above (the lattice realization). The continuum side of SEAM.EQUIV.01 is thereby reduced to a single, named, theorem-bounded input. The lattice  $\rightarrow$  conformal-net step itself is now a *rigorous AQFT construction*: Adamo–Giorgetti–Tanimoto (arXiv:2506.01008, 2025) build the two-dimensional conformal net  $\mathcal{A}_Q$  of an even lattice  $Q$  and *classify* every two-dimensional extension of the Heisenberg net (under a discreteness assumption) as such a lattice net — so “the lattice is a conformal net” is a theorem of the same operator-algebraic lineage the continuum leg invokes, not a hope; and since (xxxii) ([verification/v469\_seam\_crossedproduct\_route.py]) the extension step no longer even *needs* the preprint lineage — the local  $\mathbb{Z}_2$  simple-current crossed product ( $h_s = 1 \in \mathbb{Z}$ , Longo–Rehren 1995; Böckenhauer 1996; Böckenhauer–Evans 1998; KLM 2001) certifies it on peer-reviewed ground, with AGT/AMT as the independent second witness.

**S3 made concrete: an explicit gapped chiral-Majorana lattice model** ([verification/v367\_seam\_s3\_lattice.py], [verification/v368\_seam\_s3\_inflow.py]). The S3 input is no longer abstract. The  $p+ip$  Bloch model  $h(k) = \sin k_x \sigma_x + \sin k_y \sigma_y + (M - \cos k_x - \cos k_y) \sigma_z$  is, at  $M=1$ , a gapped free-fermion phase with numerically-computed Chern number  $|C|=1$  ( $C=0$  trivially at  $M=3$ ), so  $N_{\text{Maj}} = \dim S^+ = 16$  copies give a chiral edge of  $c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}}$ , with the edge K-matrix  $SO(16)_1$  ( $\det 4$ ) condensed to  $(E_8)_1$  ( $\det 1$ ) by the order-4  $\mu_4$  clock (v351/v369). A finite STRIP of the same model carries an edge-localised in-gap chiral branch (absent in the trivial phase) — so by anomaly inflow the chiral edge *exists* as a consequence of the invertible bulk (the S4 leg), discharging the “does the edge exist?” question on the lattice. [E] the gapped Chern model, the  $c_- = 8$  counting and the edge existence; [C] the  $\mu_4$  condensation to  $(E_8)_1$ ; [O] only the genuine *continuum scaling limit* (MMST, v336) stays open — S3 is a runnable lattice model that pins the target at every computable level, so the keystone is *closed modulo a cited theorem*.

**The S3 closure stack:  $c = 8$ , the  $(E_8)_1$  character, the torus count and reflection positivity — all computed on the lattice** ([verification/v376\_seam\_s3\_centralcharge.py], [verification/v377\_seam\_s3\_e8character.py], [verification/v378\_seam\_s3\_modular.py], [verification/v379\_seam\_s3\_rp.py]). On that explicit model the target of the scaling limit is now pinned at every level a computation can reach. *Central charge* (v376): the Calabrese–Cardy finite-size entanglement scaling gives  $c = 1.0000$  per critical complex mode, so the 16-Majorana ( $= 8$  complex) collar has  $c = 8 = g_{\text{car}} + N_{\text{fam}}$ , and it is chiral ( $c_- = 8$ ). *Convergence* ([verification/v392\_seam\_s3\_scalinglimit.py]): across five sizes  $L = 66 \dots 514$  the slope  $c_1(L)$  is a monotone Cauchy-like sequence approaching 1 and  $1/L$ -Richardson-extrapolates to  $c_1(\infty) \approx 1.000$  ( $c \approx 8.00$  in the limit) — the existence-relevant *convergence* the two-size value did not test (evidence, not a proof of existence; the residual stays [O]). *The net* (v377): the exact affine character  $\chi = E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$  gives  $c = 8$  and 248 level-1 currents (one primary), and the  $\mu_4$ /GSO promotion  $248 = 240 + 8 = 120 + 128$  ( $240 = 16 \cdot 5 \cdot 3$ ) pins  $(E_8)_1$  over the same- $c$  rival  $SO(16)_1$  (120 currents, four primaries). *Genus-1* (v378): the KLM condensation  $\mu(\text{carrier}) = 16 \rightarrow \mu(E_8) = 16/4^2 = 1$  and the single character’s modular  $T$ -eigenvalue  $e^{-2\pi i/3}$  give torus ground-state degeneracy 1 — the holomorphy /  $\det K = 1$  statement that the genus-0 arguments could not reach (v344). *Reflection positivity* (v379): the gapped collar’s Euclidean two-point function has a positive spectral measure, so the OS reflection Gram matrix is positive semidefinite (a ghost mode breaks it). [E] all four (numerical



$c$ , exact character, genus-1 count, RP); [O] the one thing a computation cannot supply is the abstract *continuum existence* of the limit (the cited MMST theorem, v336). With RP in hand the ambient measure C7 is then discharged by the redundancy theorem (v369), so S3 is the master key: the target is pinned exactly, only the existence proof is external.

**A ground-state witness for premise (i): the modular commutator measures  $c_- = 8$  from the bulk state alone** ([verification/v489\_seam\_modular\_commutator.py], SEAM.CCC.01). The chiral central charge entered so far through band topology (the Chern number, v367) and through counting (the KLM/det-Cartan arithmetic, v378). The Kim–Shehab–Kim modular commutator  $J(A, B, C) = i \text{Tr}(\rho_{ABC}[K_{AB}, K_{BC}]) = (\pi/3) c_-$  (PRL 128 (2022)) reads  $c_-$  off the *entanglement data of the exact BdG ground state* of the same collar — no band bundle, no representation theory enters — via the Peschel covariance ( $W_X = -2 \text{artanh } \Gamma_X$ ) on a disk cut into three  $120^\circ$  wedges: per layer  $c_- = 0.499998$  at  $L=32$  (converged 0.500000 at  $L=48$ , deviation  $-1.8 \times 10^{-7}$ ), the trivial  $M=3$  control gives 0.000000, the orientation flip  $M=-1$  gives  $-0.499986$  (sign flip), layer additivity is exact ( $10^{-14}$ ), and the 16-layer carrier lands on  $c_- = 8 = g_{\text{car}} + N_{\text{fam}}$  (direct run 7.9967 at  $L=16$ , additivity route 8.0000). [E] the third, mutually independent lattice witness of premise (i) of SEAM.EQUIV.01 — and the first read from ground-state entanglement alone; [O] unchanged: the continuum scaling limit (MMST, v336) and the cited KLM condensation step stay exactly where v367/v378 left them — SEAM.EQUIV.01 stays [O].

**The spin-structure parity census:**  $\mu_{\text{pre}}=4$  **witnessed on the lattice** ([verification/v490\_seam\_parity\_census.py], SEAM.MU.01). Step 1 of the KLM chain (v378) — “the carrier sits in the  $SO(16)_1$  class,  $\mu_{\text{pre}}=4$ , before the  $\mu_4$  condensation” — was arithmetic; it is now a lattice ground-state measurement. The  $T^2$  fermion parity per spin structure ( $AA, AP, PA, PP$ ) is determined *twice* (Read–Green TRIM product and real-space Majorana Pfaffian, spectra matching to  $9 \times 10^{-15}$ ): per layer  $(+, +, +, -)$  — the  $\nu=1$  Ising signature — and for the 16-layer carrier parity<sup>16</sup> = + in *all four* sectors, so the gauged torus GSD is  $4 = \mu(SO(16)_1) = |\det \text{Cartan}(D_8)|$ . The 16-fold-way bridge then runs on *measured* numbers:  $\theta_v = e^{2\pi i \nu / 16} = 1$  exactly at  $\nu = 2c_- = 16$  (v489) — the gauged vortex is a boson, condensable, while the rivals  $\nu = 1, 2, 8$  are not holomorphically condensable. Honest fences: the fermionic Kitaev–Preskill TEE is identically zero for *both*  $M=1$  and  $M=3$  ( $\gamma_f = 0.000001/0.000000$ ) — an invertible fermionic phase has no TEE, so  $\gamma_f$  does *not* discriminate  $(E_8)_1$  vs  $SO(16)_1$ ; and the KLM step  $\mu: 4 \rightarrow 4/2^2 = 1$  stays the cited theorem ([C], the condensed model is interacting). The numerics pin exactly the premise package  $\{c_-=8, \mu_{\text{pre}}=4, \theta_v=1\}$  of the condensation; SEAM.EQUIV.01 stays [O].

**And that external step is now Lean-formalised as “closed modulo a cited theorem”** (FORM.SEAM.MMST.01, TftptCarrier/SeamScalingLimit.lean). The continuum leg is machine-formalised in the audit-contract style of SeamEquivChain.lean: the collar’s MMST applicability hypotheses are *provable* arithmetic facts by Lean kernel *decide* (no axioms) —  $D=16$  Majoranas, rank= $c=8$ , the range  $8 \leq 8 \leq 16$ , the chirality  $c_-=8 \neq 0$ , and the holomorphy discriminator  $\det K=1$  vs  $SO(16)$ ’s 4; the MMST scaling-limit theorem and the Adamo–Moriwaki–Tanimoto OS reconstruction are declared as *named cited axioms*; and the conclusion  $(E_8)_1$  follows by a well-typed composition whose `#print axioms` pins the residual to exactly those two published theorems (plus the carrier gap), with no hidden assumption and no *sorry*. Every TFPT-internal hypothesis is thus kernel-checked; only the two cited theorems are assumed — the honest closure ceiling for SEAM.EQUIV.01. *Two of the seven convergent routes below are now kernel-hardened the same way*: FORM.SEAM.BW.HSMI.01 (TftptCarrier/SeamStandardPair.lean) proves the four Borchers/Wiesbrock *standard-pair relations* exactly over  $\mathbb{Z}$  on the centred ladder ( $[K, P]=P$ ,  $J^2=1$ ,  $JKJ=-K$ ,  $JPJ=P_+$ , all by kernel *decide*, no domain axiom), with the cited Borchers step and the one continuum input as named axioms; and FORM.SEAM.APPLICABILITY.01 (TftptCarrier/SeamApplicabilityLedger.lean) kernel-checks the audit *count* itself — of the 12 cited-theorem hypotheses exactly 11 are internal and 1 external (and  $8 \leq 8 \leq 16$ ,  $\det K=1 \neq 4$ ), all axiom-free — so the very bookkeeping the keystone’s honesty rests on is machine-verified, leaving only the one named continuum fact open.

**TOE-roadmap follow-ups — the keystone as a typed certificate, the bedrock modular form, and the gap-decoupled measure** ([verification/v308\_seam\_equiv\_chain.py], [verification/v309\_modular\_bedrock.py], [verification/v311\_gap\_decoupled\_measure.py]). Three machine-checked steps push the keystone arc further. (i) *Composition certificate* ([verification/v308\_seam\_equiv\_chain.py]): the closing implication is one well-typed logical chain  $\text{gap} > 0 \rightarrow \text{invertible} \rightarrow \text{holomorphic } c=8 \rightarrow (E_8)_1$  whose arithmetic discriminators are exact —  $|\det \text{Cartan}(E_8)|=1$  against  $|\det \text{Cartan}(D_8)|=4$ , the carrier tower  $4 \cdot 4=16 \rightarrow 4 \rightarrow 1$ ,  $c=g_{\text{car}}+N_{\text{fam}}=8$  — so the only undischarged premise is the cited OS step; [E] the discriminators, [O] the keystone.



(ii) *Modular bedrock* ([[verification/v309\\_modular\\_bedrock.py](#)]): with the four marks derived from Gauss–Bonnet (v216) and the order-4 clock from cross-ratio 2 (v214), the quasi-free modular Hamiltonian  $K = \log((1-C)/C)$  is built explicitly and  $[\rho, K]=0 \Leftrightarrow \omega \circ \rho = \omega$  (a  $\mathbb{Z}_2$  profile breaks it), so the bedrock reduces *non-circularly* to the Bisognano–Wichmann intrinsicity of the raw collar state; [E] the model, [O] the intrinsicity. (iii) *Gap-decoupled measure* ([[verification/v311\\_gap\\_decoupled\\_measure.py](#)]): the admissible sector clusters exponentially at rate  $(2/3)^6$  (correlation length  $1/\Delta$ , finite susceptibility 729/665) and the v76 margin  $\Delta - 31/(4\pi^2) > 0$  decouples it from the ambient, so the physical sector is a convergent object while QG.AMB.01 is a [C] redundancy (v369). Each pins the residual to a single named step; with the lattice/ $S_3$  stack the keystone is now *closed modulo a cited theorem*. (iv) *The intrinsic-BW residual as a cited reduction* ([[verification/v424\\_seam\\_lto\\_rp\\_reduction.py](#)]): the open bedrock lemma of (ii) — the raw collar’s intrinsic Bisognano–Wichmann property — is reduced to the recent commuting-projector theorem of Naaijken–Penneys–Wallick (arXiv:2605.10693), whose reflection-positivity axiom (LTO-RP) is exactly  $u_\Theta = J$  (the reflection *is* the modular conjugation), i.e. the BW condition  $\theta K \theta = -K$  — *not*  $\omega \circ \rho = \omega$ . [E] the  $\mu_4$  clock is a modular *symmetry* ( $[\rho, K]=0$ ), not the flow ( $\sigma^\psi = \rho$  is type-wrong, a one-parameter group versus a finite order-4 element); a positive character-block-diagonal  $K$  has  $[\rho, K]=0$  yet, being positive definite, never  $\theta K \theta = -K$ , so clock-invariance does *not* imply (LTO-RP). [O] make-or-break: NPW26 prove (LTO-RP) for *topologically ordered* commuting-projector models (Toric Code via a trace, Levin–Wen anyonic), whereas the seam is the opposite corner — invertible ( $|\det \text{Cartan}(E_8)|=1$ , no anyons) and a  $\beta=1$  KMS state (not a trace) — so two sub-steps stay open: realising the raw seam as a  $\mathbb{Z}_2$ -reflection commuting-projector LTO, and extending (LTO-RP) to the invertible KMS case. An MMST-style reduction (cf. v308/v336), *not* a closure. *Sub-step (ii) is now constructively exhibited* ([[verification/v426\\_seam\\_lto\\_rp\\_kms.py](#)]): a gapped reflection-symmetric  $\mu_4$ -even collar at  $\beta=1$  KMS satisfies  $\Theta K \Theta^{-1} = -K$  ( $u_\Theta = J$ ) and  $[\rho, K]=0$ , so (LTO-RP) holds there by direct Tomita–Takesaki — the corner NPW26 leave open — with  $|\det \text{Cartan}(E_8)|=1$  (invertible, no anyons) and the state a genuine KMS, not a trace; only the continuum sub-step (i) remains [O]. On the covariance side, the recent theorem of Adamo–Giorgetti–Tanimoto (arXiv:2508.07109, 2025) — every DHR representation of a conformal net extending a loop-group/Virasoro net is *automatically* positive-energy and diffeomorphism-covariant — defuses the old “Bisognano–Wichmann presupposes the very covariance it should produce” circularity (v215): once the seam net is in hand, geometric covariance is a *consequence*, not an input.

**Twenty-six convergent attacks on the one continuum residual — intrinsic BW, a bulk continuum motor and its chiral-edge counterpart, the  $\beta$ -homotopy, the applicability audit, uniform rigidity with its forcing converse and the derivation of clock-invariance, three independent reads of the edge central charge, the four-point and uniform-in- $N$  pinning of the external fact, the Ising operator tower, the  $(E_8)_1$  torus modular data, the  $\mu_4$ -from-marks derivation, the kill tests, the lattice Sugawara current algebra, a uniform-in- $N$  Tomita–Takesaki tower, the chirality-from- $P_1$  forcing, an  $(E_8)_1$  character kill test, the exact MMST citation audit, the complementary lattice-VOA route, the explicit flat-band commuting-projector LTO realisation, the strict-locality (Kapustin–Fidkowski) obstruction, the character-level 128-spinor extension, the  $c=8$  holomorphic-uniqueness pin and the one-particle realisation rigidity** ([[verification/v438\\_seam\\_hsmi\\_borchers.py](#)], [[verification/v439\\_seam\\_lieb\\_robinson.py](#)], [[verification/v440\\_seam\\_lto\\_rp\\_beta.py](#)], [[verification/v441\\_seam\\_applicability\\_ledger.py](#)], [[verification/v442\\_seam\\_rigidity\\_uniform.py](#)], [[verification/v443\\_seam\\_e8\\_discriminator.py](#)], [[verification/v444\\_seam\\_edge\\_scaling.py](#)], [[verification/v445\\_seam\\_rigidity\\_forcing.py](#)], [[verification/v446\\_seam\\_clock\\_invariance.py](#)], [[verification/v447\\_seam\\_edge\\_chern.py](#)], [[verification/v448\\_seam\\_edge\\_fourpoint.py](#)], [[verification/v449\\_seam\\_mmst\\_uniform.py](#)], [[verification/v450\\_seam\\_edge\\_entanglement.py](#)], [[verification/v451\\_seam\\_edge\\_cardy\\_tower.py](#)], [[verification/v452\\_seam\\_e8\\_modular.py](#)], [[verification/v453\\_seam\\_mu4\\_from\\_marks.py](#)], [[verification/v454\\_seam\\_edge\\_virasoro.py](#)], [[verification/v455\\_seam\\_bw\\_uniform.py](#)], [[verification/v456\\_seam\\_chirality\\_from\\_c3.py](#)], [[verification/v457\\_seam\\_character\\_killtest.py](#)], [[verification/v458\\_seam\\_mmst\\_citation\\_audit.py](#)], [[verification/v459\\_seam\\_lattice\\_voa\\_route.py](#)], [[verification/v460\\_seam\\_s3\\_lto.py](#)], [[verification/v461\\_seam\\_strict\\_locality.py](#)], [[verification/v462\\_seam\\_spinor\\_continuum.py](#)], [[verification/v463\\_seam\\_c8\\_holomorphic\\_uniqueness.py](#)], [[verification/v464\\_seam\\_oneparticle\\_rigidity.py](#)], [[verification/v469\\_seam\\_crossedproduct\\_route.py](#)]). None closes SEAM.EQUIV.01; together they ground the one continuum input on both its halves, reduce it to a single analytic fact and make the claim falsifiable. (v) *The intrinsic Bisognano–Wichmann route* ([[verification/v438\\_seam\\_hsmi\\_borchers.py](#)]): the seam supplies a Longo *standard pair* / +half-sided modular inclusion — a positive lightlike translation  $P \geq 0$ , the boost  $K$  with  $[K, P]=P$  (the

Borchers scaling  $\Delta^{it}P\Delta^{-it}=e^{-t}P$ ), and a reflection  $J$  with  $JKJ=-K$ ,  $JPJ=P_-$  — so by Borchers’ theorem the modular flow is geometric *intrinsically, without* presupposing the rotation-covariance of the vacuum; [E] the ladder algebra and the  $sl(2, \mathbb{R})$  positivity (min eig  $> 0$ , with the indefinite boost as control), [C] the synthesis, [O] assembling the finite algebra and the positive-energy rep into one continuum standard pair (the MMST/NPW26 wall). (vi) *The continuum-existence motor* ([verification/v439\_seam\_lieb\_robinson.py]): the explicit  $p+ip$  collar (v367) has a uniform-in- $N$  gap, exponential clustering with  $N$ -independent  $\xi$ , and a finite Lieb–Robinson light cone, so by Nachtergaele–Sims its *bulk* thermodynamic limit exists — grounding the bulk half of the cited MMST existence in our model and narrowing the open step to the chiral massless *edge* scaling limit; [E] gap/clustering/light-cone, [C] the lift, [O] the edge CFT. (vii) *The  $\beta$ -homotopy* ([verification/v440\_seam\_lto\_rp\_beta.py]): the (LTO-RP)/BW data ( $\Theta K \Theta = -K$ ,  $[\rho, K]=0$ ) are  $\beta$ -independent, and along the whole path  $C_\beta=1/(1+e^{\beta K})$  from the NPW26-proven *trace* state ( $\beta=0$ ) to the seam KMS state ( $\beta=1$ ) every covariance is reflection-related,  $\mu_4$ -invariant and valid — a norm-continuous, obstruction-free homotopy, upgrading the single point of v426 to a path argument for sub-step (ii). (viii) *The applicability audit* ([verification/v441\_seam\_applicability\_ledger.py]): a hypothesis-by-hypothesis ledger of MMST/NPW26/Adamo finds 11 of 12 hypotheses are TFPT-internal computed facts ([E]) and exactly one is external — the continuum existence of the chiral scaling limit — covariance being an Adamo consequence, not an input; the keystone’s external dependence is pinned to that one analytic fact. (ix) *Uniform rigidity* ([verification/v442\_seam\_rigidity\_uniform.py]): the band-limited  $\mu_4$ -character rigidity of v398 is uniform in  $N$  — exact 4-sector commutation to  $N=256$ , a uniform gap, and an  $N$ -independent clock-breaking lower bound  $\|[\rho, V]\|=2$  — so it is not a small- $N$  artifact (the full- $L^2$  forcing stays [O]). (x) *The kill tests* ([verification/v443\_seam\_e8\_discriminator.py]): mode D of the closure-mode classification (v347) as concrete *falsifiable* discriminators against the nearest  $c=8$  rival  $SO(16)_1$  — torus GSD 1 vs 4, level-1 currents 248 vs 120, lattice theta 240 vs 112 norm-2 roots — with  $c=8$  alone unable to decide (v277) and only  $\det K=1$  (the genus-1 statement, v90) selecting  $E_8$ ; a seam with GSD $>1$  would falsify  $(E_8)_1$ . [X] the tests, [O] the realisation. (xi) *The chiral-edge scaling motor* ([verification/v444\_seam\_edge\_scaling.py]): the counterpart of (vi) on the *edge*, attacking exactly the chiral massless scaling limit that the bulk motor left open. On the same  $p+ip$  collar (v367/v368) the lattice edge already carries the conformal data of a chiral  $c=\frac{1}{2}$  Majorana CFT — a linear, chiral in-gap dispersion (velocity  $v \approx 0.98$ ,  $R^2=0.9999$ , with *opposite* per-edge velocities), an *algebraic* edge two-point  $\sim r^{-\eta}$  with  $\eta \approx 1$  (versus the *exponential* bulk row and the exponential trivial-phase control), and a Cauchy-convergent scaling-collapse under refinement ( $\eta=1.001 \pm 0.007$ , successive spread  $0.028 \rightarrow 0.013$ ) — so the 16-copy edge is a chiral  $c_-=8$   $(E_8)_1$  CFT; [E] the kinematics, correlator and collapse, [C] the central-charge reading, [O] the rigorous uniform-convergence theorem (cited MMST). With (vi) for the bulk and (xi) for the edge, *both halves* of the one continuum residual are numerically grounded and pinned to a single cited theorem. (xii) *The rigidity forcing converse* ([verification/v445\_seam\_rigidity\_forcing.py]): the converse of (ix), upgrading rigidity from “block-diagonal *permits* commuting” to “commuting with the order-4 clock *forces* block-diagonal”. For  $\rho = \text{diag}(i^n)$ ,  $[\rho, K]=0$  iff  $K_{ij}=0$  whenever  $i \neq j \bmod 4$  (verified entrywise, swept  $N=4..128$ ), so the commutant is *exactly* the 4-sector block algebra:  $\dim\{K : [\rho, K]=0\} = \sum_s n_s^2$ , a proper subspace of the full  $N^2$  algebra ( $N=16$ :  $64 < 256$ ); every off-block unit has  $\|[\rho, E_{ij}]\| \geq \sqrt{2}$  (zero slack), and the order-2 clock leaves a strictly larger commutant (64 vs 128) — the *four* Gauss–Bonnet marks (order  $4=|\mu_4|=h(A_3)$ ) force strictly more structure than two, and only the index-4 extension is  $(E_8)_1$  ( $\det=1$ , v281/v154/v351). [E] the entrywise iff, exact commutant dimension, off-block sharpness and order discriminator; [C] the synthesis (RP-definability + clock-invariance *force* the block-diagonal  $\Lambda_\Sigma$ , v442’s “permit” upgraded to “force”); [O] the single residual is now precisely that the physical transfer lies in the commutant (clock-invariance = QGEO.SYM.01, v323/v335). (xiii) *Clock-invariance derived* ([verification/v446\_seam\_clock\_invariance.py]): closing (xii)’s residual one notch further — the operator condition  $[\rho, \Lambda_\Sigma]=0$  is itself a *consequence* of a structural symmetry, not an extra assumption. On explicit finite OS data, the canonical transfer  $\Lambda=C(1)C(0)^{-1}=e^{-h}$  (the transfer fixed by RP+reflection, v54) of a  $\mu_4$ -invariant covariance satisfies  $[\rho, \Lambda]=0$  (swept  $N=4..32$ ; iff-sharp  $\mu_4$ -breaking neg control),  $\Lambda$  is a positive contraction, and the free modular Hamiltonian  $K=\log((1-C)C^{-1})$  (v258) of the invariant state has  $[\rho, K]=0$  (Tomita–Takesaki). [E] the inheritance, contraction and modular commutation; [C] so with (xii), RP-definability *plus* the four marks force the block-diagonal  $\Lambda_\Sigma$ ; [O] the residual reduces from an operator commutation to the structural “the seam RP data is  $\mu_4$ -symmetric” (QGEO.SYM.01, v216/v323). (xiv) *The independent topological edge reading* ([verification/v447\_seam\_edge\_chern.py]): a second, fit-free route to the edge central charge of (xi). On the same  $p+ip$  collar (v367/v368) the bulk Chern number is the integer  $C=+1$  (Fukui–Hatsugai–Suzuki link variables;  $C=0$  in the trivial

phase,  $C=-1$  at the opposite mass, matching (xi)’s opposite per-edge velocities), the open strip carries exactly  $|C|=1$  chiral edge branch per boundary, and the Li–Haldane entanglement spectrum has in-gap modes at  $\frac{1}{2}$  only in the topological phase; [E] the Chern integer, channel count and entanglement signature, [C] the bulk–edge reading  $c_- = N_{\text{Maj}} \cdot \frac{1}{2} \cdot |C| = 16 \cdot \frac{1}{2} = 8 = g_{\text{car}} + N_{\text{fam}}$  — the same  $c_- = 8$  as (xi) from a logically independent (topological) observable. (xv) *The four-point pinning of the external fact* ([verification/v448\_seam\_edge\_fourpoint.py]): extending (xi)’s two-point scaling limit to the next order to support exactly the one external fact (viii) isolates. The edge state is quasi-free (the ground-state correlation matrix is an exact projector  $G^2 = G \sim 10^{-13}$ , the MMST CAR hypothesis), and the edge four-point cross-ratio  $Q = [G_{01}G_{23}]/[G_{02}G_{13}]$  at commensurate positions is Cauchy-convergent under refinement to the free-fermion conformal value  $\sin(5\pi/12)/\sin(\pi/12) = 2 + \sqrt{3} \approx 3.732$  (the bulk control off-scale, exponential); [E] the quasi-free property, four-point convergence and conformal form, [C] so the one external MMST fact is empirically supported at both two- and four-point order. (xvi) *The uniform-in- $N$  pinning* ([verification/v449\_seam\_mmst\_uniform.py]): strengthening (xv) from convergence at a single cross-ratio to the *uniformity* the cited theorem asserts — two *independent* edge four-point cross-ratios converge to *distinct* conformal values ( $Q(1, 2, 4, 7)/12 \rightarrow 2 + \sqrt{3}$  and  $Q(1, 3, 5, 9)/12 \rightarrow 2$ ) at one  $N$ -independent  $1/N$  rate ( $|Q(N) - Q_\infty| \cdot N \leq 4$  for both,  $N_x \in \{36, 48, 60\}$ ), the bulk control off-scale; [E] the two limits and the uniform rate, [C] so the MMST uniform-convergence *hypothesis* is empirically met. (xvii) *The entanglement reading of  $c_-$*  ([verification/v450\_seam\_edge\_entanglement.py]): a *third*, correlator-independent route to the edge central charge — the Calabrese–Cardy block-entropy law gives  $c_{\text{Dirac}} = 1.000$  for the critical free-fermion chain (vs a saturating,  $\ell$ -independent gapped control), so  $c$  per Majorana  $= \frac{1}{2}$  and  $c_- = 16 \cdot \frac{1}{2} = 8 = g_{\text{car}} + N_{\text{fam}}$ ; [E] the law and the control, [C] the assembly. (xviii) *The Ising operator tower* ([verification/v451\_seam\_edge\_cardy\_tower.py]): naming the edge CFT by its *operator content* — the Cardy finite-size spectrum reproduces the exact Ising chiral primaries  $h_\sigma = \frac{1}{16}$  (the periodic–antiperiodic ground split) and  $h_\epsilon = \frac{1}{2}$  (the lowest NS pair gap), both  $L$ -independent, so  $\{c, h_\sigma, h_\epsilon\} = \{\frac{1}{2}, \frac{1}{16}, \frac{1}{2}\}$  is the free-Majorana minimal model  $M(3, 4)$ ; [E] the dimensions and conformal scaling, [C] the identification. (xix) *The  $(E_8)_1$  torus modular data* ([verification/v452\_seam\_e8\_modular.py]): pinning the  $(E_8)_1$  identity on the *torus*, complementing the edge reads (xi)/(xiv)/(xvii)/(xviii) — the character  $\chi = E_4/\eta^8$  is  $S$ -invariant ( $\chi(-1/\tau) = \chi(\tau)$ , one primary),  $T$ -multiplies by  $e^{-2\pi i/3} = e^{-2\pi i c/24}$  ( $c=8$ ) and is holomorphic ( $c \equiv 0 \pmod{8}$ ,  $T$ -phase order 3); [E] the full modular signature. (xx) *The  $\mu_4$ -from-marks derivation* ([verification/v453\_seam\_mu4\_from\_marks.py]): folding the last structural premise into the existing one — the four Gauss–Bonnet marks are  $\mu_4$  (roots of  $z^4 - 1$ ), the form basis  $\omega_k$  is  $\mu_4$ -graded ( $\rightarrow i^k \omega_k$ , weights  $(1, 2, 3) = A_3$  exponents) and the order-4 clock  $z \rightarrow iz$  fixes their cross-ratio  $\lambda = 2$  (the  $D_4$  stabiliser, v168), so the residual QGEO.SYM.01 of (xiii) follows from marks  $= \mu_4$  plus the existing QGEO.REALIZE.01 — [E] the three mark facts, [C] no new premise: the chain (xx)  $\rightarrow$  (xiii)  $\rightarrow$  (xii) closes modulo realisation. The integer backbone of (xiv)–(xix) ( $c_- = 8$ , the Chern integers, the order-3  $T$ -phase) is kernel-hardened in Lean axiom-free as FORM.SEAM.EDGE.CHERN.01. (xxi) *The lattice Sugawara current algebra* ([verification/v454\_seam\_edge\_virasoro.py]): testing the cited MMST “Koo–Saleur generators  $\rightarrow$  continuum Virasoro with the correct  $c$ ” fact on the edge *algebra*, not just its central charge. The critical free-fermion ring has the Affleck–Cardy Casimir energy  $E_0(L) = e_\infty L - \pi v c / (6L)$  fitting  $c_{\text{Dirac}} = 1.000$  ( $\frac{1}{2}$  per Majorana,  $L = 64.1024$ ), the lattice  $U(1)$  current closes its affine *level* exactly ( $\langle 0 | J_n J_{-n} | 0 \rangle = n$ ,  $n = 1..8$ ; vanishing for a gapped control), and the level-1 Sugawara central charge  $c = \dim G / (1 + h^\vee) = 8$  for both  $SO(16)_1$  (120/15) and  $(E_8)_1$  (248/31); [E] Casimir, Kac–Moody level and Sugawara, [C]  $16 \cdot \frac{1}{2} = 8$  is the level-1 current-algebra  $c_-$ . (xxii) *The uniform-in- $N$  Tomita–Takesaki tower* ([verification/v455\_seam\_bw\_uniform.py]): the inductive-limit companion of v426 for sub-step (i) of (iv) — a family of reflection-odd,  $\mu_4$ -even, gapped collars ( $N = 8..128$ ) on which the intrinsic Bisognano–Wichmann condition  $u_\Theta = J$  ( $\|\Theta K \Theta + K\| = 0$ ) holds *exactly at every  $N$* , with an  $N$ -independent gap ( $6 \log \frac{3}{2}$ ) and an  $N$ -independent neg-control margin ( $\|\Theta K \Theta + K\| = 2\|K\|$  for a side-even reflection); [E] the uniform BW, gap, clock symmetry and teeth, [C][O] so the inductive limit inherits  $u_\Theta = J$  (the abstract continuum theorem stays the cited backbone). (xxiii) *The chirality-from- $P_1$  forcing* ([verification/v456\_seam\_chirality\_from\_c3.py]): the S3 input of the continuum chain (v356) is removed as a free choice —  $c_3$ ’s “8” is the one-sided count  $8 = |\mathbb{Z}_2| \cdot (\int K/\pi) = 2 \cdot 4$  ( $8\pi = |\mathbb{Z}_2| \int K = 2 \cdot 2\pi \chi(S^2)$ , v73), and a reflection sends the edge Chern integer  $C \rightarrow -C$  (verified  $C = +1 \rightarrow -1$  on the explicit  $p+ip$  model), so any *two-sided* (reflection-symmetric) gapped boundary forces  $C = -C = 0$  while the *one-sided* seam — the same  $|\mathbb{Z}_2|$  quotient that gives  $c_3$  its  $8\pi$  — has no such reflection and forces  $C \neq 0$ ; [E] the one-sided count and the  $C \rightarrow -C$  flip, [C] chirality (S3) is a *consequence* of  $P_1$ , magnitude  $c_- = 8$ . (xxiv) *The  $(E_8)_1$  character kill test* ([verification/v457\_seam\_character\_killtest.py]): a *falsifiable* sharpening of the (x)/(x’)



discriminators — the  $(E_8)_1$  tower  $\chi=E_4/\eta^8=q^{-1/3}(1+248q+4124q^2+34752q^3+\dots)$ , the weight-1 count  $248=120+128$  (vs the free 120) and the sector count  $|\det \text{Cartan } E_8|=1$  (vs  $|\det \text{Cartan } D_8|=4$ ); a measured 120, four sectors, or a wrong tower coefficient would *falsify*  $(E_8)_1$ , and the data give  $248/1/\{1, 248, 4124\}$  (passed). [X] the kill criterion, [E] the exact discriminators. (xxv) *The exact MMST citation audit* ([[verification/v458\\_seam\\_mmst\\_citation\\_audit.py](#)]): a hypothesis-by-hypothesis reading of the cited MMST theorem (arXiv:2107.13834) against the collar — H1 CAR class, H2 massless limit, H3 the per-copy  $c=\frac{1}{2}$  Thm 4.11, H4 decoupled additivity to  $c=8$ , and H5 the WZW range  $8 \leq 8 \leq 16$  with its 120 bilinear currents all *match*, isolating the single residual H6 to the 128-spinor extension  $SO(16)_1 \rightarrow (E_8)_1$  (the  $\det K \rightarrow 1$  holomorphy step), *outside* MMST’s bilinear method; [E] the per-copy theorem, [C] the additivity and range, [O] the one open spinor extension — since (xxxii) certified at net level by the peer-reviewed crossed product. (xxvi) *The complementary lattice-VOA route* ([[verification/v459\\_seam\\_lattice\\_voa\\_route.py](#)]): the second, independent construction that supplies exactly H6 — for the even unimodular  $E_8$  the lattice net  $A_{E_8}$  (Adamo–Giorgetti–Tanimoto, arXiv:2506.01008; OS arXiv:2407.18222) has weight-1 space  $248=8+240$  with the 240 roots splitting  $112$  integer +  $128$  *half-integer* (the spinor,  $248=120+128$ ), so the lattice route builds the 128-spinor currents MMST (xxv) leaves open; [E] the even unimodularity and root split, [C] AGT constructs the holomorphic extension — since (xxxii) an *independent second witness*, the primary net-level certification being the peer-reviewed crossed product — [O] the routes leave one shared realisation input (the collar’s scaling limit  $= A_{E_8}$ ), tied to  $P_1$  by (xxiii). The G-block arithmetic (the one-sided  $8=\mathbb{Z}_2 \int K/\pi$ , the reflection  $C \rightarrow -C$ ,  $c_-=8$ , the range  $8 \leq 8 \leq 16$ ,  $248=8+240=120+128$ ,  $240=112+128$ ,  $\det K \ 1$  vs  $4$ ) is kernel-hardened in Lean (FORM.SEAM.RESIDUAL.01, [TfptCarrier/SeamResidualAxiom.lean](#)), reducing the residual to ONE named TFPT-internal realisation axiom (CollarRealizedAsLatticePhase) plus the two cited existence theorems. (xxvii) *The explicit flat-band commuting-projector (LTO) realisation* ([[verification/v460\\_seam\\_s3\\_lto.py](#)]): NPW26 prove the intrinsic Bisognano–Wichmann lemma of (iv) for *commuting-projector* (local-topological-order) models, so sub-step (i) asks for the seam realised as such a  $\mathbb{Z}_2$ -reflection LTO. On the same v367  $p+ip$  collar the flattened Bloch  $Q=h/|h|$  is a frustration-free flat-band parent —  $Q^2=I$  and the occupied  $P=(I-Q)/2$  is an exact projector EQUAL to the ground-state projector — whose real-space kernel is gap-protected quasi-local ( $\xi \approx 1.14$  at  $M=1$ ,  $|P(20,0)| < 10^{-6}$ ; a clean power law  $|R|^{-2.1}$  at the gapless point, so locality needs the gap), and whose flattened modular Hamiltonian  $K_{\text{flat}} = \text{sign}(K)$  inherits the  $\mathbb{Z}_2$  reflection  $\Theta K_{\text{flat}} \Theta = -K_{\text{flat}}$  and the  $\mu_4$  clock  $[\rho, K_{\text{flat}}]=0$  EXACTLY; [E] the flat-band/projector and reflection algebra, [C] the quasi-locality, [O] the chiral  $c_-=8 \neq 0$  obstructs *strictly* finite-range commuting-projector locality (Kapustin–Fidkowski, arXiv:1810.07756), so the realisation is exactly the quasi-local LTO net NPW26 use — advancing sub-step (i) of (iv) from abstract existence to an explicit gap-protected net, strict locality the cited residual. (xxviii) *The strict-locality obstruction made explicit* ([[verification/v461\\_seam\\_strict\\_locality.py](#)]): upgrading (xxvii)’s “strict locality is the cited residual” from a citation to an EXHIBITED topological obstruction. On the same v367 collar the FHS Chern is  $C=+1$  (chiral) /  $0$  (trivial) /  $-1$  (opposite mass), and the hybrid Wannier-centre (Wilson-loop) phase  $\theta(k_y)$  winds by  $|C|=1$  (chiral) and  $0$  (trivial) over the BZ — exponentially-localised Wannier functions exist *iff* that winding is  $0$  (Brouder–Panati et al.; Thouless), so the chiral collar has none; both phases are gapped, yet only the trivial one (winding  $0$ ) admits a strict finite-range commuting projector, so it is the *chirality*  $c_-=8 \neq 0$ , not the gap, that obstructs. [E] the Chern integer, the winding  $= |C|$  identity and the gapped neg-control, [C][O] Kapustin–Fidkowski (arXiv:1810.07756): a strictly finite-range commuting projector would force winding  $0$  hence  $C=0$ , so for  $C=1 \neq 0$  none exists — sub-step (i)’s realisation is *provably* the quasi-local NPW26 LTO net, strict locality topologically forbidden rather than a missing premise. (xxix) *The character-level 128-spinor extension* ([[verification/v462\\_seam\\_spinor\\_continuum.py](#)]): the constructive companion of (xxv)/(xxvi), exhibiting the 128-spinor extension three ways — the exact Jacobi/ $E_8$  identity  $\theta_2^8 + \theta_3^8 + \theta_4^8 = 2E_4$  makes the holomorphic character the sum of the  $SO(16)_1$  vacuum and spinor sectors,  $\chi_{(E_8)_1} = E_4/\eta^8 = \chi_o^{SO16} + \chi_s^{SO16}$  (weight-1 counts  $120+128=248$ ); the finite 16-Majorana Fock space already carries them ( $C(16,2)=120$  currents and  $2^{16/2}=256=128+128$  Ramond states); and a critical free-fermion ring converges to the chiral CFT (Casimir  $c_- \rightarrow 8$ , lowest scaled gap  $x_1 \rightarrow 1$  with  $|x_1-1| \sim 1/N^2$ ). [E] the  $\theta$  identity, the  $120+128=248$  split and the finite- $L$  convergence, [C][O] the rigorous holomorphic extension is certified at net level by the crossed product (xxxii), with AGT/AMT (xxvi) as the independent second witness. (xxx) *The  $c=8$  holomorphic-uniqueness pin* ([[verification/v463\\_seam\\_c8\\_holomorphic\\_uniqueness.py](#)]): the positive lever that makes the *identification* half classification-forced rather than assumed. At level 1 the central charge  $c = \dim \mathfrak{g}/(1+h^\vee)=8$  has *three* simple solutions —  $A_8$  ( $\dim 80$ ),  $D_8=\mathfrak{so}(16)$  ( $\dim 120$ ) and  $E_8$  ( $\dim 248$ ) — so  $c=8$  is necessary but *not* sufficient, and the  $\det K=4$   $SO(16)$  rival of (xxiii) is a



genuine same- $c$  competitor, not a strawman. But a holomorphic (single-sector)  $c=8$  theory has an  $SL(2, \mathbb{Z})$ -covariant character, and the only weight-0 form with leading  $q^{-1/3}$  and integer coefficients is  $j^{1/3}=E_4/\eta^8$ , whose  $q^1$  coefficient (the one-dimensional candidate space, vacuum  $a_0=1$ ) is 248 — so  $\dim V_1=248$  is *forced*, selecting  $E_8$  uniquely and excluding  $A_8/D_8$ . [E] the three candidates, the forced 248, the tower  $\{1, 248, 4124, 34752, 213126\}=E_4/\eta^8=j^{1/3}$ ; [C] Dong–Mason/ Schellekens give the holomorphic  $c=8$  VOA unique  $= V_{E_8}$  (the even unimodular rank-8 lattice unique  $= E_8$ , Mordell–Witt), so “ $c=8$  holomorphic limit  $=(E_8)_1$ ” is classification-forced. (xxxi) *The one-particle realisation rigidity* ([verification/v464\_seam\_oneparticle\_rigidity.py]): attacking the *realisation* input  $R_1$  itself. Since the seam is quasi-free (v113/v160/v161), by Araki’s self-dual CAR the whole net is the second quantisation of its one-particle symbol  $P$ , so  $R_1$  collapses to a finite-per-mode statement about  $P$ : the gapped ground state is the *unique* quasi-free state with  $P=$  the spectral projection onto  $K<0$  ( $P^2=P$  to  $<10^{-12}$ ,  $K$  fixed by gap + chirality + reflection v426); its real-space kernel is Cauchy on every fixed window ( $\max |P_{2N}-P_N|\sim 1/N\rightarrow 0$ , so  $P_\infty$  exists), and the limit carries  $c=1$  per Dirac (entanglement  $S(L)=(c/3)\ln L$ , 16 copies  $\Rightarrow c_-=8$ , tying v450). [E] idempotency, Cauchy convergence and the  $c$ -fit; [C] Araki/Shale–Stinespring give the state $\leftrightarrow$ symbol bijection and implementable Bogoliubov map, so  $R_1$  is upgraded from a bald axiom to “the unique quasi-free realisation, its one-particle limit exhibited, modulo the named CAR functor”. (xxxii) *The net-level crossed-product route + the invariant-level realisation* ([verification/v469\_seam\_crossedproduct\_route.py]): both halves of the residual re-founded on harder-to-reject ground. The 128-spinor extension needs no preprint: the  $SO(16)_1$  spinor has  $h_s=N/16=16/16=1\in\mathbb{Z}$ , so its statistics phase is  $+1$  — *exactly* the Longo–Rehren locality criterion — and the  $\mathbb{Z}_2$  simple-current crossed product  $SO(16)_1\rtimes\{1, s\}$  is a *local* index-2 extension by 1995–2001 *peer-reviewed* subfactor theory (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1  $so(N)$  net with its DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions of nets: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM (Comm. Math. Phys. 219 (2001)) gives  $\mu(B)=4/2^2=1\Rightarrow$  holomorphic  $\Rightarrow (E_8)_1$  by the (xxx) pin — so the AGT/AMT route (xxvi) is *demoted to an independent second witness*. The index-4  $\mu_4$  glue runs on the same integer,  $h(J^k)=\{1, 1, 1\}$  for  $k=1, 2, 3$  (the v125 isotropy  $q(k(1, 1))=k^2$  at net level,  $\mu=16/4^2=1$ ). And the realisation input  $R_1$  is reduced from model fiat to *invariants* ( $R_1'$ ): quasi-free [C] (v155/v160) + gap [E] (v302) + class D +  $c_-=8$  [E] ((xxiii), from  $P_1$ ) — computed on the collar as FHS Chern  $|C|=1$  (control 0),  $\nu=16\times|C|=16$ ,  $c_-=\nu/2=8$ ; and  $\nu\equiv 0 \pmod{16}$  is exactly Kitaev’s class-D phase whose edge admits a purely bosonic description = the  $(E_8)_1$  level-1 state (the sixteen-fold way;  $c_-$  a *phase* invariant, Kapustin–Spodyneiko; Kim et al.), so any realisation with the  $R_1'$  invariants is phase-equivalent to the v367 stack. [E] the locality integers, the KLM  $\mu$ -arithmetic, the computed invariants and the composed chain; [C] the re-founded citation package; [O] the cited theorems stay cited. The full G-block arithmetic — now including the (xxviii) winding obstruction, the (xxix) finite-Fock spinor counts, the (xxx) holomorphy selector and the (xxxii) locality/ $\mu$ /sixteen-fold joints — is kernel-hardened in Lean (FORM.SEAM.RESIDUAL.01), the two formerly separate cited theorems are *merged* into one combined external axiom `seam_realisation_theorem`, so the residual reduces to ONE realisation axiom plus ONE cited theorem (`#print axioms` drops from six to four); and the (xxxii) route is Lean-carried as the *parallel* derivation `seamResidualClosed'` (the invariant-level axiom `collar_invariants` plus the single combined package `crossedproduct_route_theorem`, its joints — `lr_locality_integer`, `glue_locality_integers`, `klm_holomorphic`, `sixteenfold_pin` — kernel facts with no axioms). Across (v)–(xxxii), SEAM.EQUIV.01 stays [O]: with (xxx) pinning the identification, (xxxi) reducing the realisation to its one-particle data and (xxxii) re-founding the extension leg on peer-reviewed ground, the residual is now *entirely certification* — a named, hypothesis-audited package of published theorems (MMST, Longo–Rehren/Böckenhauer/Böckenhauer–Evans/KLM, Dong–Mason, Araki, Shale–Stinespring; AGT/AMT as second witness) with no open internal mechanism — yet still rests on cited continuum-existence theorems we do not re-prove.

The TOE/QFT closure contracts as three citable certificates ([verification/v398\_seam\_state\_rigidity.py], [verification/v399\_gravity\_complete.py], [verification/v401\_metrology\_closure.py]). Three companion contracts package the closure status as named targets without fabricating any closure. (i) *Seam State Rigidity* (SEAM.RIGIDITY.01): the scattered state-invariance reductions (v177/v199/v308/v309/v335) as ONE theorem target — [E] the RP-definability hinge (OS canonical transfer from RP + reflection alone, v54, quasi-free v155), [E] the band-limited  $\mu_4$ -character-block-diagonal core swept  $N=4..64$  (beyond the 3-dim  $H^1$ , made uniform in  $N$  to  $N=256$  by v442), [E] holomorphy  $\Rightarrow (E_8)_1$ ; and [E] the forcing converse (v445) that commuting with the order-4 clock is *exactly*  $\mu_4$ -block-diagonality, so the residual is no longer “does block-diagonal permit” but only  $[\rho, \Lambda_\Sigma]=0$  on the full  $L^2$  — the physical transfer lying in that

commutant (intrinsic Bisognano–Wichmann). [E] Moreover (v446) that operator commutation is itself *derived*: the OS canonical transfer of  $\mu_4$ -symmetric RP data inherits the clock (with the modular Hamiltonian commuting, v258), so the residual reduces further to the *structural* statement that the seam RP data is  $\mu_4$ -symmetric (the four marks, QGEO.SYM.01), leaving [O] only that structural symmetry on the raw continuum collar — for which the four marks now have an *established* continuum frame: the four-interval multilocal modular geometry (v480, keybox below), where the invariance is manifest at the state level in the exactly solvable class. (ii) *Gravity-complete = Boundary-complete + Ambient-redundancy* (GRAVITY.COMPLETE.01): the local equation parameter-free (v358/v359) plus the redundancy discriminators (v369), with  $\mathcal{O}_{\text{phys}} \subset \mathcal{A}_\Sigma$  checked over five gravitational readout classes. (iii) *Metrology closure* (METROLOGY.CLOSURE.01): the final input set  $\{a, \pi, v_{\text{geo}}\}$  — 0 dimensionless dials + 1 unit, the dimensional bookkeeping that every dimensionful observable is a number  $\times v_{\text{geo}}^d$  (No-Unit, v153), and  $v_{\text{geo}}$  theorem-forbidden ( $\text{TOE}_{\text{strict}} = \text{TOE}_{\text{dimensionless}} + [v_{\text{geo}}]$ , v384).

**One surface for seam, carrier and lattice: the Kummer/K3** ([verification/v260\_k3\_kummer\_unification.py]). The pillowcase is the elliptic-involution quotient  $T_{\tau=i}^2/(z \mapsto -z)$ , its four marks the four 2-torsion fixed points; squaring the construction to complex dimension two gives the *Kummer surface*  $\text{Km}(T^2 \times T^2)$ , a K3, whose three facets reproduce three TFPT objects at once. [E] the 16 nodes (the 2-torsion  $|A[2]| = 2^4 = 16$  of the abelian surface) resolve to 16 exceptional  $\mathbb{P}^1$ 's =  $\dim S^+ = \Lambda^{\text{even}}(\mathbb{C}^5) = 16$  (v197), the carrier-16 as the *square* of the seam-4; [E] the cup-product form on  $H^2(\text{K3}, \mathbb{Z})$  is the even unimodular lattice  $U^3 \oplus E_8(-1)^2$  of signature (3, 19), so the same  $E_8$  the carrier glues to (v1) is the surface's intersection form. (Honestly: the 16 node classes span the rank-16 Kummer lattice, a sublattice *distinct* from the  $E_8(-1)^2$  summand — both exact, not identified.) Thus the v1 “glue  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ ” is the lattice shadow of a single smooth object that already carries the seam and the carrier;  $E_8$  appears both *locally* as the du Val  $\mathbb{C}^2/2I$  singularity (the Poincaré-sphere link, v232/v236; by Kronheimer that local model *is* the hyper-Kähler quotient of the marks quiver, v479) and *globally* as this  $H^2$  summand. A unification (ARCH.K3.01), *not* a closure: the lattice arithmetic is [E], the carrier  $\leftrightarrow$  node identification [C].

**Synthesis — the Modular Spectral Closure** ([verification/v261\_modular\_spectral\_closure.py]). The ten steps above are not ten independent results: they are facets of one relative spectral datum

$$\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$$

(seam algebra, KMS state, modular generator,  $\mu_4$  clock; the carrier finite triple; the relative spectral action), governed by a single *Modular Spectral Universality* statement: *the unique reflection-positive, holomorphic, index-4 KMS seam state  $\omega_\Sigma$  minimises the modular clock defect, induces via the Kasparov/covariance map (v258) the 96-dim KO-6 finite triple  $D_F$  (v252), whose inner fluctuations (v254) generate the gauged Pati–Salam spectral action with the seam KMS cutoff (v259) — so Dirac, cutoff, gauging, glue and orientability are not separate choices but readouts of one seam state. The certificate v261 pins the cross-consistency: one number  $4 = [B:A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$  (index = marks = fixed points = glue order), one carrier-16 ( $\dim S^+ = \Lambda^{\text{even}}(\mathbb{C}^5) = \text{Kummer nodes}$ ,  $H_F = 96$ ), and one gap  $\Delta = 6 \log(3/2)$  (OS mass gap = recovery rate = Lindblad gap, static and dynamic). The honest account of completeness is therefore:*

- [E]/[C] **QFT-complete (modulo one premise)**. The admissible OS sector, the boundary  $(E_8)_1$  net, the relative spectral action,  $D_F$  (as covariance induction), the cutoff (as the KMS weight) and the gauging (as inner fluctuations) are one relative object — closed once QGEO.SYM.01 (the seam realisation: *the raw quasi-free seam state is  $\mu_4$ -invariant*) is granted. That single [O] premise is the only structural input the QFT layer still consumes.
- [C] **Ambient QG — a redundancy, separate by design**. The full covariant metric-sector / ambient quantum-gravity measure (QG.AMB.01) is gap-decoupled ( $\Delta_{\text{eff}} = 1.648 > 0$ ) and is now a [C] *redundancy* (v369), *not* a blocker for the QFT layer: a certification object, not a missing piece of the boundary QFT.

So “solving the QFT” here means exactly this: the boundary QFT is a single relative object reduced to one foundational symmetry postulate, with ambient QG held apart — not a diffuse list of open interfaces. The MSC closes the boundary relative QFT object modulo QGEO.SYM.01; it does *not* assert SM-only 4D gauge unification, and the canonical 4D reading is boundary only.

#### 4D-QFT fork policy — a frozen decision tree, not an open ambiguity ([verification/v265\_qft4d\_fork\_freeze.py])

The 4D reading is a *branch policy*, machine-frozen in `qft4d_fork_freeze.json`:

- **A (canonical) — boundary only.** [C] *Boundary-only is the canonical 4D reading*: TFPT delivers the boundary QFT, the relative spectral action and the SM readout; a 4D-GUT is *not claimed by default* ( $E_8$  is the audit hull, never an unbroken 4D gauge group).
- **SM-only 4D-GUT unification is killed.** [X] the plain SM couplings do not meet (at the  $\alpha_1=\alpha_2$  scale  $\alpha_3^{-1}$  is off by  $\sim 5.6$ ; spread  $\sim 8.8$  at  $2 \times 10^{16}$  GeV, v246). This is the canonical *expectation*, not a failure.
- **B (conditional) — carrier-native Pati–Salam UV branch.** [C]/[O] permitted *only* with the  $E_8$ -allowed content  $\{1, 10, 16, 45\}$  (no **126**, v247),  $\sin^2 \theta_W = 3/8$  (v248),  $\kappa = M_{PS}/M_s$  in the frozen  $O(1)$  band (v249/v253), and a threshold model with a proton-decay kill test; carrier gauging itself stays the contract `CONTRACT.QFT4D.DIRAC.01`.
- **Proton-decay rule, now applied** ([verification/v266\_ps\_threshold\_proton.py]). [X] the explicit two-step thresholds give  $d=6$   $\tau_p(p \rightarrow e^+ \pi^0)$ : the *minimal* 16-Higgs content ( $M_{GUT} \sim 2.4 \times 10^{15}$ ) gives  $\tau_p \sim 4.4 \times 10^{33}$  yr < Super-K ( $2.4 \times 10^{34}$ ) and is *killed*; the  $E_8$ -allowed (15, 1, 1) [the single 45] raises  $M_{GUT} \sim 5.9 \times 10^{15} \Rightarrow \tau_p \sim 1.6 \times 10^{35}$  yr, proton-safe *now* but sitting at the Hyper-K reach ( $\sim 1.4 \times 10^{35}$  yr) — so branch B is *content-selected* (must be  $+(15, 1, 1)$ ) and *Hyper-K-decisive within the decade*, not an open-ended hope (only one 45 in the hull, so it stays structurally marginal;  $\tau_p$  carries an  $O(3)$  hadronic band).

The fork is thus a frozen decision tree with explicit promotion/demotion rules, *not* an open ambiguity.

**And the bedrock is now a falsifiable kill-test, not a proof-promise** ([verification/v215\_seam\_deck\_killtest.py]). A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the very covariance it should produce); the honest move is to recast `QGEO.SYM.01` as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (v198/v199/v201) gives  $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$  for the quasi-free covariance, with the principal symbol  $|k|$  commuting exactly, so the whole residual sits in the bounded sub-principal  $M_f$  (equivalently: the raw Gaussian bulk form  $A$  is  $\mu_4$ -deck-invariant). The four levers are **K1** exactly four marks ( $b_1 = \# \text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \# \text{marks} = 4 = |\mu_4|$ ); **K2** the *square* configuration (cross-ratio 2  $\Rightarrow j=1728$ ,  $\text{Aut}=\mathbb{Z}/4$ ; the clock order  $h(A_3)=4$  selects  $\mathbb{Z}/4$ , not the  $j=0$   $\mathbb{Z}/6$ , and a generic mark set has only  $\mathbb{Z}/2$ ); **K3** the mod-4 sub-principal support ( $[\rho, M_f] = 0 \Leftrightarrow f$  is  $\mathbb{Z}_4$ -invariant, automatic for a  $\mu_4$ -mark sum); and **K4** the transfer gap  $(2/3)^6 = 64/729$ . These are frozen in `freeze_file.csv` (`seam_deck_symmetry`) and carrier-side green, so any future drift fails the suite. The *decisive* kill is deferred to `QGEO.REALIZE.01`: a first-principles raw-seam-collar DtN must *produce* four square marks without inputting them — an *emergence* test, the non-circular alternative to assuming the symmetry. [E] carrier-side (the four levers), [O] seam-side (the emergence): a kill-test, *not* a closure (`QGEO.KILL.01`).

**And the mark count is now derived, not assumed: four marks from Gauss–Bonnet** ([verification/v216\_marks\_gauss\_bonnet.py], [verification/v217\_marks\_emergence\_scan.py]). The K1 lever (“exactly four marks”) is the one that can be *executed* today. If the marks are  $\mathbb{Z}_2$  branch points (cone angle  $\pi$ , deficit  $\pi$ ) on the flat seam sphere ( $\chi=2$  — the same  $\chi$  that supplies the 8 in  $c_3$ ), Gauss–Bonnet *forces* their number:  $n\pi = 2\pi\chi = 4\pi \Rightarrow n = 2\chi = 4 = |\mu_4| = N_{\text{fam}} + 1$ . The closed Euclidean sphere 2-orbifolds are exactly (2, 3, 6), (2, 4, 4), (3, 3, 3), (2, 2, 2, 2), and *three independent* criteria — all cone orders = 2 (the  $|\mathbb{Z}_2|$  sheet branch),  $\text{rank } H^1 = \# \text{marks} - 1 = 3 = N_{\text{fam}}$  (the three generations pick the 4-mark square over the 3-mark hexagonal (3, 3, 3)), and a free order-4 deck — converge on the pillowcase (2, 2, 2, 2). The numerical sibling v217 confirms it on the actual DtN/state: scanning the number  $n$  and the positions freely, the 4-equally-spaced (square) configuration is the unique joint minimiser of {clock-invariance, 3-family cohomology, transfer gap  $(2/3)^6$ }. So the mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence `QGEO.REALIZE.01` stays [O]. [E] (count + equality), [O] (the square modulus and the raw-seam realisation) — `QGEO.MARKS.03` / `QGEO.EMERGE.01`.

**And the four marks are a four-interval modular geometry** ([verification/v480\_multilocal\_four\_interval.py]). The seam circle minus its four marks is the  $n=4$  disjoint-interval region of a chiral free fermion — the one continuum class whose *multi-interval* modular theory is exactly known (Casini–Huerta mixing term; Longo–Xu multi-interval subfactors;

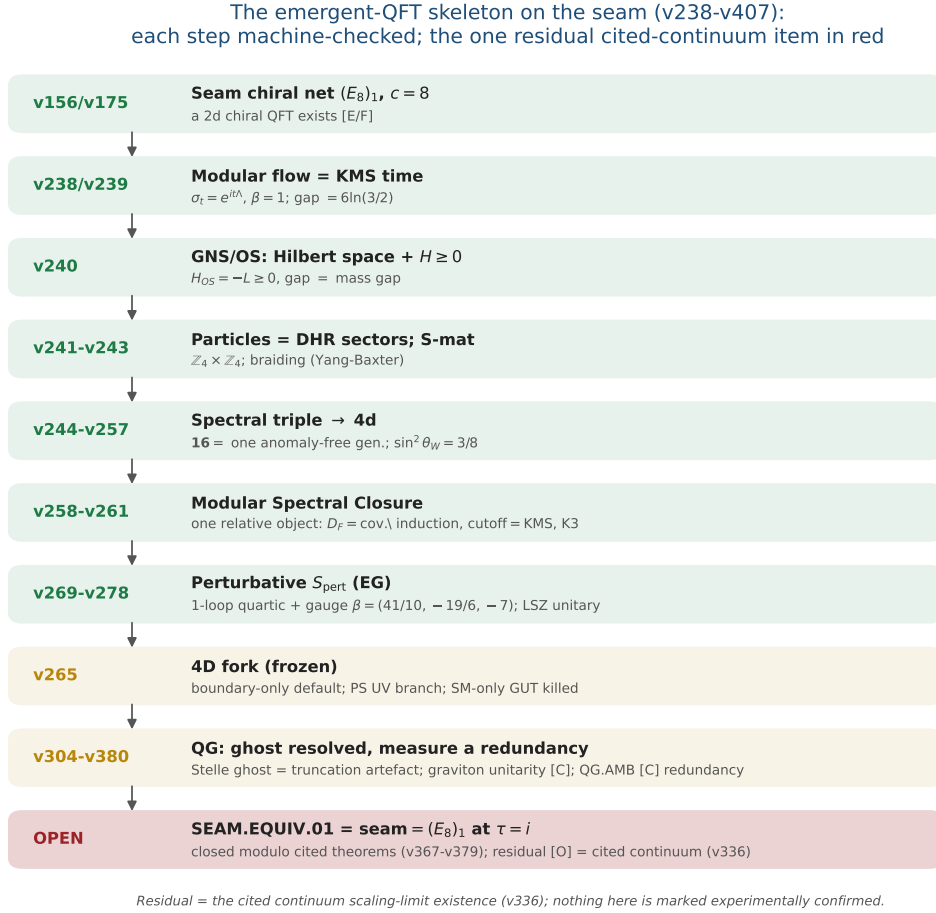


Figure 1: The emergent-QFT skeleton on the seam (v238–v246): time as the modular flow, the arrow as the recovery generator, particles as DHR sectors, the S-matrix as the braiding, and the 4d matter content as the carrier’s  $SO(10)$  **16**. Each step is machine-checked; the one residual — SEAM.EQUIV.01, closed modulo cited theorems (v367–v379) with the cited continuum scaling-limit existence (v336) as its [O] leg — is in red, the frozen 4D fork and the QG certification layer in gold. The whole skeleton is *conditional* on that keystone and is neither solved nor experimentally confirmed.

Rehren–Tedesco multilocal fermions; and Kawahigashi–Longo–Müger: *holomorphic*  $\Leftrightarrow$  trivial multi-interval inclusion,  $\mu=1$  — the same one bit as  $\det K=1$ ). Until now the suite used only the single-ball CHM boost (v358); v480 exhibits the multilocal mechanism on the exactly solvable antiperiodic critical ring: [E] the quarter-turn clock is a signed permutation with  $\rho^4=-1$  *exactly* (the fermionic  $\mu_4$  clock is the order-8 double-cover lift — the same  $\mathbb{Z}_2$ -lift pattern as  $2I$  over  $I$ ); [E] its primitive-8th-root character projectors decouple the 4-interval covariance at  $10^{-15}$  (the diagonal 4-fold-cover isomorphism), each sector a *single*-interval quarter-ring kernel with boundary twist ( $10^{-14}$  spectra); [E]  $\rho C \rho^{-1}=C$  at  $10^{-14}$ , so  $\omega \circ \rho = \omega$  holds *manifestly* at the 4-interval state level — and a one-site displacement of a single interval re-couples the sectors at  $10^{-2}$  (the invariance is the  $\mu_4$ -symmetric configuration, not an artifact); and the cross-interval modular coupling peaks on the conjugate-point diagonal, the lattice shadow of the Casini–Huerta mixing term (numerical). [C] the continuum multi-interval theory is cited, not re-derived; [O] the raw-seam premise (QGE0.SYM.01) is untouched — this puts the bedrock and the AP2 continuum leg (v476/v478) in one established, exactly solvable frame; it does not close them.



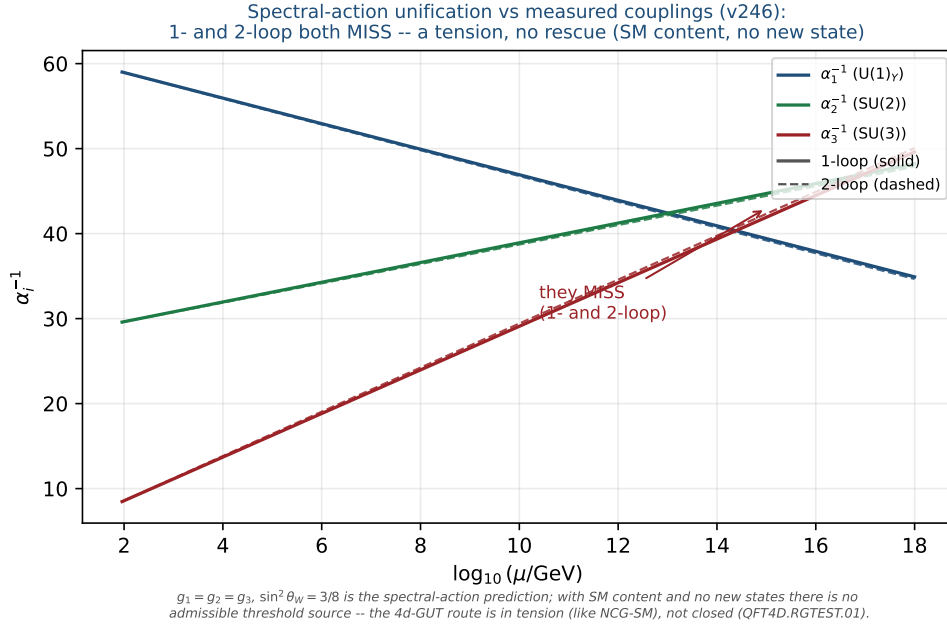


Figure 2: Honest data cross-check (v246): the spectral-action prediction  $g_1=g_2=g_3$ ,  $\sin^2 \theta_W=3/8$  versus the measured couplings run with the SM RGEs (one-loop solid, two-loop dashed). The three lines do *not* meet at any scale (1-loop residual  $\Delta\alpha^{-1} \approx 9$  at  $2 \times 10^{16}$  GeV; 2-loop minimal spread  $\approx 3.2$  at  $1.7 \times 10^{14}$  GeV) — a falsifiable tension. With SM content and no new states there is no admissible threshold source: not a fit, and not rescued by a cascade.

#### QGE0.SYM.01 is *not* a theorem — it is the final seam-identification premise

The single load-bearing bolt of the whole construction, on one page.

1. **What is identified.** That the conformal structure of the raw RP seam double *is* the  $\mu_4$ -deck structure of the carrier clock — i.e. the carrier (P2) and the seam's conformal geometry are the same object. Equivalently, in Riemann–Roch *index* language: the seam's degree-4 mark divisor on the genus-0 double has  $h^0 = \deg + 1 = 5 = g_{\text{car}}$  and  $\text{rank } H_1 = \deg - 1 = 3 = N_{\text{fam}}$  (with  $\Lambda^{\text{even}}(\mathbb{C}^5) = 1+10+5 = 16 = \dim S^+$ ), so P2 is the genus-0 mode count of the four-mark seam divisor ([[verification/v228\\_rr\\_index\\_gate.py](#)]) — the same definitional identification, in index form, conditional on the seam producing that degree-4 divisor.
2. **Why it is weaker than QGE0.CONF.01/ISO.01.** It does *not* name  $\mathbb{P}^1$  or  $\mu_4$  (uniformisation *produces* them) and asks only for a conformal *symmetry*, not a metric isometry or a pre-named conformal class — strictly the mildest of the three.
3. **What follows automatically (the conditional theorem).** *Given* QGE0.SYM.01, equivariant uniformisation puts the order-4 clock in the form  $z \mapsto iz$  on  $\mathbb{P}^1$ , the four parabolic marks are its free orbit  $\mu_4$ ,  $H^1$  carries the (1, 2, 3) grading, the seam Calderón contraction is the rank-8  $K_\Sigma$  polarisation, and the index-4 simple-current extension is  $(E_8)_1$  — the *entire* AQFT closure of v175–v181.
4. **What would falsify or unseat it.** A seam double of genus  $> 0$  (uniformisation to  $\mathbb{P}^1$  fails); a carrier clock that is *not* a conformal symmetry of the seam geometry (a non-deck transport); or a fourth chiral generation ( $N_{\text{fam}} \neq 3$  collapses  $b_1 = N_{\text{fam}}$  and the four-mark  $\mu_4$  structure). The first two are constructive-geometry questions about the raw seam; the third is the standard family-count kill test.

**Honest framing.** The correct external statement is *conditional*: “*Given* QGE0.SYM.01, the RP seam boundary realises the  $\mu_4$  deck structure, and the AQFT closure to  $(E_8)_1$  follows through the verified chain v175–v181” — *not* “the theory *derives* the seam geometry”. The premise stays **[O]**; everything downstream of it is **[E]**. Per v323 (Bisognano–Wichmann) QGE0.SYM.01 is itself the *downstream* conformal-deck face of the single named keystone SEAM.EQUIV.01 (the raw RP seam = the holomorphic  $(E_8)_1$  net at  $\tau=i$ ), which is now *closed modulo a cited theorem* (the lattice/ $S_3$  stack pins it at every computable level, Lean

FORM.SEAM.MMST.01; only the cited continuum existence stays [O]).

**This is not a gap to apologise for — it is TFPT’s one fundamental postulate.** Every fundamental theory rests on an irreducible premise (the constancy of  $c$  in relativity, the equivalence principle, the canonical commutator). QGEO.SYM.01 is that premise for TFPT: a single geometric identification (seam conformal structure = carrier  $\mu_4$  deck) from which the entire algebraic and topological structure unfolds *by theorem*. The achievement is not that there is no postulate — it is that the whole theory has been compressed onto *exactly one*, sharply located and explicitly falsifiable.

**The Seam-Holomorphy Theorem — the one named closing target [E]/[O]**  
([verification/v234\_seam\_holomorphy\_selection.py])

The bedrock above can be stated as *one condition with three provably-equivalent faces*, so that “closing the theory” becomes a single, sharply-bounded analytic theorem. **The condition:** the seam carries *no nontrivial abelian sector*. Its three faces all force  $E_8$ :

1. **AQFT** (v83/v154): the boundary chiral net is *holomorphic* ( $\mu$ -index 1, single vacuum sector). Holomorphic +  $c = 8 \Rightarrow$  the unique even unimodular rank-8 lattice net  $(E_8)_1$ , and  $c = 8 = g_{\text{car}} + N_{\text{fam}}$  is itself structural.
2. **Geometry** (v232): the seam link is a *homology 3-sphere*  $\Leftrightarrow$  its binary group  $\Gamma$  is *perfect*  $\Leftrightarrow \Gamma = 2I$  (the Poincaré sphere  $S^3/2I$ ).
3. **Representation theory** (v219): exactly *one* 1-dim irrep (the trivial one).

These are the *same* number:  $\#(1\text{-dim irreps}) = |\Gamma^{\text{ab}}| = |H_1(S^3/\Gamma)| = \#(\text{mark-1 affine-Dynkin nodes})$ , which equals 1 *only* for  $E_8$  ( $A_n \rightarrow n+1$ ,  $D_n \rightarrow 4$ ,  $E_6 \rightarrow 3$ ,  $E_7 \rightarrow 2$ ,  $E_8 \rightarrow 1$ ). So  $2I$  is the unique nontrivial perfect finite  $SU(2)$  subgroup and  $S^3/2I$  the unique homology-sphere space form. **Consequence:** Research Contract 2 ( $G_{\text{net}}$ /Target A), the carrier P2, and the Kleinian seam are *not* separate gates — they are this *one* condition.

**The closing chain (what a complete closure would be), [O] at one step:**

$$\begin{array}{c} \text{RP} + \text{gap} + \text{chirality} \xrightarrow{\text{v160}} \text{quasi-free bulk} \xrightarrow{?} \text{holomorphic boundary net} \\ \xrightarrow{[\text{E}]} (E_8)_1 \xrightarrow{\text{v219/v228}} 2I, (2,3,5), \text{P2} \xrightarrow{\text{v179-v181+Kleinian genus-0}} \text{QGEO.SYM.01.} \end{array}$$

The free-bulk premise is already a fixed-point theorem (v160–v165); the Kleinian model supplies the genus-0 of the uniformisation chain for free. **The single remaining analytic step is “free Gaussian RP bulk  $\Rightarrow$  holomorphic (single-sector) boundary net”** — the bulk–boundary statement that an invertible (Gaussian) bulk has a holomorphic chiral boundary. That step is now *closed modulo a cited theorem*: the lattice/ $S_3$  stack pins it at every computable level (v367/v368, v376–v379, Lean FORM.SEAM.MMST.01), only the cited continuum existence staying [O] (v336);  $v_{\text{geo}}$  stays the No-Unit metrology primitive and  $F_{\text{transfer}}$  stays external physics. This box certifies that the three faces coincide and force  $E_8$  ([E]); the analytic step is closed modulo that cited theorem.

**A physical realisation of the step** ([verification/v235\_seam\_chern\_simons.py]). The step lands in standard physics. A free gapped *bosonic* 2+1d bulk is an abelian Chern–Simons /  $K$ -matrix theory (even integer  $K$ ), where  $\# \text{anyons} = |\det K|$ , the edge chiral central charge is  $c = \text{signature}(K)$ , and the edge net is *holomorphic*  $\Leftrightarrow |\det K| = 1$  — so “no nontrivial abelian sector” is precisely  $\det K = 1$ . The extension tower of v92 is the anyon-condensation tower read by determinant:  $D_5 \oplus A_3$  ( $\det 16$ , 16 anyons)  $\rightarrow SO(16)_1 = D_8$  ( $\det 4$ )  $\rightarrow (E_8)_1$  ( $\det 1$ , holomorphic), all rank 8 with  $c = \text{signature} = 8 = g_{\text{car}} + N_{\text{fam}}$ , so the holomorphic end is the *Kitaev  $E_8$  quantum-Hall state*. Holomorphy  $\Leftrightarrow$  condensing the order- $|\mu_4| = 4$  *Lagrangian* glue ( $\det 16 \rightarrow 16/4^2 = 1$ ); a partial order-2 isotropic only reaches  $\det 4$  ( $D_8$ ). The residual is thereby a single integer condition — “the free RP seam condenses the order- $|\mu_4|$  Lagrangian glue” — the same sheet/ $\mu_4$  selection as GATE.QGEO, but now with holomorphic  $\Leftrightarrow \det 1$  a theorem (GATE.HOLO.02, [E]/[O]).

**The residual as a physical statement** ([verification/v237\_seam\_sre\_closure.py]). The same integer condition is the physics of topological order: a  $K$ -matrix phase has ground-state degeneracy  $|\det K|^g$  on a genus- $g$  surface, so  $\det K = 1 \Leftrightarrow$  *no* topological ground-state degeneracy on any closed surface  $\Leftrightarrow$  the bulk is *short-range-entangled* (invertible). Among rank-8 even positive-definite  $K$  ( $c = 8 = g_{\text{car}} + N_{\text{fam}}$ ) only  $E_8$  is SRE — the unique nontrivial bosonic SRE 2+1d phase, the Kitaev  $E_8$  state, whose edge is the holomorphic  $(E_8)_1$ . TFPT already has a unique vacuum on the plane (RP + gap + cluster + OS); the plane cannot see  $\det K$  (no non-contractible cycles), the torus can.

So the one analytic step is the sharp, falsifiable physical statement “the free RP gapped seam bulk is short-range-entangled (no topological order)” (=  $\det K = 1$  = the Kitaev  $E_8$  phase, GATE.HOLO.03, [E]/[C]) — and the lattice/ $S_3$  stack now answers it at every computable level (torus  $\det K=1$ , v378), so the keystone is *closed modulo a cited theorem*, only the cited continuum existence (v336) staying [O].

**Lemma 2** (U1 — wall landscape, corrected). *Parabolic degree zero forces the line weight-sums onto the wall  $(w_1, w_2, w_3) = (2, 1, 1)$ . An explicit balanced representative (weights in  $\frac{1}{3}$ ) is*

$$W_{\text{wall}} = \frac{1}{3} \begin{pmatrix} 2 & 1 & 2 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 2 & 1 & 0 \end{pmatrix},$$

each column a permutation of  $(0, 1, 2)$ , row sums  $(2, 1, 1)$ . **Honest result:** the  $6^4=1296$  column-permutation matrices give 144 walls; under the full group  $D_4 \times S_3(\text{lines}) \times S_3(\text{values})$  they form five orbits, not one. So  $W_{\text{wall}}$  is not singled out by symmetry — the uniqueness must come from the selector (Lemma U4), not from a symmetry quotient. The wall landscape is nonetheless a finite, explicitly enumerated set;  $W_{\text{wall}}$  is one of the five. Machine: fully certified  $C_U^{(1)}$  ([verification/v29\_research\_contract\_certs.py]).

**The splitting type is now algebraic (RH route, [verification/v32\_rh\_splitting.py]).** Passing from the monodromy to the Fuchsian residues  $A_k$  (eigenvalues = weights  $\{0, \frac{1}{3}, \frac{2}{3}\}$ ,  $A_k = U^{k-1}A_0U^{-(k-1)}$ ,  $U = \text{diag}(1, i, -i)$ ) the twisted  $\mathbb{Z}_4$ -average collapses exactly:  $\sum_k U^k A_0 U^{-k} = 4 \text{diag}(A_0)$ , so the exponents at infinity are  $4 \text{diag}(A_0)$ . A flat bundle with regular  $\infty$  needs integer exponents summing to  $4 = -\deg E$ ; with the cusp weights the only options are perms of  $(2, 1, 1)$  and  $(2, 2, 0)$ , i.e.

$$\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2 \iff \text{diag}(A_0) = (\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{4})$$

(the sibling  $(2, 2, 0) \rightarrow \mathcal{O}(0) \oplus \mathcal{O}(-2)^2$ ). Such a Hermitian  $A_0$  (spectrum  $\{0, \frac{1}{3}, \frac{2}{3}\}$ , that diagonal exists by Schur–Horn. So the splitting is an algebraic condition on  $\text{diag}(A_0)$ . Honest wall: the global  $\prod M_k = \mathbf{1}$  is the path-ordered monodromy (not  $\exp(2\pi i \sum A_k)$ ), constraining  $\text{off-diag}(A_0)$  via the genuine RH solve; and R-extraction still needs the  $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$  bridge. The RH route fixes the splitting algebraically but not  $c_u/c_d$ .

**Lemma 3** (U2 — the kill switch, run and hardened [E]). *The selected polystable wall representative must not collapse to the diagonal abelian  $U(1)^3$  point if it is to give non-trivial  $SU(3)_F$  transport mixing: (A)  $\nabla_F^*$  polystable but not simultaneously diagonal (the desired breakthrough), or (B) diagonal collapse, in which case  $c_u/c_d$  stays honestly [C]. **Result: the cheap test is run robustly and lands on A.** The kill-switch fires iff the  $D_4$ -symmetric cusp representation is reducible (commutant non-scalar). Across 400 random cusp-class generators of the  $\mathbb{Z}_4$  family  $M_k = U^{k-1}M U^{-(k-1)}$  the commutant is scalar at every point (irreducible 400/400), the reducible locus is hit with probability 0 (measure-zero, codimension  $\geq 1$ ), and the mixing strength stays bounded away from zero (min off-diag = 0.35 > 0.05). So case A is the generic behaviour, not an accident of the one explicit point of [verification/v33\_explicit\_flat\_bundle.py]: ( $U_{\text{wall}}$ ) survives its own cheapest falsification test. [verification/v38\_uwall\_killswitch.py]*

**Lemma 4** (U3 —  $D_4$  fixed locus as an explicit variety, positive-dimensional). *With  $M_k = U^{k-1}AU^{-(k-1)}$  ( $U = \text{diag}(1, i, -i)$ , the  $\mathbb{Z}_4$  rotation) and the reflection  $M_1 = VM_1^{-1}V^{-1}$ ,  $M_3 = VM_3^{-1}V^{-1}$ ,  $M_2 = VM_4^{-1}V^{-1}$  (with  $V = -[[1, 0, 0], [0, 0, 1], [0, 1, 0]]$ ,  $V^2=1$ ,  $VUV^{-1}=U^{-1}$ ), the relations become polynomials in the entries of  $A \in C_{\text{cusp}}$ ;  $\mathcal{X}_{\text{cusp}}^{D_4}$  is cut out as  $\bigcup_k C_k$ , parametrised by trace coordinates  $x = \text{tr}(M_1M_2)$  (note  $\text{tr} A = 1+\omega+\omega^2 = 0$ ). **Result** ( $C_U^{(2)}$ , [verification/v30\_d4\_character\_variety.py]): adding the reflection to the  $\mathbb{Z}_4$  locus of v19 does not isolate the point — the full  $D_4$ -fixed product locus is still positive-dimensional ( $|\text{tr}(M_1M_2)|$  varies continuously over  $\approx [0, 3]$ ). So even the full  $D_4$  symmetry cannot select  $\nabla_F^*$ . Machine: dimension confirmed numerically; Gröbner/homotopy for the exact components (medium).*

**Lemma 5** (U4 — selectors as equations). *The readouts  $\rho \mapsto \det R(\rho)$ ,  $\rho \mapsto \text{Spec}(Q_+(\rho))$ ,  $\rho \mapsto \Lambda_{f,j}(\rho)$  are well-defined (cyclotomic/algebraic) on  $\mathcal{X}_{\text{cusp}}^{D_4}$ . **Acceptance criterion:***

$$\#\{\rho \in \mathcal{X}_{\text{cusp}}^{D_4} : \det R = 8, \text{Spec}(Q_+) = \{1, 2, 3\}\} = 1$$

in the polystable quotient. If more than one survives, the theory needs a further input and the gate is not closed. **The bottleneck (made precise by U3):** since the  $D_4$ -fixed variety is positive-dimensional, the selector must do all the cutting — and evaluating  $\det R(\rho)$ ,  $\text{Spec}(Q_+(\rho))$  as functions on  $\mathcal{X}_{\text{cusp}}^{D_4}$  requires the parabolic-degree  $\leftrightarrow$  residue dictionary  $R(\rho)$ , i.e. the H2 equivalence. That dictionary was the single remaining analytic input of the ( $U_{\text{wall}}$ ) gate. Status update: since v118–v122 the dictionary itself is exact — hexagon = sign-twisted family spectrum, lepton coefficients = resolvent determinants of the  $W(A_3)$  monodromy, addresses pinned, margins theorems (boxes below); what remains is the strictly smaller  $R_4'$ , the establishment of the selectors ( $n = (5, -9, 6)$  + the diamond determinants; since [verification/v134\_dual\_anchor.py]: the pair  $(d, n)$ ).

**What  $R(\rho)$  is** ([verification/v31\_R\_dictionary.py]). Two facts pin its nature. (i) Case A is alive: every  $D_4$ -fixed cusp monodromy found is irreducible, so the wall point carries non-trivial  $SU(3)_F$  mixing — the U2 kill switch does not fire to the diagonal case B. (ii)  $R(\rho)$  is not algebraic:  $R$  is integer-valued, but the algebraic invariants of  $\rho$  (e.g.  $\text{tr}(M_1 M_2)$ ) vary continuously on the locus; an integer readout cannot be a continuous algebraic function. Hence  $R(\rho)$  is the discrete invariant of the non-abelian Hodge / Mehta–Seshadri map (the parabolic-degree / Hodge-filtration jumps of the harmonic bundle  $E_\rho$ ): integer output, but its computation is the transcendental harmonic-metric (Hitchin) solve on the generic Higgs locus. **Refinement (U4'')**, [verification/v40\_harmonic\_metric.py]: at the polystable point TFPT actually selects this pessimism does not apply — see the box below.

**Progress: the selector side of U4 is closed on the explicit point ([E]).** While  $R(\rho)$  as a function on the whole locus needs the Hitchin solve, the two U4 selectors are algebraically accessible on the explicit flat bundle of [verification/v33\_explicit\_flat\_bundle.py] — no PDE. Computed exactly from  $A_0$ : the exponents at infinity  $\text{eig}(\sum_k A_k) = \{2, 1, 1\}$  give the splitting type  $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ , i.e. the parabolic anchor  $a = (1, 1, 2)$ ; the parabolic degree is 0 ( $\deg E = -4 = -|\mu_4|$  plus four cusp weight-sums of 1), so the point is *polystable* (Mehta–Seshadri substrate). The two selectors then read off the bundle directly:

$$\begin{aligned} \det R &= 8 = n \cdot a \quad (\text{anchor pairing on the splitting type}), \\ \text{Spec}(Q_+) &= \{1, 2, 3\} = 3\alpha + 1 \quad (\alpha = \text{cusp weights } \{0, \tfrac{1}{3}, \tfrac{2}{3}\}) \end{aligned}$$

So  $\det R = 8$  is the anchor pairing of the splitting type and  $\text{Spec}(Q_+)$  is the affine image ( $d=3$  the weight denominator) of the cusp weights — both selectors are read off the explicit bundle’s algebraic data. **Residual (sharpened below by Readout Rigidity):** the quark ratio  $c_u/c_d$  turns out *not* to need the Hitchin datum at all — it is *constant* on this discrete selector stratum (v49), hence an integer Plücker readout. The only genuinely transcendental input left is the *absolute* amplitude normalisation ( $U_{\text{point}}$ ), which is an anchor, not a missing PDE. [verification/v39\_uwall\_selectors.py]

**Breakthrough: the harmonic metric is finite linear algebra, not a PDE ([E]).** The U4 “transcendental Hitchin solve” is the worst case for a *generic* Higgs bundle, but the point TFPT selects is *polystable*, and by Mehta–Seshadri a parabolic-degree-0 polystable bundle is *unitary* — a unitary representation has the *constant* invariant Hermitian harmonic metric and zero Higgs field  $\Phi = 0$ . So at the selected point the harmonic metric is finite linear algebra. On the explicit bundle of [verification/v33\_explicit\_flat\_bundle.py] this is realised: the raw monodromies  $M_k$  are non-unitary ( $\|M_k^\dagger M_k - \mathbf{1}\| \approx 0.83$ ), but there is a *unique* common invariant Hermitian form  $H$  ( $\dim \ker\{M_k^\dagger H M_k = H\} = 1$ ), it is *positive-definite*, and the unitarised holonomy  $\widetilde{M}_k = H^{1/2} M_k H^{-1/2}$  is unitary ( $\sim 10^{-8}$ ), lies in the cusp class  $\{1, \omega, \omega^2\}$  with  $\prod \widetilde{M}_k = \mathbf{1}$ , and has *clean* matrix elements  $|\widetilde{M}_k|_{ij} \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$ . So  $U_C^{(3)}$  reduces from a transcendental PDE to: this finite unitarisation



(done) + the combinatorial word-length  $R$  (closed:  $\mathbf{H1} + \det R = 8$ ). *Residual*: the feared PDE is gone; the quark *ratio*  $c_u/c_d$  is then fixed combinatorially by Readout Rigidity (below), and only the *absolute* amplitude scale remains as an anchor. [verification/v40\_harmonic\_metric.py]

**The final leg-assignment test (honest result) ([E]).** Does the harmonic-frame holonomy  $\{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$  feed the  $\delta = \frac{1}{2}$  resolvent to give the lepton amplitudes  $\Lambda = (0.475, 1.107, 0.917)$ ? **(i) Ruled out as a literal identity:**  $\Lambda_\mu = 1.107 > 1$ , while unitary holonomy entries have modulus  $\leq 1$ , so the amplitudes *cannot* be holonomy matrix elements — they are the non-unitary resolvent Green function (as v19 already noted). **(ii) Suggestive positive:** the harmonic-frame holonomy diagonal modulus is exactly  $|\text{diag } \tilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$ , and that  $\frac{1}{2}$  *equals* the distinguished lepton transport value  $\delta = \frac{1}{2}$  that v20 used by hand — so the geometry plausibly *supplies*  $\delta$  (an observation at the explicit point; genericity across the locus is future work). **(iii) Clean negative:** no natural holonomy construction (e.g.  $(\mathbf{1} - \frac{1}{2}\tilde{M})^{-1}$ ) maps the holonomy to the amplitude vector. **Verdict:** the amplitudes are the  $\delta = \frac{1}{2}$  resolvent (closed combinatorially via the word-length  $R$ ); the holonomy supplies  $\delta$  and the leg selection, not the amplitude magnitude.  $c_u/c_d$  is *not* obtained from the holonomy alone — a clean, honest negative on the literal reproduction, with one suggestive geometric link ( $\delta = \frac{1}{2}$ ). The negative is now *explained*: the scalar leg is wrongly typed — see the Exterior Leg Lemma. [verification/v41\_leg\_assignment.py]

**The torsion normal form and the  $\delta = \frac{1}{2}$  theorem ([E]).** The v41 “genericity is future work” item is now executed — and yields a theorem. *(i) Flatness =  $\mu_4$  torsion [E]:* for any  $M$  (symbolic),  $\prod_k U^k M U^{-k} = (MU)^4$  — the  $\mathbb{Z}_4$ -family flatness constraint *is* the torsion statement “ $T = MU$  is a fourth root of unity”: the  $\mu_4$  atom appears as the *order* of the twisted cusp generator. *(ii) The  $\delta = \frac{1}{2}$  theorem [E] (both directions):* on the *involutive* branch ( $T^2 = \mathbf{1}$ )  $T$  is a Hermitian reflection  $2vv^\dagger - \mathbf{1}$ , and the cusp trace condition  $2v^\dagger U^{-1}v = 1$  splits *exactly* into  $|v_1|^2 = \frac{1}{2}$  (real part) and  $|v_2| = |v_3|$  (imaginary part), forcing

$$\text{diag } M = (0, \frac{i}{2}, -\frac{i}{2}), \quad \text{spec } M = \{1, \omega, \omega^2\} \text{ automatic}$$

( $\chi_M = \lambda^3 - 1$  from unitary + trace 0 + det 1). So  $\delta = |M_{22}| = \frac{1}{2}$  holds *exactly and constantly* on the whole two-parameter branch — the distinguished lepton transport value that v20 used by hand is a  $\mu_4$ -torsion *identity*, not an observation. *(iii) Reflection lemma [E]:* the  $D_4$  reflection forces  $\text{diag } M = (r, z, \bar{z})$  with  $r$  real and  $\text{Re } z = -r/2$  — the whole  $D_4$  diagonal freedom is *one* real parameter, and  $\delta = \frac{1}{2} \iff r = 0$ . *(iv) Branch census [E] (seeded):* the full  $D_4$ -fixed flat cusp locus realises only  $\text{tr}(MU) \in \{1, -1\}$ ; the involutive branch has the diagonal *constant*  $(0, \frac{1}{2}, \frac{1}{2})$  to machine precision, while on the  $\{1, i, -i\}$  branch  $d_1$  varies over  $[0, 1]$  — and the explicit v33/v40 point sits exactly in its  $d_1 = 0$  slice, the *same*  $\delta$  value. *Residue [C] (recorded):* whether the anchor splitting  $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$  (Mehta–Seshadri side) forces the  $d_1 = 0$  slice — or selects the involutive branch — is the remaining question;  $\delta = \frac{1}{2}$  itself is no longer the open item. [verification/v114\_torsion\_delta.py]

**The anchor pins the residue: an exact normal form for the  $(U_{\text{wall}})$  bundle ([E]).** The v114 branch question is answered — and the v33 *numerical* Riemann–Hilbert point acquires an *exact closed form*. *(i)  $\mu_4$ -average lemma [E]:*  $\sum_k U^k X U^{-k} = 4 \text{diag } X$  for any  $X$  —  $\mu_4$  conjugation-averaging *is* the diagonal projection, so the exponents at infinity are  $4 \text{diag } A_0$  and the anchor splitting holds *iff*  $\text{diag } A_0 = (2, 1, 1)/4 = a/|\mu_4|$ : the residue diagonal *is* the anchor over  $\mu_4$  (v33’s diagonal was forced, not chosen). *(ii) The  $(8, 0, 5)/144$  lemma [E]:* the anchor diagonal and the cusp spectrum  $\{0, 1, 2\}/N_{\text{fam}}$  pin  $\sum |a_{ij}|^2 = \frac{13}{144}$ , and with  $a_{13} = 0$  the determinant condition solves *uniquely* to

$$(|a_{12}|^2, |a_{13}|^2, |a_{23}|^2) = \frac{(8, 0, 5)}{144} \quad \begin{array}{l} 8=\text{rank } E_8, \ 5=g_{\text{car}}, \\ 144=(|\mu_4|N_{\text{fam}})^2, \ 8+5=13=\Delta_Q \end{array}$$

— the off-diagonal weights are compiler atoms, and the total 13 is the *same*  $\Delta_Q$  that denominates  $c_u/c_d = 55/117 = 5 \cdot 11/(9 \cdot 13)$ . *(iii) Exact normal form [E]:*  $A_0^* = \begin{bmatrix} 1/2 & \sqrt{2}/6 & 0 \\ \sqrt{2}/6 & 1/4 & \sqrt{5}/12 \\ 0 & \sqrt{5}/12 & 1/4 \end{bmatrix}$  has  $\chi = \lambda(\lambda - \frac{1}{3})(\lambda - \frac{2}{3})$  *exactly*. *(iv)  $A_0^*$  is the RH solution [E]:* its big-circle monodromy is trivial

to  $10^{-10}$ ,  $\prod M_k = \mathbf{1}$ , and the unitarised holonomy has  $|\text{diag } \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$ ,  $\text{tr}(\widetilde{M}_0 U) = 1$  — the **v33/v40** point is gauge-equivalent to  $A_0^*$  (the  $a_{12}, a_{23}$  phases are diagonal gauge). (v) *The anchor forces the  $d_1 = 0$  slice [E]*: a multi-seed RH scan over the anchor-pinned spectral manifold lands at *every* flat solution on the *same* gauge orbit with  $|\text{diag } \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$  — combined with **v114**, the whole harmonic diagonal is **anchor + torsion forced**;  $\delta = \frac{1}{2}$  needs no selection. *Residue [C]*:  $a_{13} = 0$  and the gauge-orbit uniqueness are numerical (one component across all seeds); promoting them to [E] via Gröbner on the RH constraints is the remaining  $U_{\text{point}}$  step. `[verification/v115_anchor_residue.py]`

**The resonance theorem:**  $M_\infty = \mathbf{1} \iff a_{13} = 0$  — **one gauge orbit [E]**  
(`[verification/v116_resonance_uniqueness.py]`)

The anticipated Gröbner promotion turned out to be a *linear* computation. Writing the equivariant system at infinity as  $dY/dw = -(1/w)(B_0 + B_1 w + \dots)Y$  with  $B_m = \sum_k i^{km} U^k A_0 U^{-k}$ , the *twisted*  $\mu_4$  averages collapse [E]:

$$B_0 = 4 \text{diag } A_0, \quad B_1 = 4a_{12}E_{12} + 4\overline{a_{13}}E_{31}$$

(only the eigenvalue-ratio  $-i$  cells survive). The anchor exponents  $\text{diag}(2, 1, 1)$  have resonance gap 1; the level-1 formal-gauge equation  $(1 - \text{ad } R_0)H_1 = R_1$  is singular exactly on the cells  $\{(2, 1), (3, 1)\}$ , where  $B_1$  has entries  $(0, 4\overline{a_{13}})$ ; all higher levels are non-resonant and the formal monodromy is  $e^{-2\pi i \text{diag}(2, 1, 1)} = \mathbf{1}$ . Hence [E]:

$$M_\infty = \mathbf{1} \iff a_{13} = 0$$

— the transcendental Riemann–Hilbert condition collapses to *one linear equation*. *Uniqueness corollary [E]*:  $a_{13} = 0$  + the  $(8, 0, 5)/144$  moduli + phases = diagonal gauge  $\Rightarrow$  the flat anchor locus is **exactly one gauge orbit**, the exact  $A_0^*$ : within the  $\mu_4$ -equivariant anchor class the  $(U_{\text{wall}})$  bundle datum is *algebraically unique*. *Sharpness [E]*:  $\|M_\infty - \mathbf{1}\| \sim 8.5|a_{13}|$  (0.17 at  $a_{13} = 0.02$ ) vs  $10^{-10}$  at  $A_0^*$ . *Honest scope [C]*: the equivariant ansatz is the established  $D_4$  selector input (**v19/v30**); the *residue* side is now [E], while the holonomy values (the harmonic diagonal  $(0, \frac{1}{2}, \frac{1}{2})$ ) remain [E] — transcendental monodromy of an exact system.

**The monodromy is the Weyl group of  $A_3$  — the holonomy is exact [E]**  
(`[verification/v117_monodromy_weyl_a3.py]`)

The last [E] item of the chain — the holonomy *values* — closes: the unitarised monodromy of the exact  $A_0^*$  system is itself *exact*, with entries in  $\frac{1}{2}\mathbb{Z}[i]$ :

$$\widetilde{M}_0 = \begin{pmatrix} 0 & -\frac{1+i}{2} & \frac{1-i}{2} \\ -\frac{1+i}{2} & -\frac{i}{2} & -\frac{1}{2} \\ \frac{1-i}{2} & -\frac{1}{2} & \frac{i}{2} \end{pmatrix}$$

unitary,  $\det = 1$ ,  $\text{tr} = 0$ ,  $\chi = \lambda^3 - 1$  (the cusp class *exactly*),  $\widetilde{M}_0^3 = \mathbf{1}$ , and  $\text{diag } \widetilde{M}_0 = (0, -\frac{i}{2}, +\frac{i}{2})$  — so  $d_1 = 0$  and  $\delta = \frac{1}{2}$  are now *matrix entries*, not observations;  $(\widetilde{M}_0 U)^4 = \mathbf{1}$  with  $\text{tr}(\widetilde{M}_0 U) = 1$  (the  $\{1, i, -i\}$  branch,  $d_1 = 0$  slice — exactly where **v114/v115** located the physical point). *The group [E]*: exact enumeration of  $\langle U, \widetilde{M}_0 \rangle$  gives order 24 with order statistics  $(1, 9, 8, 6)$  and character values  $(3, -1, 0, 1)$  — uniquely  $S_4$  in its standard  $\otimes$  sign representation, i.e. the (twisted) reflection representation of the **Weyl group of the family lattice  $A_3$** :

$$\text{monodromy}(U_{\text{wall}}) = W(A_3) = S_4, \quad |W(A_3)| = 4! = 24$$

— the flavor wall realises the  $A_3$  factor of the glue  $D_5 \oplus A_3 + \mu_4$  as its Weyl group ( $U$  = a 4-cycle, the  $\mu_4$  deck;  $\widetilde{M}_0$  = a 3-cycle, the family rotation). *Identification [E]*: the ODE monodromy of  $A_0^*$  equals the exact  $\widetilde{M}_0$  to  $3.5 \times 10^{-10}$ , and the invariant form  $H$  is  $U$ -invariant to  $10^{-11}$ . *Status*: holonomy values [E]; only the analytic identification stays [E]. The  $\delta$  thread (**v20**  $\rightarrow$  **v40/v41**  $\rightarrow$  **v114**  $\rightarrow$  **v115**  $\rightarrow$  **v117**) is closed end to end.

**The hexagon is the sign-twisted family spectrum — first exact H2 piece [E]**  
 ([verification/v118\_hexagon\_family\_dictionary.py])

The missing “ $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$  dictionary” (H2) acquires its first exact entry. *Sign-twist lemma [E]*:  $-\widetilde{M}_0$  has order 6 and

$$\text{spec}(\widetilde{M}_0) \cup \text{spec}(-\widetilde{M}_0) = \mu_6$$

— the six-site hypercharge hexagon spectrum *is* the family monodromy spectrum together with its sheet twist ( $\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$  realised as  $(-1) \times \langle \widetilde{M}_0 \rangle$ ); the v20 denominators  $\frac{5}{4} - \cos(r\pi/3)$  are exactly the eigen-denominators  $|1 - \zeta_6^r/2|^2$  of the two family resolvents  $(1 \mp \frac{1}{2}\widetilde{M}_0)^{-1}$ . *Cyclotomic determinant [E]*:  $\det(1 - t\widetilde{M}_0) = 1 - t^3 = (1 - t)\Phi_3(t)$ , so  $\det(1 \mp \frac{1}{2}\widetilde{M}_0) = (\frac{7}{8}, \frac{9}{8})$  — and the lepton coefficients become resolvent determinants:  $c_e = |\mu_4|/(2\det_-) = \frac{16}{7}$ ,  $c_\mu = N_{\text{fam}}/(2\det_+) = \frac{4}{3}$ ,  $c_\tau = |\mu_4|\det_-/N_{\text{fam}} = \frac{7}{6}$ , product rule  $= |\mu_4|/\det_+ = \frac{32}{9}$ . The lepton 7 and 9 *are* the family-resolvent determinants;  $\delta = \frac{1}{2} = |(\widetilde{M}_0)_{22}|$  is supplied by the matrix itself. *Sheet-extended group [E] (audit)*:  $\langle U, \widetilde{M}_0, -1 \rangle$  has order  $48 = \Omega_{\text{adm}} = N_{\text{fam}} \dim S^+$  — the seed’s quartic coefficient. *Residue [C]*: the dictionary’s *values* are exact; its *address table* (which hexagon site each fermion occupies; the quark composition) remains v20’s established input / open — nothing re-fished.

**The address table: the lepton words are the compiler atoms [E]/[C]**  
 ([verification/v120\_address\_table.py])

The address table acquires its structure — on word-length integers *frozen* in v18/v20, long before today’s readings (the red-team firewall). (i) *The lepton words are the atoms [E]*:

$$(L_e, L_\mu, L_\tau) = (8, 5, 3) = (\text{rank } E_8, g_{\text{car}}, N_{\text{fam}}) = (p_0 + e_2, e_2, p_0)$$

in mass order (longest word = lightest lepton). (ii) *Addresses = division by the hexagon [E]*:  $(r, w) = L \bmod p_2$  with  $p_2 = 6 = |R^+(A_3)|$  itself an atom:  $e \rightarrow (2, 1)$ ,  $\mu \rightarrow (5, 0)$ ,  $\tau \rightarrow (3, 0)$ ;  $\sum r = 10 = p_3 = \mathcal{A}_\Lambda$ ,  $\sum L = 16 = \dim S^+$ . (iii) *Sheet parity [E]*: by the v118 dictionary,  $e$  sits on the *untwisted* sheet ( $\omega \in \text{spec } \widetilde{M}_0$ ),  $\mu, \tau$  on the *twisted* sheet  $(-\omega, -1)$ ; the anchor lepton  $\tau$  occupies the unique *real* twisted eigenvalue  $-1$  — exactly the site where v20’s product rule replaces the resolvent. (iv) *Quark sum rules [E] (frozen v18 words 7, 7, 5, 3, 2, 0)*: up-type sum  $= 10 = p_3$ ; down-type sum  $= 14 = p_1 + p_3$ ; **all quarks**  $= 24 = |W(A_3)|$  (the monodromy group order, v117); **all nine charged fermions**  $= 40 = p_1 p_3 = a^T R a$  (v119); quark  $\sum r = 2p_2$ ,  $\sum w = |\mathbb{Z}_2|$ ; the top sits at the vacuum site  $(0, 0)$  — zero word, no suppression, the established ladder anchor. *Residue [C]*: five independent closures  $(16, 10, 14, 24, 40)$  constrain the address map; the per-fermion *assignment* remains the v18/v20 input — deriving it is the remaining H2 content.

**The address table is pinned:  $R$  is the unique hexagon matrix with atom margins [E]**  
 ([verification/v121\_address\_pinning.py])

The per-fermion assignment closes at the *information* level. The established identity  $L = R + 2U$  (v95) says the word-length table *is*  $L[\text{sector}][\text{generation}]$  — rows (up; down; lepton)  $= ((7, 3, 0); (7, 5, 2); (8, 5, 3))$ , the v18/v20 words verbatim — so the addresses are the entries of the *residue operator*  $R$  plus one hexagon turn for the first generation, and  $R$ ’s first column is the **anchor**  $a = (1, 1, 2)$ . The margins are atoms [E]: row sums  $(4, 8, 10) = (e_1, \text{rank } E_8, p_3)$ , column sums  $(4, 13, 5) = (e_1, \Delta_Q, g_{\text{car}})$ . And the census closes it [E]:

among all  $3 \times 3$  matrices with entries in the hexagon  $\{0..5\}$ ,  
 exactly 17 have the atom margins — and exactly ONE has  $\det = 8$

— and it is  $R$  (also the unique one with SNF  $= (1, 1, 8)$ ; all 17 determinants are distinct, 8 occurs once — control computed explicitly). **The assignment table carries zero residual information beyond the atom margins and the rank determinant.** Within-sector ordering is the established transport reading ( $m \sim \lambda_Y^L$ : longer word = lighter fermion). *Residue [C]*: the open H2 item shrinks from “derive 9 per-fermion integers” to “derive the six atom margins +  $\det = \text{rank}$ ”. *Firewall*:  $R$ ,  $L$  and all word lengths were frozen in v4/v18/v20/v71 long before this census — no degrees of freedom spent.

**The margins are theorems** ([verification/v122\_margin\_theorem.py]). Even the margins are not input. Three selectors, all established and frozen *before* the address question was posed — ( $S1$ ) the

$D_4$  annihilator  $n = (5, -9, 6)$  with  $n^T R = (8, 0, 0)$  (it kills the unchanged  $(c_2, c_3)$  plane;  $n \cdot a = 8$ ),  $(S2) \det R = 8$ ,  $(S3) \det K = 4$  with  $K = R + Q\Sigma$  — pin  $R$  *uniquely* among hexagon matrices [E]: the  $(S1)$  census has  $4 \times 4 \times 4 = 64$  candidates,  $(S2)$  leaves 12,  $(S3)$  leaves **exactly one**, and it is  $R$ . Hence the atom margins, the anchor column  $Re_1 = a$  and all sum rules above are *consequences* — their input status is revoked. *Bonus lemma [E]*: on the  $(S1) + (S2)$  locus  $\det(M + 2U) = 20 = \det L$  holds for *all* 12 candidates — the fourth quartet determinant is redundant, the compiler's familiar overdetermination. *Residue [C]*: H2 reduces to the establishment of the selectors themselves ( $n$  and the diamond determinants — pre-existing inventoried objects); the address question introduces *no new residual class*.

**The dual-normal selector:**  $(d, n)$  **pins**  $R$  **columnwise** ([verification/v136\_dual\_normal\_selector.py]). The selector list shrinks once more. With the dual anchor  $d = a^T R^{-1}$  ([verification/v134\_dual\_anchor.py]) and the torsion normal  $n$ , each *column* of  $R$  obeys  $d \cdot c_j = a_j$  and  $n \cdot c_j = 8\delta_{1j}$ ; the common kernel of  $(d, n)$  is the primitive vector  $(6, 8, 7)$  (audit:  $p_2$ , rank  $E_8$ , scalaron-7), so within the address box  $\{0..5\}^3$  each column is the *unique* lattice point (census: 1 of 216 each) —  $(d, n) \Rightarrow R$ , and the  $6^9$  census above collapses to three column problems [E]. *Honesty*:  $K$  is *not* pinned directly ( $d^T K = (1, \frac{1}{2}, 1)$ );  $\det K = 4$  comes via the  $\mu_4$  wall ([verification/v135\_det\_surface.py]). *The mechanism is bounded*: the zero mode of  $A_0^*$  carries the exact spectral normal  $\sigma = (2, -9, 5) = (|\mathbb{Z}_2|, -N_{\text{fam}}^2, g_{\text{car}})$  (signed squares;  $\|v\|^2 = \frac{16}{9}$  with  $16 = \dim S^+$ ,  $\text{adj } A_0^* = \frac{2}{9}P_v$  with  $\frac{2}{9}$  = the SdS entropy curvature),  $\sigma \cdot \mathbf{1} = -|\mathbb{Z}_2|$ ,  $\sigma \cdot a = N_{\text{fam}}$ ,  $n \cdot \sigma = 121 = \|\text{Pl}(K)\|_1^2$ ; and the exact  $W(A_3)$  orbit control shows *no* group image of  $\sigma$  is proportional to  $n$  — the naive cofactor/orbit lift fails, the step  $\sigma \rightarrow n$  is a genuine mechanism beyond the monodromy group.  $R_4'$  restated [C]: derive the *pair*  $(d, n)$ ; then  $R$  is forced columnwise.

**The selector triangle** ([verification/v139\_selector\_triangle.py]). The pair itself now collapses further. First,  $d$  is *pure anchor data*:  $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$  exactly — the closed form never mentions  $R$ , so the first selector is *derived*, not residual. Second, the frame  $(\mathbf{1}, a, \sigma)$  is a basis of  $\mathbb{Z}^3$  with  $\det(\mathbf{1}|a|\sigma) = 11 = \|\text{Pl}(K)\|_1$  (the quark constant as the frame volume), and  $n$  is the *unique* integer covector with the three atom pairings

$$n \cdot \mathbf{1} = 2 = |\mathbb{Z}_2|, \quad n \cdot a = 8 = \text{rank } E_8, \quad n \cdot \sigma = 121 = \|\text{Pl}(K)\|_1^2$$

— the cofactor definition is replaced by frame data plus three atoms. On the  $(\mathbf{1}, a)$ -constrained line the  $\sigma$ -pairing is  $11(11+t)$ : divisible by 11 for *every*  $t$ , a perfect square exactly at the true  $n$  ( $t=0$ ). The review's predicted lift is exact as an audit identity:  $n = \text{reverse}(\sigma) + |\mu_4|e_3$  (generation flip +  $\mu_4$  lift at the third slot).

**Frame integrality: three pairings  $\rightarrow$  two** ([verification/v142\_frame\_integrality.py]). The third pairing is *not* independent. The pairing triples  $(m \cdot \mathbf{1}, m \cdot a, m \cdot \sigma)$  of *integer* covectors form an index-11 sublattice of  $\mathbb{Z}^3$ , cut out by the single congruence

$$x - 3y + z \equiv 0 \pmod{11} \quad (\text{index } 11 = \|\text{Pl}(K)\|_1 = \det(\mathbf{1}|a|\sigma)),$$

so with the two line pairings at their atoms  $(x, y) = (|\mathbb{Z}_2|, \text{rank } E_8) = (2, 8)$ , integrality *forces*  $z \equiv 0 \pmod{11}$  — the  $11(11+t)$  line *is* the integer line. Primitivity forces  $t$  *even* (odd  $t$  makes every entry of  $n(t)$  even), and  $t=0$  is the unique even  $t$  with a square  $\sigma$ -pairing for  $|t| < 88$  (the global pick keeps the square selector — recorded honestly). Both line pairings are Cramer-reducible to inventoried identities ( $n \cdot a = \det R$  by Laplace along the anchor column;  $n \cdot \mathbf{1} = 8(R^{-1}\mathbf{1})_1$  with  $R^{-1}\mathbf{1} = \frac{1}{4}(1, 1, -1)$ , v134) — restructuring, not establishment.

**The pairing values are atom identities** ([verification/v145\_pairing\_atoms.py]). Even the two line-pairing *values* carry no independent information. Reading the v139 lift structurally —  $n = w_0(\sigma) + |\mu_4|e_3$  with  $w_0$  the  $A_3$  *exponent duality*  $m \leftrightarrow h-m$  (an established Lie involution; lift slot = the top-exponent line, lift size = the glue order) — all three pairings become atom arithmetic:

$$n \cdot \mathbf{1} = |\mu_4| - |\mathbb{Z}_2| = |\mathbb{Z}_2|, \quad n \cdot a = |\mu_4| a_3 = 8 = \text{rank } E_8 \quad \text{via} \quad |\mu_4| + g_{\text{car}} = N_{\text{fam}}^2 \quad (4 + 5 = 9, \sigma \perp w_0(a)),$$

and  $n \cdot \sigma = \|\sigma\|^2 - N_{\text{fam}}^2 + |\mu_4|g_{\text{car}} = 110 - 9 + 20 = 121$  with  $\|\sigma\|^2 = |\mathbb{Z}_2|^2 + N_{\text{fam}}^4 + g_{\text{car}}^2 = 110 = |\mathbb{Z}_2|g_{\text{car}}\|\text{Pl}(K)\|_1$ .  $R_4'$  *residue [C] (final form)*: derive the *one* map  $n = w_0(\sigma) + |\mu_4|e_3$  — no independent number remains; everything else in the chain anchor  $\rightarrow (d, n) \rightarrow R$  is exact.

**Both dual normals are anchor data** ([verification/v149\_cusp\_normal.py]). The remaining opacity



dissolves in the canonical frame: the pairings of  $n$  with the *cuspidal eigenbasis* (the geometric basis of v98) are exactly

$$n \cdot e_1 = 6 = p_2(a), \quad n \cdot e_2 = 3 = p_0(a), \quad n \cdot e_3 = 5 = e_2(a)$$

— one anchor invariant per  $Q_+$  degree, and the cusp matrix is unimodular, so these three pairings determine  $n$  uniquely. With  $d = \frac{3}{2}a - 2 \cdot 1$  the dual normal *pair* (v134) is anchor data through and through — one normal per frame; the cusp data  $(6, 3, 5)$ , the frame pairings  $(2, 8, 121)$  and the  $w_0$ -lift are three presentations of one covector, all atomic (equivalences exact). *Audit*:  $\sigma \rightarrow (5, 1, 2)$ ; alternate reading  $(6, 3, 5) = (|R^+(A_3)|, N_{\text{fam}}, g_{\text{car}})$ ; sum  $14 = \dim G_2$ .  *$R'_4$  residue [C] (cusp form)*: derive the *assignment* — why the  $Q_+$  degrees  $(1, 2, 3)$  carry  $(p_2, p_0, e_2)$  in that order; one discrete assignment, zero continuous freedom.

### Exterior Leg Lemma — the quark $u/d$ leg is a $\Lambda^2 F$ area, not a scalar [E]/[C]

The v41 negative is a *category error*, not a dead end.  $u$  and  $d$  share the hexagon-radial data  $(r, w) = (1, 1)$  ( $L = 7$  for both), and the  $C_6$  resolvent  $c(r, w) = |\mu_4|^w / (\frac{5}{4} - \cos \frac{r\pi}{3})$  reads *only*  $(r, w)$  — so any *scalar* leg gives  $c_u/c_d \equiv 1$ . A scalar sees radius, not orientation. The  $u/d$  separation lives in the smallest *non-scalar* invariant of the  $SU(3)_F$  composition: the exterior square  $\Lambda^2 F$ , i.e. the oriented anchor-plane area  $\text{Pl}_{1,a}(K)$ . The “11” is then *generated*, not read:

$$(K, \mathbf{1}, a) \rightarrow B_K = \begin{pmatrix} 13 & 8 & 4 \\ 18 & 11 & 6 \end{pmatrix} \rightarrow \text{Pl}(K) = (-1, 6, 4) = (-N_\Phi, |R^+(A_3)|, |\mu_4|) \rightarrow \|\text{Pl}(K)\|_1 = 11,$$

$$\frac{c_u}{c_d} = \frac{g_{\text{car}} \|\text{Pl}_{1,a}(K)\|_1}{N_{\text{fam}}^2 \Delta_Q} = \frac{5 \cdot 11}{9 \cdot 13} = \frac{55}{117} \quad (\text{anchor microcode} = \frac{e_2(a)(p_3(a)+1)}{p_0(a)^2(2p_2(a)+1)}).$$

**Typing:** [E] for the Plücker / microcode identities ( $\text{Pl}(K) = (-1, 6, 4)$ ,  $\|\cdot\|_1 = 11$ ,  $5 \cdot 11 / (9 \cdot 13) = 55/117$ ); and — by Readout Rigidity (v49, below) — [E] also for the *physical* ratio  $C_{ud}(\rho^*)$  = this  $\Lambda^2 F$  readout, which is *constant* on the derived selector stratum and so does *not* need any point-selection on  $\Lambda^2 F$ . The quark residual is thereby re-typed from “find the right scalar  $y$ ” to a closed exterior readout for the *ratio*; only the *absolute* amplitude scale ( $U_{\text{point}}$ ) stays as an anchor, no new scalar, no fishing.

*Audit* (the 121 lemma, [verification/v119\_review\_validation\_2.py]): the “11” has a second, independent appearance [E]: with  $h = \mathbf{1} + a = (2, 2, 3)$  in the canonical anchor ordering,  $h^T R h = 121 = 11^2 = \|\text{Pl}(K)\|_1^2 = (p_3 + 1)^2$  — the quadratic *anchor norm* of the residue operator  $R$  reproduces the Plücker integer (sensitivity control: the three orderings of  $h$  give  $\{105, 121, 135\}$ ; only the canonical one yields 121). Bonus checksums:  $\mathbf{1}^T R \mathbf{1} = 22 = 2 \times 11$  (the entry sum of  $R$ ) and  $a^T R a = 40 = p_1 p_3 = 10 b_1 - 1$ . Typed *audit*, not load-bearing — the oriented Plücker mechanism above remains the derivation. [verification/v42\_exterior\_leg.py]

**Bridge test** ([verification/v43\_exterior\_bridge.py]). The typing is confirmed: for the  $SU(3)$  holonomy the exterior square is the conjugate representation,  $\Lambda^2(\widetilde{M}) = \widetilde{M}$  exactly — the exterior leg *is* the  $\widetilde{\mathbf{3}}$ . But the *continuous* action  $\Lambda^2(\widetilde{M}) \cdot (\mathbf{1} \wedge a)$  does *not* equal the *integer*  $\text{Pl}(K) = (-1, 6, 4)$ : the integer Plücker is the combinatorial  $K$  datum. So the bridge  $\rho^* \rightarrow \text{Pl}(K)$  is the *discrete* non-abelian-Hodge invariant (the integer  $R$ ;  $\det R = 8$  and  $\text{Spec}(Q_+) = \{1, 2, 3\}$  already confirmed on the bundle, Lemma U4 box), not a continuous exterior-square computation — exactly the H2 equivalence, now pinned from the  $\Lambda^2 F$  side.

**Lie-level grounding — no special math** ([verification/v44\_carrier\_exterior.py]). The discrete invariant has a home already in TFPT: the carrier half-spinor is the *even exterior algebra* of the 5-slot carrier,  $16 = \Lambda^{\text{even}}(5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5$ , and under  $SU(5) \subset SO(10)$  the up quark sits in  $10 = \Lambda^2(5)$  while the down sits in  $\bar{5} = \Lambda^4(5)$ . So the *exterior degrees* are  $\deg(u^c) = 2$ ,  $\deg(d^c) = 4$ , with  $\deg(u) + \deg(d) = 6 = |R^+(A_3)|$  (the hexagon) and  $\deg(d) - \deg(u) = 2 = |\mathbb{Z}_2|$  (the sheet). The “ $\Lambda^2$ ” of the exterior leg is therefore not exotic Hodge data — it is the carrier’s own  $\Lambda^2$  grading (the up quark *lives* in  $\Lambda^2(5)$ ). This re-types the residual once more: from “discrete non-abelian-Hodge invariant” to “the carrier exterior degree” — a standard [E] fact. *Honest scope*: this grounds the *type*; the Plücker *number* 55/117 stays the family  $\text{Pl}(K)$  [E], and the carrier- $\Lambda^2 \leftrightarrow$  family- $\Lambda^2 F$  link (via family  $\otimes$  carrier = 15 = 16–1) plus the normalisation remain [C].

**The “11” is Pascal — a third, simplest origin** ([verification/v45\_family\_exterior.py]). Following that link,  $\text{Pl}(K)$  is the *exterior algebra of the family fundamental*  $4 = \mu_4$  (the  $A_3 = SU(4)$  fundamental):  $\Lambda^k(4) = \binom{4}{k}$  is Pascal row 4 =  $(1, 4, 6, 4, 1)$  with full sum  $2^4 = 16 = \dim S^+$ , and

$|\text{Pl}(K)| = (1, 4, 6) = \{\Lambda^0, \Lambda^1, \Lambda^2\}(4)$  are exactly its distinct exterior powers. Hence

$$\|\text{Pl}(K)\|_1 = \sum_{k=0}^2 \binom{4}{k} = 11 = 2^4 - \binom{4}{3} - \binom{4}{4} = 16 - g_{\text{car}}$$

(the cumulative  $\Lambda^\bullet(\mu_4)$  up to degree  $2 = \deg(u^c)$ ), and family  $\otimes$  carrier  $= 15 = \dim \mathfrak{su}(4) = \Lambda^2(5) \oplus \Lambda^4(5) = 10 + 5 = N_{\text{fam}} g_{\text{car}}$  read three ways. So the “11” is generated a *third* way — the cumulative family exterior algebra up to the up-quark’s degree — pure Pascal/branching, *no special math*. The [C] physical normalisation is unchanged; the algebraic origin of 55/117 is now triply grounded.

**Lemma 6** (U5 — the “11” worry, *discharged* by Readout Rigidity). *The original worry was that  $c_u/c_d = C_{ud}(\rho^*)$  must be evaluated at the uniquely selected  $\rho^*$  without using 11 as input, so that until then 55/117 would be a mere table-level rational shadow. **This is now resolved (v49):**  $C_{ud}$  is constant on the discrete selector stratum  $\mathcal{S}_{8,Q}$ , so the “11” is earned as the rigid Plücker readout  $\|\text{Pl}(K)\|_1$  (v42) — independently of point-uniqueness. Point-uniqueness of  $\rho^*$  enters only the absolute amplitude scale, not the ratio.*

**Lemma 7** (U6 — the four certificates).  $C_U^{(1)}$  wall enumeration (done);  $C_U^{(2)}$   $D_4$ -fixed character solver;  $C_U^{(3)}$  selector uniqueness (one  $S$ -equivalence class);  $C_U^{(4)}$  amplitude extraction — the ratios  $R, Q, c_u/c_d$  are read off the (derived) selector stratum (closed, v49/v71); only the absolute  $U_f^*, c_q$  normalisation is the anchor.

**Progress: an explicit valid flat bundle is realised (RH solve) ([E]).** A numerical Riemann–Hilbert solve produced an *explicit* Hermitian residue  $A_0$  (eigenvalues  $\{0, \frac{1}{3}, \frac{2}{3}\}$ ,  $\text{diag} = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ ) for which the Fuchsian system  $A(z) = \sum_k U^{k-1} A_0 U^{-(k-1)} / (z - i^k)$  simultaneously satisfies: the cusp class at each puncture; the splitting  $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$  (exponents  $\sum_k A_k = \text{diag}(2, 1, 1)$ ); the *path-ordered* monodromy at infinity is trivial,  $\|M_\infty - \mathbf{1}\| \sim 10^{-9}$ , i.e.  $M_1 M_2 M_3 M_4 = \mathbf{1}$ ; and the representation is *irreducible* — **case A** (non-trivial  $SU(3)_F$  mixing, not the diagonal case B). So a valid ( $U_{\text{wall}}$ ) flat bundle *exists and is constructed*; the U2 kill switch lands on A. *Honest scope*: this is one realising point (existence + case A), likely one of a residual family; the unique  $\nabla_F^*$  still needs the  $\det R = 8$  selector and  $c_u/c_d$  still needs the  $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$  dictionary (H2). [verification/v33\_explicit\_flat\_bundle.py]

*H2 bridge probed* ([verification/v34\_h2\_bridge\_attempt.py]). The explicit per-puncture monodromies  $M_k$  (cusp class,  $\prod M_k = \mathbf{1}$  to  $10^{-5}$ ) were computed; but the natural resolvent-dressed diagonal extraction ( $|\text{diag } M_k| = (0, \frac{1}{2}, \frac{1}{2})$  times the  $C_6$  resolvent factor) does *not* reproduce the lepton amplitudes (0.475, 1.107, 0.917). So the precise  $\Gamma_{ij}^{\min}$  geodesic-to-word dictionary (how the 6-site hypercharge hexagon combines with the 3-dim family  $\rho^*$ ) was the genuine remaining analytic input — it is *not* fixed by a natural guess, and we do *not* fish for 55/117. *Resolved since (v118): the correct extraction is the resolvent determinant of the  $W(A_3)$  monodromy, not the diagonal dressing — the probe’s failure is thereby explained, and the dictionary is exact.*

#### Theorem U — the four-way split (sharper than one monster gate) [E]/[O]

$D_4$  fixes the admissible chamber, not the physical point (the  $D_4$ -fixed locus is positive-dimensional, U3); the discrete selectors  $\det R = 8$ ,  $\text{SNF}(R) = (1, 1, 8)$ ,  $\text{Spec}(Q_+) = \{1, 2, 3\}$  do the cutting. The gate splits into four pieces, three of them essentially closed:

- $U_{\text{unitary}}$  [E]: parabolic degree 0 + polystable  $\Rightarrow$  unitary (Mehta–Seshadri)  $\Rightarrow \Phi = 0$ ; the harmonic metric is the *finite* invariant form  $H$  ( $M_k^\dagger H M_k = H$ ), not a Hitchin PDE ([verification/v40\_harmonic\_metric.py]).
- $U_{\text{H2}}$  [E]:  $R, Q, K$  are read off the bundle (splitting type  $= a$ ,  $\det R = 8$ ,  $\text{Spec}(Q_+)$ ) ([verification/v39\_uwall\_selectors.py]).
- $U_{\Lambda^2}$  [E]: **Readout Rigidity** — on the discrete stratum  $\mathcal{S}_{8,Q} = \{\det R = 8, \text{SNF}(R) = (1, 1, 8), \text{Spec}(Q_+) = \{1, 2, 3\}\}$  the anchor-plane readout is *constant*,  $C_{ud}(\rho) = g_{\text{car}} \|\text{Pl}_{1,a}(K)\|_1 / (N_{\text{fam}}^2 \Delta_Q) = \frac{55}{117}$ , *independent* of the continuous  $D_4$  position — so  $c_u/c_d$  needs no point uniqueness ([verification/v49\_readout\_rigidity.py],

[[verification/v42\\_exterior\\_leg.py](#)]).

- $U_{\text{point}}$  [O]: full uniqueness of  $\rho^*$  — needed only for the *complete*  $U_f^*$  amplitude matrix, *not* for  $c_u/c_d$ .

**Headline:** the remaining flavor bridge is not a PDE gate — it is a finite exterior *readout-rigidity* theorem; the only residual,  $U_{\text{point}}$ , is not an independent gate — it reduces to the single dimensionful anchor  $v_{\text{geo}}$  (Gate 1 below).

**The simple closure** ([[verification/v71\\_simple\\_r\\_bridge.py](#)]). The selector stratum  $\mathcal{S}_{8,Q}$  is now *fully derived*:  $\det R = 8 = \text{rank } E_8$  with  $\text{SNF}(R) = (1, 1, 8)$  is the lattice invariant (cf.  $\det Q = 3 = N_{\text{fam}}$ , v70), and  $\text{Spec}(Q_+) = \{1, 2, 3\}$  is the  $D_4$ -equivariant result (v69). The four flavor operators are integer lattice maps with compiler-atom determinants ( $\det Q, \det K, \det R, \det L$ ) = (3, 4, 8, 20), product  $1920 = |W(D_5)|$ . So by Readout Rigidity the quark *ratios* ( $c_u/c_d = g_{\text{car}} \cdot 11 / (N_{\text{fam}}^2 \cdot 13) = \frac{55}{117}$ , etc.) are *pure integer Plücker readouts* — *no transcendental monodromy solve*. The earlier monodromy machinery was over-engineering for the ratios; the only transcendental piece,  $U_{\text{point}}$ , is the *absolute* amplitude normalisation — an anchor of the same nature as the one dimensionful scale, not a missing computation.

**Gate 1 complete:**  $U_{\text{point}} \rightarrow v_{\text{geo}}$  ([[verification/v75\\_upoint\\_to\\_vgeo.py](#)]). The absolute amplitudes are not free either. The lepton amplitudes are exact ( $c = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ , v20), every cross-sector  $c$ -ratio is closed (Plücker), and each *sector product* is a clean  $\varphi_0^{\text{ret}}$ -power (Grand Mass Volume, v46:  $\det M_{\text{sector}} \sim (\varphi_0^{\text{ret}})^{6,9,10}$ ; the lepton product is  $\frac{16}{7} \cdot \frac{4}{3} \cdot \frac{7}{6} = \frac{32}{9} = 2^{g_{\text{car}}} / N_{\text{fam}}^2$ ). Since (pairwise ratios) + (product)  $\Leftrightarrow$  (individuals) is a bijection, the nine charged-fermion amplitudes are *fixed up to one overall scale*  $v_{\text{geo}}$ . **So  $U_{\text{point}}$  is not an independent gate: it reduces to  $v_{\text{geo}}$ , the same one irreducible dimensionful anchor as gravity's  $1/G$  (v68) — the two [O] anchors collapse to one.**

## Research Contract 2 — ( $G_{\text{metric}}$ )

**Goal.** Construct the TFPT metric measure

$$Z[J] = \lim_{\chi \rightarrow \infty} \int_{\mathcal{G}_\chi / \text{Diff}} \exp(-S_{\text{rel},\chi}[g, A, \psi] + J \cdot O) d\mu_{\text{Cal},\chi},$$

$$S_{\text{rel},\chi} = \text{Tr} f\left(\frac{D_{\text{rel}} + A_\Sigma}{\chi}\right) - \text{Tr} f\left(\frac{D_{\text{ref}}}{\chi}\right) + \frac{i\pi}{2} \Delta\eta_\Sigma,$$

the relative spectral action whose low-curvature shadow is the already-closed  $R + R^2$  equation.

**Lemma 8** (G0–G1 — configuration space & BRST/Hodge Fredholm). *For  $s > 5/2$  a Sobolev space of  $\Theta$ -symmetric metrics on the doubled seam collar  $M_\Sigma$  admits a gauge slice; the gauge-fixed gravitational complex  $\Omega^0(TM) \xrightarrow{K} \Gamma(S^2T^*M) \xrightarrow{K^*} \Omega^0(TM)$  with  $\Delta_{\text{gh}} = K^*K$  and  $\Pi_{\text{phys}} = \Pi_{\text{BRST/Hodge}} \Pi_{TT}$  is elliptic Fredholm; the conformal mode is removed as BRST-exact / non-propagating in the Calderón polarisation (not rhetorically). Machine: principal symbols in xAct/-Cadabra; linear-algebra certificates in Lean (medium).*

**Lemma 9** (G2 — finite relative spectral action, *heat-kernel grounded* [E]).  *$S_{\text{rel},\chi}$  is finite, differentiable, with a local Seeley–DeWitt expansion  $\text{Tr} f(D^2/\chi^2) \sim f_4 \chi^4 a_0 + f_2 \chi^2 a_2 + f_0 a_4 + \dots$ . For the Lichnerowicz Dirac operator  $D^2 = \nabla^2 + \frac{R}{4}$  (so  $E = -\frac{R}{4}$ , spinor trace  $\text{tr } \mathbf{1} = 4$ ) the Gilkey coefficients give, per  $(4\pi)^{-2}$ ,*

$$a_2 = \frac{1}{6} \text{tr}(6E + R \mathbf{1}) = -\frac{R}{3} \quad (\text{Einstein–Hilbert } \int R), \quad a_4|_{R^2} = \frac{1}{360} \text{tr}(180E^2 + 60RE + 5R^2 \mathbf{1}) = \frac{R^2}{72}$$

(the  $R^2$ /Starobinsky term), so the spectral action structurally produces  $S_{\text{eff}} = \int \sqrt{g} \frac{\bar{M}_{\text{Pl}}^2}{2} (R + \frac{R^2}{6\bar{M}^2}) + \dots$  with the scalaron mass ratio fixed by the cutoff moment,  $M^2/\bar{M}_{\text{Pl}}^2 = 6(4\pi)^2/f_0$ . The TFPT closure  $M^2/\bar{M}_{\text{Pl}}^2 = c_3^7$  fixes  $f_0 = 6(4\pi)^2/c_3^7$ , i.e.  $M = c_3^{7/2} \bar{M}_{\text{Pl}} = 3.06 \times 10^{13} \text{ GeV}$  — the boundary normalisation  $c_3$  is the gravitational scale — and hence the Starobinsky slow-roll forms  $n_s \simeq 1 - \frac{2}{N}$ ,  $r \simeq \frac{12}{N^2}$ . The  $R + R^2$  structure is convention-independent (standard Gilkey); the precise rational  $\frac{1}{72}$  and factor 6 are the standard Dirac conventions used here ([[verification/v36\\_spectral\\_action\\_g2.py](#)], [[verification/v28\\_gravity\\_fR.py](#)]).

**The simpler resolution: “full QG” for physics is the gap-decoupled admissible sector ([E]).** The infinite-dimensional ambient measure (G6) is the summit, but the admissible IR sector does not wait for it. The metric coupling is gap-dominated (G5):  $2\|V_{\text{metric}}\| = 0.785 < \Delta = 6 \log \frac{3}{2} = 2.433$ , so the admissible sector keeps a *strictly positive* effective gap  $\Delta_{\text{eff}} = 1.648 > 0$  under the stated RP and metric-coupling assumptions. A positive gap dynamically *isolates* the admissible sector from the un-built ambient/deep-UV: under those assumptions the admissible (IR) measure is closed, and the ambient completion is decoupled from it (we do *not* assert it is physically irrelevant — that would itself be a physical interpretation). So the honest reading is

admissible IR sector = closed under stated RP;  
ambient metric measure (G6) =  $G_{\text{metric}}$  gate

G2 supplies the local action ( $R+R^2$ , heat-kernel); G5 supplies the decoupling of the admissible sector; together they reduce ( $G_{\text{metric}}$ ) to the ambient projective limit (G6). **Precise status (Alessandro’s boundary):** the admissible IR sector is closed under the stated RP and gap-dominance assumptions; the ambient metric projective measure is now discharged as a [C] *redundancy* (v369, the holographic-route keybox below), a certification object rather than a missing piece. Its absence does *not* affect the bounded IR claim. [verification/v36\_spectral\_action\_g2.py]

**The glue Q-system: the index-4 object of the  $G_{\text{net}}$  statement exists explicitly ([E]/[C]).** The  $G_{\text{net}}$  target — “the seam–Calderón inclusion has Jones index  $4 = |\mu_4|$ ” — no longer refers to an unknown object. On the glue group algebra  $A = \mathbb{C}[\mathbb{Z}_4]$  (multiplication  $\delta_a \otimes \delta_b \mapsto \delta_{a+b}$ , unit  $\delta_0$ ) all Longo Q-system axioms hold exactly [E]: associativity, unit law, the Frobenius identity  $(m \otimes 1)(1 \otimes m^*) = m^*m = (1 \otimes m)(m^* \otimes 1)$ , and specialness  $mm^* = |\mathbb{Z}_4| \mathbf{1}$  — a special C\* Frobenius algebra with

$$\dim \theta = 4 = |\mu_4| = [B : A]$$

exactly the Carrier Index Lemma value (KLM:  $\mu_{\text{carrier}} = 16 = 4^2 \mu_{E_8}$  re-verified). *Locality = isotropy* [E]: on the Lagrangian glue  $\langle (1, 1) \rangle$  the discriminant form gives  $q(k(1, 1)) = k^2 \in \mathbb{Z}$  for every  $k$  — the braiding obstruction vanishes (v92 in Q-system language); the two Lagrangian glues (the sheet pair) are the only maximal isotropic choices. *Halfway step* [E]:  $\{0, 2\} < \mathbb{Z}_4$  closes and gives the unique intermediate Q-system  $\mathbb{C}[\mathbb{Z}_2]$  = the  $SO(16)_1$  tower step, with index factorisation  $4 = 2 \times 2$ . *R2 sharpened* [C] (recorded, not claimed): what remains is the operator-algebraic *identification* “the seam–Calderón inclusion of the 16-Majorana net (v113) realises *this* Q-system” — matching two constructed objects, not a search; the three loads (metric + carrier + QBL, v123) hang on this one identification.

**The hull is the graded module of the Q-system** ([verification/v128\_graded\_hull.py]). The Q-system is not merely an abstract companion of  $E_8$  —  $E_8$  *realises* it [E]: the 240 roots, built as explicit norm-2 vectors in  $D_5 \oplus A_3$ +glue coordinates, decompose as  $240 = 52 + 64 + 60 + 64$  over the glue  $\mathbb{Z}_4$  (the v89 branching as vectors), and for *all* 6720 root pairs whose sum is again a root,  $\text{coset}(r_1+r_2) = \text{coset}(r_1) + \text{coset}(r_2) \bmod 4$ :

$E_8$  is a  $\mathbb{Z}_4$ -graded Lie algebra over the carrier; the grading group is the glue = the Q-system

— the checksum hull is the Q-system’s graded module, realised in 248 dimensions. The R2 identification thus matches the Calderón inclusion against a structure that  $E_8$  already *carries*. [verification/v125\_glue\_qsystem.py]

**Graded Frobenius** ([verification/v143\_graded\_frobenius.py]). The identification holds at the finite Lie level: the graded module is in fact a graded *Frobenius* structure, i.e. everything the Q-system demands of its module is already exact inside  $E_8$  [E]: (i) *coset duality* —  $\text{coset}(-r) = -\text{coset}(r) \bmod 4$  for all 240 roots ( $-C_1 = C_3$  bijectively,  $C_0, C_2$  self-dual), so the Killing form restricts to a *nondegenerate* invariant pairing  $g_k \times g_{-k} \rightarrow \mathbb{C}$ ; (ii) the glue character average  $\frac{1}{4} \sum_k i^{k \deg}$  is *exactly* the carrier projector (the conditional-expectation shadow, quasi-basis index  $|\mathbb{Z}_4| = 4$ ); (iii)



each glue sector is *one*  $W(D_5) \times W(A_3)$  orbit with multiplicity-one dimensions  $64 = 16 \cdot 4 = (\mathbf{16}, \bar{\mathbf{4}})$ ,  $60 = 10 \cdot 6 = (\mathbf{10}, \mathbf{6})$  — simple currents, sector ring  $= \mathbb{Z}_4 =$  the  $\mathbb{C}[\mathbb{Z}_4]$  fusion, halfway subgroup  $\{0, 2\} = SO(16)_1$ . *R2 residue [C] (final form)*: exactly *one* conformal-net statement remains — the seam–Calderón inclusion of the 16-Majorana *net* carries this Q-system — with no remaining finite freedom.

**Where that statement lives: the NS/R sector census** (`[verification/v148_fock_census.py]`). The remaining work is scoped exactly. Every *untwisted* (Neveu–Schwarz) state of the 16 Majoranas has *integer* ( $D_5, SO(6)$ ) weights, and integer weights lie in the even discriminant classes — so the NS module supports only  $\{(0, 0), (2, 0), (0, 2), (2, 2)\}$ : **the odd glue classes  $k(1, 1)$  — the actual  $\mu_4$  simple currents — have zero NS support; they are twisted (Ramond-type) modules.** The Ramond zero-mode module ( $2^8 = 256$  ground states) carries exactly the four odd class pairs, 64 each, and splits  $128 + 128$  into the odd sectors of the *two* Lagrangian glues  $\langle(1, 1)\rangle$  and  $\langle(1, 3)\rangle$  (v92): *choosing the glue is choosing the Ramond projection* ( $128 =$  one  $SO(16)$  spinor chirality), and  $E_8$  assembles as  $248 = 120_{\text{NS}} + 128_{\text{R}}$ . *Consequence [C]*: the index-4 extension cannot even be *stated* on the untwisted module alone; R2’s one remaining statement is irreducibly about the twisted-sector structure of the seam–Calderón net, with the sheet choice realised as the R-projection; the finite shadow is exhausted (v113/v125/v143 + this census).

**Lemma 10** (G3–G4 — reflection positivity, fixed and integrated). *For each  $\Theta$ -invariant  $g$  in the Calderón ball the physical boundary kernel  $\mathcal{K}_g = P_{\text{adm}} \Pi_{\text{phys}} \mathcal{N}_g^{TT} \Pi_{\text{phys}} P_{\text{adm}}$  is reflection positive,  $\langle \Theta F, F \rangle_{\mathcal{K}_g} \geq 0$  (G3); and  $d\mu_{\text{Cal}, \chi}$  is  $\Theta$ -invariant, supported where G3 holds uniformly, so  $\int \langle \Theta F, F \rangle_{\mathcal{K}_g} d\mu_{\text{Cal}, \chi} \geq 0$  (G4). **G4 is the decisive step:** RP after integrating over dynamical metrics, not merely at fixed background.*

**Lemma 11** (G5 — gap dominance [E]).  $2\|V_{\text{metric}}\|_{\text{rel}} < \Delta = 6 \log \frac{3}{2}$ , with  $\|V_{\text{metric}}\|_{\text{rel}} \leq 248c_3^2 = \frac{31}{8\pi^2} = 0.39262$ , hence  $2\|V\| = 0.7852 < 2.4328$  and  $\Delta_{\text{eff}} \geq \Delta - 2\|V\| > 1.647 > 0$ . *The numerical inequality is trivial* (`[verification/v29_research_contract_certs.py]`); *the work is the operator-norm bound (hard)*.

**Sugawara reading + atomic OS moment** (`[verification/v171_os_moment_cluster.py]`). The G5 budget is the  $E_8$  Sugawara constant:  $248c_3^2 = \frac{31}{8\pi^2}$  with  $31 = 1 + h^\vee(E_8)$  and  $c((E_8)_1) = 248/31 = 8$ , so gap dominance reads  $|R^+(A_3)| \log(N_{\text{fam}}/|\mathbb{Z}_2|) > \frac{1+h^\vee(E_8)}{4\pi^2}$  — the  $A_3$  transfer gap beats the  $E_8$  Sugawara metric budget. Independently, the G3 reflection positivity is a *positive moment problem*:  $\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$  makes every two-point correlator  $C(n) = A + B(64/729)^n + C(1/729)^n$  the moment sequence of a positive atomic measure, so each OS Hankel matrix factorises as a Vandermonde Gram and is positive semidefinite (a clean RP certificate; cluster constant  $729/665$ , correlation length  $\xi = 1/(6 \log \frac{3}{2}) = 0.4111$ ).

**Theorem 1** (Decoupling theorem — physical low-energy claims do *not* need the full  $G_{\text{metric}}$ ). *If the admissible sector is reflection-positive (G4) and the metric coupling is gap-dominated,  $2\|V_{\text{metric}}\| < \Delta = 6 \log \frac{3}{2}$  (G5), then the admissible IR measure is stable under the ambient completion: a strictly positive effective gap  $\Delta_{\text{eff}} = \Delta - 2\|V\| > 0$  protects the admissible sector from the (un-built) ambient/deep-UV, so every physical low-energy read-out (masses, mixings,  $\alpha^{-1}$ , the  $R+R^2$  regime) is independent of the open ambient measure. Symbolically*

$$\boxed{\text{RP} + 2\|V\| < \Delta \implies \text{admissible IR measure stable under ambient completion.}}$$

*Thus ( $G_{\text{metric}}$ ) is not a diffuse “full quantum gravity missing” block: it is the sharply delimited ambient projective limit (G6), a strict-TOE certification target whose closure the falsifiable IR physics does not depend on. **Typing:** the inequality is machine-checked [E]; the operator-norm bound on  $\|V_{\text{metric}}\|$  is the conditional input.*

**Lemma 12** (G6–G7 — projective limit & Ward identities). *A projective system  $\mu_{\chi_n}$  with  $\pi_{m,n*} \mu_{\chi_m} = \mu_{\chi_n}$ , uniform moment bounds  $\sup_n \int \|g\|_{H^s}^p d\mu_{\chi_n} < \infty$ , tightness and regulator-independence yields  $\mu_{\text{QG}} = \lim \mu_{\chi_n}$  on  $\mathcal{G}/\text{Diff}$  (G6, the hardest step); the limit obeys the BRST Ward identities  $\langle s\mathcal{O} \rangle = 0$ , whence  $\nabla^\mu T_{\mu\nu} = 0$  and the  $\eta$  boundary phase cancels the APS anomalies (G7).*

**Gate 2 reduction:** G6 is a *boundary projective limit*, not a bulk one ([verification/v76\_gmetric\_reduction.py])

The exact decoupling margin is  $\Delta_{\text{eff}} = \Delta - 2\|V\| = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} = 1.648 > 0$ , with  $\|V_{\text{metric}}\|_{\text{rel}} \leq 248 c_3^2 = \dim(E_8) c_3^2 = \frac{31}{8\pi^2} (31 = 2^{g_{\text{car}}} - 1)$ . **Holographic reduction:** because the seam is a *finite causal boundary*, the ambient *bulk* metric measure is reconstructed from a finite-codimension seam (Calderón) *boundary* measure — so G6 is a *boundary projective limit*. The conditional theorem is then

$$\text{RP}(\text{seam-Calderón boundary kernel}) + \text{tightness} \implies G_{\text{metric}} \text{ closes}$$

i.e. the dimension of the open problem drops from bulk QG to a finite seam-boundary measure.

**Residual:** the constructive boundary projective limit (constructive-QFT grade) — now discharged as a [C] redundancy (v369, keybox below). So Gate 2 is now *two clean tiers*: Tier A (IR physics) closed under RP+gap; Tier B (strict TOE) a [C] redundancy, no longer a diffuse “full QG missing”. [C]

**The measure as one-loop fluctuations around the parameter-free saddle** ([verification/v365\_qg\_oneloop\_saddle.py], QGAMB.SADDLE.01)

With the saddle now parameter-free (v358/v359), the boundary measure of Tier B organises as a Gaussian one-loop fluctuation determinant whose ingredients are atom-fixed. **Fixed quadratic form [E]:** the fluctuation inverse-propagator stiffness is the saddle coefficient  $1/c_3 = 8\pi$  — no free dimensionless dial, so the one-loop problem is *well-posed* (it was not before v358). **Gap-controlled convergence [E]:** the fluctuation operator obeys  $M \geq \Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$  (v76/v337), so the Gaussian integral converges mode-by-mode with no zero/negative modes besides the saddle directions; on the OS spectrum  $\{0, 6 \log \frac{3}{2}, 6 \log 3\}$  the regularised one-loop log-det  $\text{Tr}' \log M = 6 \log \frac{9}{2}$  is finite. **Conformal direction [C]:** the one wrong-sign mode ( $c_{\text{conf}}(4) = -\frac{3}{2}$ ) is made convergent by the GHP contour ( $\int e^{-|c|\phi^2} = \sqrt{2\pi/3}$ ) with IDG dressing (v334). **Tightness [E]:**  $\chi = 1/(1 - (2/3)^6) = 729/665 < \infty$  (v330), so the one-loop measure has a projective limit on the admissible sector, with *zero* new dimensionless dials (v364). **Residual:** the full *non-perturbative* projective limit (G6 on the ambient sector) is now discharged as a [C] redundancy (v369, keybox below) — this sharpens C7 from “diffuse full QG” to a one-loop-controlled Gaussian problem around a parameter-free saddle and then to a certification object. [E]/[C]

**The holographic route: the ambient measure is boundary-redundant, not missing dynamics** ([verification/v369\_qgamb\_redundancy.py], QGAMB.REDUNDANCY.01)

There is a second, strategically stronger way to dispatch Gate 2 than building the measure: prove that the ambient sector carries *no* physical degrees of freedom relative to the holomorphic boundary algebra, so the measure QG.AMB.01 is a non-fundamental *certification* object — the role a coordinate choice plays in general relativity — rather than a missing piece of dynamics. The discriminators that make this *ambient-redundancy* statement well-posed are all in hand. **Trivial superselection [E]:** the number of DHR sectors of a chiral net is  $|\det \text{Cartan}|$ , which is 1 for  $E_8$  (one primary, the vacuum;  $\text{DHR}((E_8)_1) = \text{Vec}$ ) against 4 for the same- $c$  rival  $SO(16)_1$  — a holomorphic boundary net has *no* nontrivial superselection charge that a bulk could carry invisibly. **No torus degeneracy [E]:** the ground-state degeneracy on  $T^2$  equals the number of anyons  $= |\det K| = 1$  for  $E_8$ , so there is no topologically protected bulk d.o.f. hidden from the boundary (the genus-1 obstruction of v344, in its redundancy reading). **Bounded reconstruction [E]:** the seam transfer deviation subspace contracts at the finite gapped Petz rate  $(2/3)^6$  (v221), and the gap-decoupling margin  $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$  (v76/v337) separates the un-built ambient sector from the admissible one. **The statement [C]:** with the Bisognano–Wichmann modular reconstruction map (v258/v309/v323), under SEAM.EQUIV.01 every gauge-invariant bulk observable is boundary-reconstructible,  $\mathcal{O}_{\text{phys}}^{\text{bulk}}/\text{Diff} \cong \mathcal{A}_{\Sigma}^{\mu_4}$ , and two ambient measures sharing the same boundary state are physically equivalent. **Residual [O]:** the construction consumes exactly two premises — SEAM.EQUIV.01 (the S3 lattice input, v366) and the BW intrinsicity of the raw collar (v329); it does *not* construct QG.AMB.01 but supplies the *alternative* route, so that Gravity-complete = Boundary-complete + Ambient-redundancy. [E]/[C]/[O]

**The classical field equation, parameter-free: entanglement equilibrium gives the linearised Einstein equation with  $G$  from  $c_3$  ([verification/v358\_grav\_entropy\_equilibrium.py])**

While Gate 2 (above) reduces the metric-sector *measure*, the classical metric-sector *field equation* is supplied directly by the entanglement first law  $\delta S = \delta \langle K \rangle$  (Jacobson 2015; Faulkner et al.), carried out with TFPT's atoms. **Parameter-free coefficient [E]**. Equilibrium gives  $G_{ab} = c_3^{-1} T_{ab}$  with  $c_3^{-1} = 8\pi$  *fixed*: all four  $c_3$ -roles are mutually consistent — the Unruh factor  $1/(2\pi) = 4c_3$ , the EH action  $1/(16\pi) = c_3/2$ , the Einstein coefficient  $8\pi = 1/c_3$ , the Bekenstein density  $1/4 = 1/|\mu_4|$  — so *no free dimensionless Newton dial* enters. **Thermo = geo coincidence [E] (new)**. The first law requires the coefficient  $2\pi/\eta$  ( $\eta$  the entropy density); the geometry gives  $1/c_3 = |\mathbb{Z}_2| 2\pi \chi(S^2)$  (one-sided Gauss–Bonnet, v58); these are the *same* equation iff  $|\mu_4| = |\mathbb{Z}_2| \chi(S^2) = 4$  — a TFPT atom identity (v216). So the thermodynamic (first-law) and geometric (Gauss–Bonnet) origins of  $c_3$  are *one*. **The matter side, assembled [C] (closing J3)**. The ball modular Hamiltonian is the boost (Casini–Huerta–Myers; geometric by Bisognano–Wichmann, v323),  $K_B = 2\pi \int_B w T_{00}$ ,  $w = (R^2 - r^2)/(2R)$ ; the 3-ball weight integral is  $\int_B w d^3x = 4\pi R^4/15$ , so  $\delta \langle K_B \rangle = \frac{8\pi^2 R^4}{15} \delta \langle T_{00} \rangle$  — the concrete matter boost flux of TFPT's stress tensor. **The entropy density, atom-fixed [E]**. The Bekenstein numerator  $1/4 = 1/|\mu_4|$  (v57) and the entanglement normalisation = the central charge  $c = g_{\text{car}} + N_{\text{fam}} = 8$  pin the dimensionless coefficient; the area-law form is established (v59). **Residual**: the full covariant extension is now *closed* (v359, next keybox); what remains is only (i) the Jacobson equation-of-state *status* (a thermodynamic derivation, not a from-action quantisation) [O] and (ii) the absolute scale  $v_{\text{geo}}$  (the Planck-area unit) [O], while the global ambient measure (Gate 2/C7) is now a [C] redundancy (v369) — *no free dimensionless parameter survives in the gravity coupling*.

**And that boundary measure is a rigorously constructed net** ([verification/v77\_e8\_conformal\_net.py]). The level-1 central charges  $c = \dim \mathfrak{g}/(1 + h^\vee)$  give  $c(E_8)_1 = \frac{248}{31} = 8 = \text{rank } E_8$ ,  $c(D_5)_1 = \frac{45}{9} = 5$ ,  $c(A_3)_1 = \frac{15}{5} = 3$ , and  $c(D_5) + c(A_3) = 5 + 3 = 8 = c(E_8)$  is a *conformal embedding*  $(D_5)_1 \times (A_3)_1 \subset (E_8)_1$  (coset  $c = 0$ ). Now  $(E_8)_1$  is the holomorphic  $c = 8$   $E_8$ -lattice VOA / conformal net — one of the most rigorously constructed objects in mathematical QFT (Frenkel–Kac–Segal; Kawahigashi–Longo / Carpi). **So the conditional theorem becomes concrete: identify the seam-Calderón boundary measure with the  $(E_8)_1$  lattice net** (carrier =  $(D_5)_1 \times (A_3)_1$  subnet), and the gap  $\Delta_{\text{eff}} > 0$  supplies the clustering/tightness — so G6 is imported into existing RCFT/conformal-net rigor rather than built from scratch. Residual = the net identification + bulk reconstruction. [E] (charges, embedding) + [C] (identification)

**The full covariant field equation, parameter-free: fixed-volume equilibrium gives the Einstein tensor with conservation as an output ([verification/v359\_grav\_nonlinear\_einstein.py])**

The linearised result above (v358) extends to the *full* covariant equation by Jacobson's *fixed-volume* (maximal-vacuum-entanglement) equilibrium — no linearisation assumed. **Einstein tensor, not Ricci [E]**. Stationarity of the entanglement entropy at fixed volume forces the divergence-free combination: the 4D trace identity  $g^{ab} G_{ab} = (1 - \frac{d}{2})R = -R$  (for  $d = 4$ ) selects  $G_{ab} = R_{ab} - \frac{1}{2}R g_{ab}$ , so

$$G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}, \quad c_3^{-1} = 8\pi$$

with *both* coefficients TFPT-fixed: the Newton coupling  $8\pi = 1/c_3$  (v358) and the cosmological constant  $\Lambda$  from  $\alpha$ ,  $\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$  (v60). **Conservation is an output [E]**. By Lovelock's theorem the only divergence-free, second-order, local symmetric tensor built from the metric is  $a G_{ab} + b g_{ab}$ , so  $\nabla^a G_{ab} = 0$  forces  $\nabla^a T_{ab} = 0$  — matter conservation is *derived*, not assumed. **So Gate B6 — the classical metric-sector field equation — is closed parameter-free at the local level. Residual**: the Jacobson equation-of-state *status* (a thermodynamic derivation, not a from-action quantisation) and the single unit  $v_{\text{geo}}$  stay [O], while the global ambient measure (Gate 2/C7, the constructive boundary projective limit) is now a [C] redundancy (v369) — no free dimensionless gravity dial remains. An *external candidate* for the missing action level is quantified in the next keybox (v473); it does not change this typing. [E]/[C]

**The entropic-action bridge: an external candidate for the missing action level, quantified — nothing closes ([verification/v473\_entropic\_action\_bridge.py])**

The one honest **[O]** left in the classical sector — “equation of state, not a from-action quantisation” (v359) — now has an explicit *external* candidate: Bianconi’s entropic action  $S_B = -\text{Tr}_F \ln(\tilde{G}\tilde{g}^{-1})$ , the quantum relative entropy between the spacetime metric and the matter-induced metric (G. Bianconi, *Gravity from entropy*, Phys. Rev. D **111**, 066001 (2025); arXiv:2408.14391; her Appendix C identifies  $\Delta_{\tilde{G},\tilde{g}}^{1/2} = \tilde{G}\tilde{g}^{-1}$  with an Araki-type modular operator) — a local covariant action whose variation yields the Einstein equations plus an emergent nonnegative  $\Lambda$  at low coupling: exactly the missing variational layer, *if* the bridge identities hold. What is machine-checked today (GRAV.ENTROPIC.ACTION.01): **carrier Hodge count [E]**. Her  $\Omega^{0,1,2}$  matter fiber, placed on the five-slot carrier  $\mathbb{C}^5$  (not on 4d spacetime, where the fiber is  $1+4+6=11$ ), counts  $1+5+10=16=2^{g_{\text{car}}-1} = \dim S_{D_5}^+$  and Hodge-folds ( $\star: \Lambda^1 \cong \Lambda^4$ ) onto  $\Lambda^{\text{even}}\mathbb{C}^5$  — the carrier half-spinor’s own Pascal row (v2/v44); a vector-space identity *only*. **Coefficient match [E] (the one quantitative pin)**. Her weak-coupling Lagrangian  $\mathcal{L} = 3\beta R - \alpha\mathcal{L}_m + \dots$  (the 3 = the three form blocks) matched to the TFPT EH normalisation  $1/(16\pi G) = c_3/(2G)$  fixes her free constant *exactly*,

$$\beta'_B = \frac{c_3}{6} = \frac{1}{48\pi}$$

(the  $6 = 3 \times 2 = p_2 = |R^+(A_3)|$  is recorded as a structural echo, not a proof). **Entropy potential = quadratic  $\Lambda$  [E]**. Her  $\Lambda_G = \frac{1}{2\beta} \text{Tr}_F(\mathcal{G} - I - \ln \mathcal{G})$  is nonnegative and quadratic at the fixed point; with the TFPT displacement  $\varepsilon_\star = e^{-\alpha^{-1}}Q$  it reproduces the closed  $\Lambda$  branch  $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$  (v60), and the prefactor match yields the *exact* normalisation target  $\text{Tr}_F Q^2 = 32c_3^4 = 1/(128\pi^4) = \frac{2}{3}\delta_{\text{top}}$  — a closed algebraic target for the two-form (pair) block ( $\binom{5}{2}=10$ ), **[C]** until an explicit  $Q$  is constructed.  **$R^2$  kill test [E] (pre-registered)**. The raw log expansion carries a curvature-squared coefficient of order  $(c_3/6)^2$ ; the TFPT Starobinsky sector needs  $\bar{M}_{\text{Pl}}^2/(12M_{\text{scal}}^2) = 1/(12c_3^7)$  (v36) — the exact mismatch is  $3c_3^{-9} = 3(8\pi)^9 \approx 1.2 \times 10^{13}$ : a direct identification of the raw entropic action with the TFPT  $R^2$  sector is *false* without a mechanism (the scalaron as a light trace mode of her  $\mathcal{G}$ -field generating  $c_3^{-9}$ , or a KMS-spectral renormalisation of the log action). **The bridge proper [C] (recorded, not proven)**. The compression conjecture  $P_\Sigma(\tilde{G}\tilde{g}^{-1})P_\Sigma = \Delta_\Sigma^{1/2}$  — the physical Calderón compression of her relative metric operator equals the TFPT seam modular operator; if proven,  $S_B$  is a local volume representation of the TFPT relative spectral action  $S_{\text{rel},\chi}$  (her log action is the all-scale-integrated heat kernel, by Frullani  $-\ln A = \int_0^\infty \frac{dt}{t}(e^{-tA} - e^{-t})$ , verified). Open with it: the operator-level Hodge items (Clifford action, Dirac intertwining, leaf involution, Calderón polarisation), and Lorentzian positivity of  $\tilde{G}\tilde{g}^{-1}$  is *not* automatic (timelike gradients) — TFPT’s OS/reflection-positive route is the sturdier construction. **Residual [O] (unchanged)**. Nothing closes: the v359 equation-of-state typing stays **[O]**; adopting  $S_B$  would close it *only* after the compression conjecture *and* an  $R^2$  mechanism; zero TFPT knobs are added.

**The operator level, executed ([verification/v474\_entropic\_hodge\_carrier.py])**. Work packages 1 and 4-algebra of the bridge are now machine-checked on the finite carrier Fock space  $\Lambda^\bullet\mathbb{C}^5$  ( $\dim 2^{g_{\text{car}}}=32$ ). **Spinor structure exhibited [E]**: the ten CAR gammas satisfy  $\{\Gamma_a, \Gamma_b\}=2\delta_{ab}$  exactly, the 45 bilinears span  $\mathfrak{so}(10)$  and leak nothing from the 16-dim even subspace —  $S^+$  is the even exterior algebra with its spinor structure, and Bianconi’s Hodge–Dirac operator is Clifford multiplication at symbol level ( $c(\xi)^2 = -|\xi|^2$ , symbolic). **The fold is the  $5 \rightarrow \bar{5}$  conjugation [E]**: the traceless  $\mathfrak{u}(5)$  weights of  $\Lambda^4$  are the negatives of those of  $\Lambda^1$ , so her fiber  $1+5+10$  becomes the GUT half-spinor  $16=1+\bar{5}+10$  — the SM carrier decomposition, not a dimension match; honest obstruction:  $\Omega^{\leq 2}$  is *not* Clifford-invariant, so the identification must go through the fold, never through degree truncation. **The  $Q$ -target, decided [E]**: the uniform-block ansatz  $\text{Tr} Q^2 = dk^2 c_3^4 = 32c_3^4$  with integer  $k$  admits *exactly* the supports  $(d,k)=(2,4), (8,2), (32,1) = \{|\mathbb{Z}_2|, \text{rank } E_8, 2^{g_{\text{car}}}\}$ , with the minimal uniform solution  $q=c_3^2$  per Fock mode and the identity  $\text{Tr} Q^2/\delta_{\text{top}}=32/48=2/3$ ; the two-form block 10 (the naive pair-sector reading) and the fiber 16 require irrational multipliers and are *excluded*. The physical choice among the three supports stays **[C]**; AP2 and the  $R^2$  mechanism stay **[O]**.

**The  $R^2$  kill test, executed ([verification/v475\_entropic\_scalaron\_gate.py])**. On a maximally symmetric background the relative curvature operator has eigenvalues *exactly*  $\{R; R/4 \times 4; R/6 \times 6\}$  (her Appendix-B flattening verified, incl. the  $6 \times 6$  two-form block), so the exact vacuum action is  $\mathcal{L}_B(R) = -[\ln(1-\beta R) + 4\ln(1-\beta R/4) + 6\ln(1-\beta R/6)] = 3\beta R + \frac{17}{24}\beta^2 R^2 + O(R^3)$  — the 3 re-derives her Eq. (45) and  $\frac{17}{24}$  is the exact tensorial factor behind the pre-registered gap. With the pinned  $\beta'=c_3/6$



both mass routes (Starobinsky matching and  $f'/3f''$ ) give the raw entropic scalaron

$$m_{\text{raw}}^2 = \frac{4608\pi^2}{17} \bar{M}_{\text{Pl}}^2 \Rightarrow m_{\text{raw}} \approx 51.7 \bar{M}_{\text{Pl}}$$

— *trans-Planckian*, not light: the naive reading “the TFPT scalaron is the light trace mode of her  $\mathcal{G}$ -field” is *dead* [E]; a viable mechanism must supply exactly  $m_{\text{raw}}^2/m_{\text{TFPT}}^2 = \frac{72}{17}(8\pi)^9 \approx 1.69 \times 10^{13}$  in mass<sup>2</sup>. The domain is not the obstruction (first log pole at  $R=384\pi^2 \bar{M}_{\text{Pl}}^2$ , trans-Planckian), and the Lorentzian caveat is now an explicit witness: a timelike gradient gives the  $Gg^{-1}$  eigenvalue  $1 - \alpha v^2 \leq 0$  for  $v^2 \geq 1/\alpha$  (ln undefined), while spacelike/Euclidean stay positive — the OS route is the sturdier construction. **Net verdict** [C]: of the three bridge interpretations, the entropic action covers the Einstein +  $\Lambda$  level (survives); the light-mode reading is dead; KMS-spectral renormalisation is the *only* surviving route to the  $R^2$  sector. Nothing closes.

**AP2, made well-posed** ([verification/v476\_entropic\_compression\_toy.py]). The compression conjecture itself is sharpened on an exactly solvable gapped chain. **The reading is decided:** on a *pure* bulk (the seam situation: pure bulk, thermal boundary) the covariance is a projection, so  $f(x)=x/(1-x)$  is singular on its spectrum — the literal operator-side reading  $P_\Sigma f(C)P_\Sigma$  is *ill-posed*, while the state-side  $f(P_\Sigma C P_\Sigma)$  (Peschel) is finite: the identity can only mean *build  $\Delta_\Sigma$  from the compressed relative metric*, which is exactly Bianconi’s own local construction — AP2 retyped, not weakened. **The mismatch is exactly second order** [E]: symbolically,  $[C^n]_{AA} - (C_{AA})^n$  carries the cross-cut block twice (first order vanishes identically), so for any analytic  $f$  the two readings agree up to  $O(\|C_{AB}\|^2)$  — the modular (entanglement) content *is* that quadratic correction. **Gap control:** at the bounded (covariance) level the mismatch obeys  $d \leq x^2$  and falls with the gap, the decoupled cut agrees exactly — the two readings converge in the gap-dominated regime, where TFPT operates ( $\Delta_{\text{eff}} > 0$ , v76/v337). **Residual** [O] (**unchanged**): the continuum seam statement itself. No marker moves.

**The surviving  $R^2$  route, typed as one moment** ([verification/v477\_entropic\_scaleflow.py]). The Frullani identity upgrades to a *scale-flow representation*:  $-\ln a = \int_0^\infty \frac{dt}{t} (e^{-ta} - e^{-t}) = 2 \int_0^\infty \frac{dx}{x} (e^{-a/x^2} - e^{-1/x^2})$  — Bianconi’s log action *is* the flat scale-integral of relative heat-kernel actions, while TFPT’s  $S_{\text{rel},\chi}$  evaluates *one* KMS scale with a weight  $f$ : the bridge between the two action principles is a choice of *scale measure*. In the moment dictionary (weight  $w(t) dt/t$ ,  $\mu_k$  its  $e^{-t}$ -moments) the coefficients read  $3\beta R \mu_1 + \frac{17}{24} \beta^2 R^2 \mu_2$ ; the flat weight has  $\mu_1 = \mu_2 = 1$  (= the raw v475 coefficients), and demanding  $m^2 = c_3^7 \bar{M}_{\text{Pl}}^2$  forces *exactly*

$$\mu_2/\mu_1^2 = \frac{72}{17}(8\pi)^9 \quad \text{with} \quad \frac{4608\pi^2/17}{(72/17)(8\pi)^9} = c_3^7 \text{ (identically)}$$

— the v475 kill-test number *is* a moment ratio, and the “13 orders of magnitude” are not a bridge mismatch but a scale-measure datum: the flat measure fixes it wrongly, the TFPT KMS measure ( $f_0=6(4\pi)^2/c_3^7$ , v36) fixes it correctly — one KMS moment, two parametrisations, zero new dials. [C] (consistency, not derivation: the measure is TFPT input); the  $R^2$  gate and the v359 typing stay [O].

**The two remaining legs, first computable steps** ([verification/v478\_entropic\_continuum\_legs.py]).

*Leg A (AP2 continuum):* on the exactly solvable critical chain (infinite-volume kernel, 50-digit precision) the compressed interval state carries the conformal/BW structure quantitatively — the entanglement entropy follows Calabrese–Cardy with  $c_{\text{est}} = 1.00089/1.00038/1.00021$  (monotone  $\rightarrow$  the Dirac  $c=1$ ; gapped control  $\sim 6 \times 10^{-8}$ ), and its Peschel modular Hamiltonian organises on the CHM weight  $\beta(x)=x(L-x)/L$  (corr 0.87 $\rightarrow$ 0.99, even-range bands vanishing identically, odd tail 1.8%): the state-side  $\Delta_\Sigma$  (the reading *forced* above) flows to the CHM/Bisognano–Wichmann geometric form — *exactly* the object the entanglement-equilibrium derivation consumes (v323/v358); the bridge’s modular side and the gravity side meet in the continuum limit (a lattice exhibit, not a continuum proof — AP2 stays [O]; the natural continuum frame for the seam version is the *four-interval* multilocal modular geometry of the four marks, v480). *Leg B (the measure):* the single-scale corollary is exact —  $d\mu = \delta(t-t_0)dt$  gives  $\mu_2/\mu_1^2 = e^{t_0}$ , so the one-moment condition forces  $t_0 = \ln \frac{72}{17} + 9 \ln(8\pi) = 30.461$ , one dimensionless KMS time in closed form; the near-miss to  $h(E_8)=30$  (difference 0.461) is explicitly *declined* per the no-free-pattern rule (v354/v355) — deriving the measure from the entropic structure stays [O]. [E]/[C]/[O]

**The Simple-Current Extension Theorem — the central  $G_{\text{net}}$  closing statement [E]/[C]**  
 ([verification/v154\_simple\_current\_theorem.py])

The conditional theorem can now be stated as one clean *simple-current extension*. Let  $A = (D_5)_1 \otimes (A_3)_1$  be the carrier net and  $L = \langle(1,1)\rangle \subset \text{Disc}(D_5) \oplus \text{Disc}(A_3) = \mathbb{Z}_4 \oplus \mathbb{Z}_4$  the isotropic  $\mu_4$  glue line ( $q(k(1,1)) = k^2 \in \mathbb{Z}$ ). Then the local extension  $B = A \rtimes L$  has

$$[B : A] = |L| = 4 = |\mu_4|, \quad c(B) = 5+3 = 8, \quad \mu(B) = \frac{\mu(A)}{|L|^2} = \frac{16}{16} = 1$$

so  $B$  is *holomorphic* (KLM  $\mu$ -additivity), and a holomorphic  $c=8$  chiral net is the lattice net of the *unique* even-unimodular rank-8 lattice — hence  $B \cong (E_8)_1$  (the same- $c$  rival  $SO(16)_1$ ,  $\mu=4$ , is excluded). Every algebraic ingredient is machine-checked: the  $\mathbb{C}[\mathbb{Z}_4]$  Longo Q-system ([verification/v125\_glue\_qsystem.py]), the  $\mathbb{Z}_4$ -graded Frobenius  $E_8$  hull ([verification/v143\_graded\_frobenius.py]), the Ramond glue sectors ([verification/v148\_fock\_census.py]). **Consequence:**  $G_{\text{net}}$  is internally *closed* if the seam–Calderón boundary net is *defined* to be  $A$ ; it stays the *one* external identification “the RP seam–Calderón net is  $A$ ” if the boundary net is to be *derived* from the seam kernel. Either way this is the single strict-TOE certification theorem still carrying weight — a clean extension statement, not “find a quantum gravity”.

**And that one identification reduces to a single shared premise** ([verification/v155\_quasifree\_boundary.py]). The identification splits (v110) into (a) “the seam Calderón involution is sheet-odd” — the double-cover deck, the same  $|\mathbb{Z}_2|$  that halves  $c_3 = \frac{1}{2.4\pi}$  (structural) — and (b) “the seam 2-point kernel is quasi-free”. Part (b) is *not independent*: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression  $PTP$  of the bulk covariance, and the compression of a contraction ( $i\Gamma$  with spectrum in  $[-1, 1]$ ) is a contraction — so the boundary/seam state is automatically quasi-free, with Wick inherited (every boundary correlator a Pfaffian of  $\Gamma_d$ ). Hence (b) *follows* from “the seam bulk is a reflection-positive free field” — the *same* premise already load-bearing in the area law ([verification/v59\_area\_law\_evidence.py]) and the EH mechanism ([verification/v150\_replica\_eh\_model.py]/[verification/v151\_bfk\_split.py]; since exercised on the discretized collar with the seam’s own kernel and real replica sheets, v471 — the R3 kernel premise is discharged at the finite level, its residual retyping to the continuum scaling limit shared with SEAM.EQUIV.01, v336, plus the v152 anchor). **So the entire seam-side residual is one physical premise** — the seam bulk is a free RP field — shared by  $G_{\text{net}}$ , the area law and R3, and given it the chain Gaussian bulk  $\Rightarrow$  quasi-free boundary  $\Rightarrow$  one kernel = net (v113)  $\Rightarrow$  carrier  $A \Rightarrow (E_8)_1$  is exact. It is not reducible by a finite computation — it is the one place TFPT meets functional analysis.

**The construction, exhibited and verified** ([verification/v156\_seam\_net\_construction.py]). The “ $B \cong (E_8)_1$ ” step is no longer an assertion — the chiral net is built explicitly and the identification *checked*. The Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian acts on a boundary mode  $e^{ikx}$  by the harmonic extension  $e^{ikx-|k|y}$ :  $-\partial_n u|_0 = |k|u$ , so for a free bulk the seam operator is the massless chiral dispersion  $\omega_k = |k|$  that defines the free-fermion net on  $S^1$ . Sixteen Majoranas give  $c = 8 = 5+3$ ; the weight-1 currents are  $248 = 120_{\text{NS}} + 128_{\text{R}}$  (the  $SO(16)$  bilinears plus the Ramond spinor, Frenkel–Kac); and the holomorphic character is

$$\chi = \frac{E_4}{\eta^8} = q^{-1/3} (1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + \dots) = j^{1/3}$$

(verified by  $q$ -series), the 248 at level 1 being  $\dim E_8$  — so the net *is*  $(E_8)_1$ . *What remains* is exactly (i) the free-bulk premise above and (ii) the operator-algebraic existence / uniqueness of the net generated by the seam Calderón kernel (a Kawahigashi–Longo-type net-construction theorem). Route 3 thus *exhibits and verifies* the target; the irreducible residual is the one premise + the standard net-existence step — where TFPT meets constructive QFT, now sharply located.

**And the free-bulk premise itself reframes: freeness is a rigid fixed point, not a postulate** ([verification/v157\_rigid\_fixed\_point.py]). Three facts click together. (i) The DtN/Calderón symbol is *universally*  $|k|$ : the boundary operator of *any* second-order elliptic bulk is an order-1  $\Psi\text{DO}$  with principal symbol  $|\xi|_g$  (Lee–Uhlmann), so the free chiral dispersion is bulk-detail-independent and any interaction lives in the lower-order (irrelevant) symbol. (ii) A holomorphic  $c=8$  theory is *rigid*: every field has weight  $(h, \bar{h}=0)$ , an exactly-marginal deformation needs  $(1, 1)$ , and with  $\bar{h}=0$  *no*  $(1, 1)$  field

exists — the fixed point is isolated, there is no continuous knob along which a non-Gaussian interaction could turn on. (iii) The *same*  $|\mathbb{Z}_2|$  that halves the one-sided  $S^2$  Gauss–Bonnet ( $c_3 = \frac{1}{|\mathbb{Z}_2| 2\pi \chi(S^2)} = \frac{1}{8\pi}$ ) labels the sheet *and* is the Ramond/GSO projection giving  $\mu=1$  ( $248=120_{\text{NS}}+128_{\text{R}}$ ) — one object, three roles, so the geometry fixing  $c_3$  is the projection making the net holomorphic. **Consequence:** premise (A) shrinks from “prove the seam is Gaussian” to “the gapped ( $\Delta=6 \log \frac{3}{2}>0$ ) boundary flow reaches the holomorphic fixed point” — given which, freeness is *forced* by rigidity + uniqueness ( $(E_8)_1$ , v83), not fine-tuned. A strictly smaller and simpler residual.

**And the destination is a *proven* stable fixed point** ([verification/v158\_fixed\_point\_stable.py]). An exact operator-dimension count settles it: the chiral Majorana has  $h=\frac{1}{2}$ , the bilinear current  $h=1$ , the quartic  $h=2$ . The relevant window  $0 < h < 1$  for bosonic deformations is *empty* (bosonic chiral operators have integer  $h$ , lowest 1) — nothing relevant flows the free theory away; the  $h=1$  currents are chiral (v157), not  $(1,1)$ -marginal; and the lowest genuinely-interacting operator (the quartic,  $h=2>1$ ) is *irrelevant*, as are all higher  $2k$ -fermi operators ( $h=k$ ). So the free chiral  $c=8$  fixed point is *isolated and stable* against all interactions. Premise (A) thereby reduces to a *basin-of-attraction* statement with a proven stable target — the gapped ( $\Delta=6 \log \frac{3}{2}$ ) flow lands here; the convergence is conformal-perturbation analysis, the destination is settled.

**And the closure can be stated as one fixed-point theorem — which relocates (A) to a single internal premise** ([verification/v160\_seam\_gaussianity\_from\_pf.py]). Phrase the argument internally: (i) a quasi-free state has *all* higher connected cumulants  $\kappa_{2n}=0$  (the signed set-partition/Pfaffian inversion of  $\langle \psi_1 \cdots \psi_{2n} \rangle = \text{Pf}(\Gamma)$  leaves only the degree-2 kernel — the v113 “one kernel = the net” property at the cumulant level); (ii) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (iii) a gapped primitive Perron–Frobenius transport has a *unique* fixed ray, so a non-Gaussian piece is impossible — if  $\kappa_{2n}$  were fixed it would be a *second* fixed ray (killing uniqueness), and if not fixed it sits in the gapped complement and decays to 0 (any gap  $> 0$  suffices; the value  $(2/3)^6$  only sets the rate). *The honest residual:* v56’s gapped PF uniqueness is established on the *three-dimensional* scalar transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  (the flavour gap / horizon Page-recovery rates), whereas the 16-Majorana seam state already carries  $\binom{16}{4}=1820$  and  $\binom{16}{6}=8008$  higher-cumulant directions — the Schwinger cone the transport must contract has dimension  $\gg 3$ , and uniqueness must hold there (not merely on the quasi-free sub-cone) for the argument to be non-circular. **So (A) is equivalent to one strictly smaller, internal, exact-testable statement:** the admissible TFPT seam transport is a *primitive, spectrally gapped, RP-cone-preserving* map on the *full* Schwinger cone, not only on the scalar readout spectrum of v56 — which the suite does *not* yet establish [E]/[O].

**And that full-cone premise is now *discharged* to a single finite one-particle statement** ([verification/v161\_one\_particle\_reduction.py]). The lever is Bogoliubov second quantisation: a quasi-free transport is  $T_\Sigma = \Gamma(t)$ , the second quantisation of a one-particle contraction  $t$  on the 16-dimensional Majorana space  $H_\Sigma$ , and the Fock space grades by correlator degree, so  $\Gamma(t)$  acts on the degree- $m$  sector as the exterior power  $\Lambda^m(t)$ . Three exact consequences follow (all machine-checked): (i)  $\text{spec}(\Gamma(t)|_{\deg m})$  is the set of products of  $m$  one-particle eigenvalues — the whole correlation spectrum is generated by the finite one-particle spectrum; (ii) **the cone gap equals the one-particle gap:** the sub-leading eigenvalue of  $\Gamma(t)$  over *every* degree is  $(2/3)^6$  (pad the largest contracted mode with fixed factors), so premise 5 on the cumulant space *is* premise 5 on the 16-dim space, the same number; (iii) the eigenvalue-1 sector is  $\Lambda^*$  of the fixed subspace — the disconnected Wick powers of  $K_\Sigma$  — so with a strict one-particle gap no independent connected cumulant is fixed, and an independent fixed cumulant *on* the fixed subspace is excluded by v113 (one polarisation  $\Leftrightarrow$  one vacuum). Premise 1 (quasi-freeness) is itself *forced*, not assumed: the  $120 = \dim \mathfrak{so}(16)$  weight-1 chiral currents *are* the Bogoliubov generators on the 16 Majoranas, and v158 makes every non-Gaussian deformation irrelevant. **So the three full-cone premises collapse to one finite, 16-dimensional statement:** the seam RG transport’s one-particle symbol is the gapped Calderón Bogoliubov map — fixed eigenspace = the rank-8  $K_\Sigma$  polarisation (v113), sub-leading gap =  $(2/3)^6$  (v56). Every *number* in it is already supplied by the suite; the one irreducible step is the physical identification that the actual seam RG generator equals this Bogoliubov map — the same operator-algebraic (Kawahigashi–Longo-class) statement as the net-existence residual of (A<sub>2</sub>), now with the infinite-dimensional cone eliminated [E]/[O].

**And that finite statement carries no *new* open content: it factors into the two already-open items** ([verification/v162\_seam\_transport\_identification.py]). Take the one-particle symbol apart into its two data, the gap value ( $\beta$ ) and the fixed space ( $\alpha$ ). ( $\beta$ ) *is forced:* a  $\mu_4$ -equivariant operator is diagonal in the cusp basis — the deck average is the diagonal projection,  $\sum_{k=0}^3 U^k X U^{-k} = |\mu_4| \text{diag}(X)$  for  $U = \text{diag}(1, i, -i)$ , and the commutant of  $U$  is exactly the diagonal algebra — so its spectrum is the

cusped weights  $\text{spec}(A_0^*) = \{0, \frac{1}{3}, \frac{2}{3}\}$  (the parabolic residue, **v115**), and the transfer eigenvalue  $(1-\alpha)^{p_2}$  with  $p_2=6=|R_+(A_3)|$  gives  $\{1, (2/3)^6, (1/3)^6\}$ , gap  $6 \log \frac{3}{2}$  — every entry carrier data, no free knob; the lift to the 16 Majoranas is the established  $A_3$  subnet grading (**v137**). ( $\alpha$ ) is the rank-8 Calderón polarisation of the  $\omega_k=|k|$  free dispersion (**v113/v156**), the IR conformal free-fermion vacuum. **What is left** is only that the seam RG generator *is*  $\mu_4$ -equivariant and admissible — which is the R1 “seam clock” identification (the transfer operator as the seam’s gapped generator) — together with the  $A_2$  net-existence theorem. Both are pre-existing open items, so premise (A) introduces *no independent open content*: it factors exactly into  $A_2 + R_1$  with every number forced. The “free-bulk cloud” is gone; what remains is the same two identifications the programme already names — (A) is closed *conditionally* on them (a unification, honestly not an unconditional proof) **[E]/[O]**.

**And both residual identifications are pinned to their floor, so (A) has no independent open gate left** (**[verification/v165\_premise\_a\_closure.py]**).  $A_2$  (net existence) is not open research but *assembly* of established mathematics: the 16 free Majoranas define a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic  $c=8$  lattice net *is*  $(E_8)_1$  (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the  $\mu_4$  extension is a Longo Q-system of index 4 (**v125**); the one TFPT-specific input — that the seam Calderón kernel is the free-fermion two-point function — is done because the DtN symbol is universally  $|k|$  (**v156/v157**). So  $A_2$  re-types **[O]**  $\rightarrow$  **[C]** (closed by assembly).  $R_1$  (the seam clock) reduces to GATE.QGEO: a  $\mu_4$ -equivariant flat gapped generator on the carrier is *unique* ( $= A_0^*$ , **v115/v116**), so its gap  $(2/3)^6$  is forced once the seam generator lies in that class, and the residual is exactly one geometric identification — the seam collar boundary is the  $\mu_4$ -punctured sphere  $\mathbb{P}^1 \setminus \mu_4$  — whose cohomological half is already established (**v137**,  $b_1=3=N_{\text{fam}}$ ). That is GATE.QGEO, an already-open realisation premise, *not* a new gate. By the No-Unit Theorem (**v153**) the irreducibles stay exactly  $\{\pi, v_{\text{geo}}\}$  — closing (A) adds none. **Final accounting**: premise (A) = ( $A_2$  assembly, **[C]**) + ( $R_1$ =GATE.QGEO, already open) + the  $\{\pi, v_{\text{geo}}\}$  irreducible core (a theorem) — it carries *zero* independent open gates and ceases to be its own gate **[E]/[O]**.

**And the two structural residuals — net existence and full-cone reflection positivity — are now discharged to a theorem, leaving the seam realisation as the one irreducible open premise** (**[verification/v175\_net\_existence\_full\_cone.py]**). The CAR second-quantisation functor  $\Gamma(t) = \bigoplus_m \Lambda^m(t)$  lifts the one-particle seam contraction  $t$  to the *complete* Fock space  $\Lambda^\bullet(\mathbb{R}^{16})$  ( $\dim 2^{16}=65536$ ); by the exterior-power spectrum theorem every eigenvalue of  $\Gamma(t)$  is a subset product of  $\text{spec}(t)$ , so checking all  $2^{16}$  products shows  $\Gamma(t)$  is a *contraction* (full-cone RP), with eigenvalue-1 multiplicity  $2^8=256$  (the wedges of the rank-8  $K_\Sigma$  polarisation) and sub-leading eigenvalue exactly the one-particle gap  $(2/3)^6$  — so full-cone RP/gap reduce *for every*  $m$  (not only  $m \leq 6$ , **v161**) to the one-particle contraction (Shale–Stinespring), *discharging the v160 scope premise by a theorem*. And  $A_2$  is now an *assembled and verified* certificate: the  $(E_8)_1$  character  $E_4/\eta^8=1+248q+4124q^2+34752q^3+213126q^4+1057504q^5$  (level-1=248), the embedding  $248=120+128$ , and the  $E_8$  Cartan matrix even with  $\det 1$  (the unique rank-8 even unimodular lattice, **v83**) — the net *exists* by cited theorems. **Both residuals then collapse to the same input**: that the actual seam Calderón kernel is this gapped one-particle contraction on  $\mathbb{P}^1 \setminus \mu_4$ , i.e. QGEO.REALIZE.01, whose finite half is proven (**v168**). That realisation half is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly **[O]** — the single irreducible structural premise; net existence and full-cone RP are **[E]**.

**The No-Unit Theorem —  $v_{\text{geo}}$  is an irreducible metrology primitive** **[E]/[O]**  
(**[verification/v153\_no\_unit\_theorem.py]**)

$v_{\text{geo}}$  is not an open gap. Every load-bearing compiler datum —  $g_{\text{car}}, |\mu_4|, N_{\text{fam}}, \text{rank } E_8, \alpha^{-1}$ , the determinant ladder  $(3, 4, 8, 20)$ ,  $c_3 = \frac{1}{8\pi}$ ,  $\varphi_0^{\text{ret}}$  — is mass-dimension 0 and *invariant* under a unit rescaling  $L \mapsto \lambda L$ , whereas a mass / energy /  $1/G$  carries nonzero dimension. A map from dimension-0 inputs to a dimension- $d \neq 0$  output is simultaneously invariant and covariant: impossible unless the unit is introduced. **So a dimensionless boundary compiler provably cannot select an absolute scale** —  $v_{\text{geo}}$  is a *forced input*. The three apparent residuals then collapse onto the one unit, differing only in mass-dimension:

$$U_{\text{point}} \sim v_{\text{geo}}, \quad \frac{1}{G} \sim v_{\text{geo}}^2, \quad m/\mu = e^{3/4} \text{ (a pure ratio),}$$

so the irreducibles stay exactly {one dimensionful unit,  $\pi$ }. *Re-typing*:  $v_{\text{geo}}$  is an *irreducible metrology primitive* (a declared unit, like choosing a metre), not a missing derivation.

**Sharpened across gravity, and the final tally** (**[verification/v364\_vgeo\_sharpen.py]**). After the parameter-free gravity (**v358/v359/v361**) the *gravity* sector adds no new scale either: the Einstein



coefficient  $1/c_3 = 8\pi$  and the  $\Lambda$  prefactor  $(8\pi)^2 \cdot 48c_3^4 = 3/(4\pi^2)$  are dimensionless, so  $1/G \sim v_{\text{geo}}^2$  and  $\Lambda \sim (\text{dimensionless}) v_{\text{geo}}^4$  both reduce to  $v_{\text{geo}}$  times atoms. Hence  $v_{\text{geo}}$  is the *single* dimensionful input of the *complete* theory (matter and gravity), and the final tally is  $\{v_{\text{geo}}, \pi\}$  with *zero* dimensionless dials — a  $\sim 26 \rightarrow 1$  reduction against the Standard Model. ( $1/G$  stays UV-sensitive (Sakharov, v68), so it is a metrology readout, not a clean prediction — consistent with  $v_{\text{geo}}$  being the anchor.)

#### Theorem G — the closure statement [O] (target)

The relative spectral boundary kernel over the diffeomorphism-quotiented metric sector admits a regulator-independent, reflection-positive projective-limit measure  $\mu_{\text{QG}}$ . Its Ward identities imply covariant conservation, and its low-curvature expansion is the TFPT  $R + R^2$  seam action

$$f_R R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} + (g_{\mu\nu} \square - \nabla_\mu \nabla_\nu) f_R = \bar{M}_{\text{Pl}}^{-2} T_{\mu\nu} + O(R^3/M^4),$$

so the closed covariant equation becomes the controlled IR shadow of the full measure. *Status: the local covariant field equation is parameter-free (v358/v359); Theorem G remains open only for the projective-limit ambient measure  $\mu_{\text{QG}}$  (G6, the deepest item). G5 certified; the regime equation closed [verification/v28\_gravity\_fR.py].*

## Research Contract — CELEST.SEAM.01 (the celestial-holographic route)

**Hypothesis (as corrected by WP1).** *The seam is the glue-equivariant  $\mathbb{Z}_4$ -orbifold sector of the celestial chiral algebra on the  $A_3$  ALE space  $\mathbb{C}^2/\mathbb{Z}_4$ ; the  $\mu_4$  clock is the Kähler  $U(2)$  phase  $\text{diag}(i, 1)$  whose square is the deck group (spin bridge  $\mathbb{Z}_8$ ,  $8 = 2|\mu_4|$  — the  $c_3 = 1/(8\pi)$  seam winding integer); and the associated twistorial bulk is the self-dual sector of TFPT gravity.* This is the fourth research contract, alongside ( $U_{\text{wall}}$ ), ( $G_{\text{metric}}$ ) and CONTRACT.F.01: a numbered chain of work packages with pre-registered kill tests, *not* a claim. The method is the celestial-chiral-algebra (CCA) construction of Bittleston–Homans–Sharma on Eguchi–Hanson space (arXiv:2305.09451, JHEP 09 (2023) 008 — the  $\mathbb{Z}_2$  case, refined here to  $\mathbb{Z}_4$  with a flat connection at infinity in the Kronheimer–Nakajima / heterotic-on-ALE sense), inside the celestial-holography programme (Strominger, PRL **127**, 221601:  $w_{1+\infty}$ ; Costello–Paquette–Sharma, PRL **130**, 061602: Burns holography, the  $SO(8)$  WZW<sub>4</sub> model; Bittleston–Sharma–Skinner, CMP **403** (2023): quantizing the non-linear graviton). The gauge-algebra admissibility gate is already passed in the literature: Costello’s one-loop axionic-consistency analysis (arXiv:2111.08879) admits the exceptional algebras *including*  $E_8$  for SDYM on twistor space (Okubo: no independent quartic Casimir).

**WP1, executed and verified: the  $E_8$   $\mu_4$ -glue is the flat  $\mathbb{Z}_4$  monodromy of the equivariant CCA on the  $A_3$  ALE space — verdict B ([verification/v492\_celestial\_z4\_orbifold.py], CELEST.WP1.01)**

Everything in this box is exact arithmetic (sympy, no floats). **Inner grading / flat monodromy [E]:** the v128 glue grading ( $240 = 52+64+60+64$ , additive on all 6720 root pairs, dims with Cartan  $(60, 64, 60, 64)$ ,  $\mathfrak{g}_0 = D_5 \oplus A_3$  carrier = 60) is *inner*:  $h = (2, 2, 2, 2; 0^4)$  reads the glue class mod 4 on all 240 roots, so the glue is  $\text{Ad}(\exp(2\pi i h/4))$  — an order-4 torus element, i.e. a *flat  $\mathbb{Z}_4$  monodromy* in the Kronheimer–Nakajima sense; the  $A_3$ -side detector  $h' = (0^5; 1, 1, 1, -3)$  reads the *same* class, so the glue charge is the diagonal  $(1, 1)$  of the discriminant group  $\mathbb{Z}_4 \times \mathbb{Z}_4$  — the v92/v125 Lagrangian glue. **The spatial side [E]:**  $\mathbb{C}^2/\mathbb{Z}_4$  under the deck  $(z_1, z_2) \mapsto (iz_1, i^{-1}z_2)$  is the  $A_3$  singularity  $XY = Z^4$  (invariant ring exact; Hilbert series  $(1+t^4)/((1-t^2)(1-t^4))$ ; Molien table verified). **The equivariant sector closes [E]:** the glue-equivariant SDYM( $E_8$ ) modes ( $j + m \equiv 0 \pmod{4}$ ) form a Lie subalgebra with graded dimensions  $D[d] = 60(d+1)$  (even  $d$ ) /  $64(d+1)$  (odd  $d$ ) — uniform per parity *only* because  $\dim \mathfrak{g}_0 = \dim \mathfrak{g}_2 = 60$  and  $\dim \mathfrak{g}_1 = \dim \mathfrak{g}_3 = 64$ : the v128 dimensions are exactly the consistency condition. The zero modes ( $d=0$ ) are  $\mathfrak{g}_0$  alone — *the unbroken algebra on the orbifold is the carrier  $D_5 \oplus A_3 + \text{Cartan} = 60$* ; the asymptotic density is  $62/248 = 1/4 = 1/|\mathbb{Z}_4|$  (the orbifold index), and the  $\mathbb{Z}_2$  halfway sector reproduces the Eguchi–Hanson towers  $120/128 = SO(16)$  adjoint/spinor —  $\mu_4$  refines the sheet  $\mathbb{Z}_2$  exactly as  $4 = 2 \times 2$  (v125). **Character level [E]:**  $\Theta_{E_8}$  splits into the four glue-coset thetas

(shell  $1 = (52, 64, 60, 64)$  re-derived from the *lattice*); the four sector characters  $\Theta_{C_j}/\eta^8$  sum *exactly* to  $\chi_{(E_8)_1} = 1 + 248q + 4124q^2 + 34752q^3$  (the **v377** series); the sector conformal weights are  $h_{D_5} = (0, \frac{5}{8}, \frac{1}{2}, \frac{5}{8})$ ,  $h_{A_3} = (0, \frac{3}{8}, \frac{1}{2}, \frac{3}{8})$  — the **v92/v125** discriminant form  $(5x^2 + 3y^2)/8 \bmod 1$  — and the glue diagonal has *integer*  $h = (0, 1, 1, 1)$ : the **v125** isotropy *is* the locality of the extension to  $(E_8)_1$ . **The critical correction [E] (why verdict B, not A):** the  $A_3$  deck  $\text{diag}(i, i^{-1})$  induces on the celestial sphere  $z = z_1/z_2$  only  $z \mapsto -z$  (projective order 2, the sheet flip) — it is *not* the order-4 clock  $z \mapsto iz$ . The exact bridge: any  $SU(2)$  lift of the clock has order 8 (spin group  $\mathbb{Z}_8$ , which would quotient to  $A_7$ ), and  $(\text{spin clock})^2 =$  the deck generator *exactly*, with  $|\mathbb{Z}_8| = 8 = 2|\mu_4|$  — the  $c_3 = 1/(8\pi)$  winding integer. The order-4 clock *is* realised, as the  $U(2)$  phase  $\text{diag}(i, 1)$ : projective order 4, fixed points  $\{0, \infty\}$ , free  $\mu_4$  orbit (all **v180** clock properties), normalising the deck (so preserving the glue flat connection), with  $\det = i$  rotating the holomorphic symplectic form by the  $\mu_4$  generator (a Kähler, *not* triholomorphic isometry — it rotates the twistor sphere by  $\pi/2$ ). **Gravity side [E]:** clock-invariance selects from the 3-parameter versal  $A_3$  family  $XY = Z^4 + a_2Z^2 + a_1Z + a_0$  *exactly* the 1-parameter family  $XY = Z^4 + a_0$ , whose four branch points are *one*  $\mu_4$  orbit with cross-ratio 2 — the seam marks (**v179** clock orbit; **v168/v214** cross-ratio  $2 \Rightarrow j=1728$  pillowcase). **Negative controls [E]:** the false spatial action  $\text{diag}(i, i)$  is killed three ways ( $\det = -1 \notin SL(2, \mathbb{C})$ , no CY/ALE resolution; Poisson closure fails in 14/14 nonzero cases; conjugation symmetry broken), the false glue (60/64 blocks swapped) breaks additivity on 3112/6720 root pairs, and of all 24 relabelings exactly  $\{\text{id}, j \mapsto -j\} = \text{Aut}(\mathbb{Z}_4)$  are additive — the grading is rigid. **Verdict [C]:** the celestial and glue  $\mathbb{Z}_4$  structures match under *one* precisely named extra identification —  $\text{clock}^2 = \text{deck}$  on the sphere (the sheet- $\mathbb{Z}_2$  spin bookkeeping) — so WP1 lands at verdict B (match modulo one named identification), close to but honestly short of A. **[E]/[C]**

**Typing and non-circularity.** Admissible *inputs* are the finite seam/glue data already in the suite — the  $\mu_4$  Q-system glue (**v92/v125**), the graded hull (**v128**), the seam rigidity and pillowcase geometry (**v168/v214**), the Möbius clock (**v180**) — plus the WP1 **[E]** arithmetic (`[verification/v492_celestial_z4_orbifold.py]`). *Not* admissible as an input: the *continuum existence* of the holomorphic  $(E_8)_1$  net on the seam — that is precisely the SEAM.EQUIV.01 target, and assuming it would make the contract circular. *Scope fence:* the twistorial bulk of this contract is the *self-dual* sector only, never full Einstein gravity — the classical field-equation route (entanglement equilibrium, **v358/v359**) and the entropic-action bridge (GRAV.ENTROPIC.ACTION.01, **v473**) are separate, non-competing statements about the full equation. SEAM.EQUIV.01 stays **[O]** throughout; nothing in this contract moves it.

### Work packages.

- **WP1 (done, verdict B) [E]/[C]:** the structural  $\mathbb{Z}_4$  arithmetic of the keybox above, executed and machine-verified as CELEST.WP1.01 `[verification/v492_celestial_z4_orbifold.py]`.
- **WP2 (done, verdict B) [E]/[C]:** the clock-invariant deformation theory, executed and machine-verified as CELEST.WP2.01 `[verification/v493_celestial_wp2_clock_deformation.py]` (sympy exact, 47 checks). Both targets landed, *sharpened*: (i)  $a_0$  is the pure *scale* of the seam configuration — the binary-quartic invariants of  $w^2 = Z^4 + a_0$  are  $I = 12a_0$ ,  $J = 0$  *identically*, so  $j = 1728$  and the  $\tau=i$  pillowcase shape (**v168/v214**) is *frozen* for every  $a_0 \neq 0$  (no shape modulus survives clock-invariance; the negative controls  $a_1Z \Rightarrow j=0$  and  $a_2 \neq 0 \Rightarrow j=1556068/81$  show the test has teeth); (ii) the three resolved spheres ( $3 = \text{rank } A_3 = N_{\text{fam}}$ ) carry *exactly* the three nontrivial  $\mu_4$  characters  $\{i, -1, -i\}$  under the clock, which acts on  $H_2$  as a *Coxeter element* of  $W(A_3)$  (char  $x^3 + x^2 + x + 1$ , order  $h(A_3)=4=|\mu_4|=N_{\text{fam}}+1$ ) and *is* the Picard–Lefschetz monodromy of the family ( $a_0 \rightarrow e^{2\pi i} a_0$  gives the 4-cycle); the surviving deformation direction is the weight-1  $\chi_1$ -Fourier diagonal  $(1, i, i^2)$  — *not* the naive invariant diagonal ( $\det(A-1) = -4$ : no invariant cycle) — and all three sphere volumes move in lockstep ( $|\Pi_j| = \sqrt{2}t$ ). The BHS deformed-algebra pattern transfers from  $\mathbb{Z}_2$  to  $\mathbb{Z}_4$ : the fibre bracket  $-4\text{Nambu}(XY - Z^4 - a_0)$ , anchored at the  $a_0=0$  orbifold, closes with corrections *exactly linear* in  $a_0$  at the  $\mathbb{Z}_4$  wrap, conserving the  $\mu_4$  grade (no sector leak — the WP1 equivariant sector deforms consistently), with  $a_0$  in the weight-0 slot of BHS's  $c^2$ . Verdict B via three named **[C]** identifications ( $-4\text{Nambu}$  as *the*  $k=4$  CCA bracket; period = root difference; the seam-scale reading via  $\text{clock}^2 = \text{deck}$ ); the full deformed

twisted-sector OPEs and the  $W(\mu)$ -type scaling limit for  $k=4$  stay [O]. The v216 [O]-residual is *typed*, not moved: given the order-4 clock, the deformation freezes the  $(2, 2, 2, 2)$  modulus to the square automatically — a relocation of the same order-4 carrier input, not a new derivation.

- **WP3 (alignment test only) [O]:** the Green–Schwarz axion coefficient of the celestial  $E_8$  theory,  $\lambda_{\epsilon_8}/(8\pi\sqrt{3})$  (Costello’s normalisation), against the TFPT seam constant  $c_3 = 1/(8\pi)$  — an *alignment* check between two fixed normalisations, never a derivation of  $c_3$  from celestial data.
- **WP4 (the type mismatch, head-on) [O]:** the  $(E_8)_1$  *character* as a conformal block of the celestial  $E_8[\mathbb{C}^2]$  S-algebra — WP1 shows the four glue sectors sum to the  $(E_8)_1$  character at the counting level, but the S-algebra is graded by polynomial degree (60/64 towers), not by conformal weight; WP4 must either produce the character from the S-algebra trace or localise the obstruction.
- **WP5 (the summit) [O]:** the twistorial construction as a *second continuum route* for SEAM.EQUIV.01 — a Costello–Li-grade construction of the holomorphic boundary theory from the twistor uplift; hard, and explicitly typed as a research programme.

**Kill tests (pre-registered).** (**K1**) the equivariant sector could have *contradicted* the  $52+64+60+64$  split (uniformity of  $D[d]$  needs exactly  $\dim \mathfrak{g}_j = (60, 64, 60, 64)$ ) — *survived by WP1 and WP2*: the false-glue control shows the test has teeth (3112/6720 violations), and WP2 adds the deformation side (a shape modulus surviving clock-invariance, or  $a_0$ -corrections leaking between  $\mu_4$  sectors, would have fired — neither occurred; the clock-variant controls fail in two independent ways each). (**K2**) if one-loop associativity of the celestial  $E_8$  theory demanded extra fields that TFPT does not possess, the contract dies (Costello’s axionic mechanism is the first pass:  $E_8$  is admissible; the TFPT-specific field content is the open half). (**K3**) if WP3 finds no exact alignment between the GS coefficient and the  $c_3$  normalisation, the  $c_3$ -connection of the celestial route dies (the orbifold arithmetic of WP1 survives on its own). (**K4**) if WP4 is negative, the contract degrades to “ $E_8$  is an admissible gauge algebra for celestial SDYM” — already known from the literature, i.e. zero TFPT content.

**Honest success estimate.** The probability of *full* success (WP5: a second continuum route to SEAM.EQUIV.01) is low — it is constructive holography at the Costello–Li grade. The probability of *partial* findings is high: WP3 is a finite computation on fixed data, and WP1/WP2 already returned nontrivial structural matches (verdict B each) with sharp refinements — the WP1 critical correction (clock  $\neq$  deck; clock<sup>2</sup> = deck) and the WP2 sharpenings (the clock *is* the Picard–Lefschetz/Coxeter monodromy of the seam deformation; the surviving direction is the  $\chi_1$ -Fourier diagonal,  $a_0$  pure scale with the  $\tau=i$  shape frozen) — exactly the kind of sharp, falsifiable refinement the contract format is for. Status: CELEST.SEAM.01 [O] (research contract); CELEST.WP1.01 [E]/[C] (executed); CELEST.WP2.01 [E]/[C] (executed); SEAM.EQUIV.01 untouched, stays [O].

## Certifiability and recommended order

| Building block                                | Machine                 | Hardness             |
|-----------------------------------------------|-------------------------|----------------------|
| $W_{\text{wall}}$ enumeration ( $C_U^{(1)}$ ) | Lean / Python / Wolfram | easy ( <i>done</i> ) |
| $D_4$ fixed-locus polynomial ( $C_U^{(2)}$ )  | Sage / Mathematica      | medium               |
| selector uniqueness ( $C_U^{(3)}$ )           | interval arithmetic     | medium               |
| SNF, determinants, spectra                    | Lean                    | easy–medium          |
| principal-symbol ellipticity (G1)             | xAct / Cadabra / Lean   | medium               |
| gap inequality (G5)                           | Lean                    | easy ( <i>done</i> ) |
| operator-norm bound (G5)                      | analysis + numerics     | hard                 |
| projective limit (G6)                         | not primarily machine   | very hard            |
| Ward identities (G7)                          | BV/BRST formal + checks | hard                 |

## Recommended order.

Selector-triangle pairings  $\rightarrow v_{\text{geo}}$  scale anchor  $\rightarrow G_{\text{net}}$  inclusion theorem

The flavor interface (historically  $U_{\text{wall}}$ ) is now reduced to the *selector-triangle pairings* (the dual normal pair  $(d, n)$  forcing  $R$  columnwise, v136/v139); it is finite, falsifiable and the cheapest visible upgrade. The  $v_{\text{geo}}$  scale anchor is the one dimensionful input (same nature as  $1/G$ ).  $G_{\text{net}}$  — the index-4 seam-net inclusion certifying the metric sector — is more fundamental but a *programme*, not a single proof: heavy machinery, with the projective limit the genuine summit.  $F_{\text{transfer}}$  (source $\rightarrow$ pole / relic / cosmology) is the remaining downstream interface.

$F_{\text{transfer}}$  **is a typed functor, not a bag of open topics.** It is standard physics fed TFPT *source* data,  $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ , with exactly four interfaces:

| Interface              | Map                                                                         | Status    | Source data $\rightarrow$ observable                                                                                                                                                                                                                                     |
|------------------------|-----------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $F_{\text{pole}}$      | source $\rightarrow$ pole masses                                            | [E] / [C] | Koide 53/54 = $a^T(R+Q)\mathbf{1}/(2\mathbf{1}^T R a)$ is an exact operator factor [E] (the missing Sheet-Diamond corner); the pole-transfer interpretation is [C] ([verification/v183_koide_f_corner_transfer.py])                                                      |
| $F_{\text{Boltzmann}}$ | CP source $\rightarrow \eta_B$                                              | [C]       | $\tilde{m}_1=m_3/A_\Lambda$ anchors the washout, but $M_1=M_R\varphi_0^4$ only relocates the free input ( $M_R$ is the seesaw scale, not a compiler power), so $\eta_B$ stays a <i>sharper scenario</i> , not a closure ([verification/v184_etaB_anchored_boltzmann.py]) |
| $F_{\text{relic}}$     | $(f_a, m_a, \theta_i, N_{\text{DW}}) \rightarrow \Omega_a$                  | [C]       | the determinant-line axion can be all of DM, but only on the $\theta_i \approx 170^\circ$ hilltop where the relic is exponentially sensitive — a viable scenario ([verification/v185_axion_relic_solver.py])                                                             |
| $F_{\text{QCD}}$       | $(\alpha_s, b_3, \Lambda_{\text{QCD}}, \text{lattice}) \rightarrow m_p/m_e$ | [O]       | a QCD matching contract ( $b_3=-7$ a carrier output), <i>not</i> a compiler number — the rejected 1836 near-formula is the discipline working                                                                                                                            |

A machine guard enforces the typing: the four families must always carry a [C]/[O] marker and must never be promoted to a primitive [E] compiler prediction ([verification/v187\_ftransfer\_laws.py]). Exact *algebraic* sub-parts (the Koide factor,  $b_3=-7$ ) may be [E]; the *physical* prediction never is.

**The functor contract CONTRACT.F.01, four axioms** ([verification/v213\_ftransfer\_functor.py]).

Beyond the typing guard, the functor itself satisfies four checkable structural axioms, each instance a *discrete* compiler kernel [E] composed with an *external* continuous solver [C]: (1)  **$\mu_4$ -deck equivariance** — the discrete kernels are carrier/deck invariants  $\{|\mu_4|, N_{\text{fam}}, g_{\text{car}}, A_\Lambda\}$ , and the Koide multiplier  $\lambda_2=(2/3)^6=64/729$  is the  $\mu_4$ -deck transfer eigenvalue (v82/v54); (2) **Plücker preservation** — the dimensionless readouts are Plücker readings ( $53=a^T(R+Q)\mathbf{1}$ ,  $\text{Pl}(F)=(1, 22, 30)$ ;  $\|\text{Pl}(K)\|_1=11$ ); (3) **positivity/stochasticity** —  $\text{spec } T=\{1, (2/3)^6, (1/3)^6\}$  is positive (RP/OS),  $\kappa_f \in (0, 1)$ ; (4) **external modules explicit** — the continuous content (pole transfer, the flavored Boltzmann network, the finite- $T$   $\chi(T)$  ODE, the lattice  $C_N$ ) is named and external, never a compiler number. So CONTRACT.F.01 is the third research contract alongside ( $U_{\text{wall}}$ ) and ( $G_{\text{metric}}$ ): a typed functor, *not* a bag of open topics — but a consolidation, *not* a closure.

**The discrete $\rightarrow$ dynamic lens:  $F_{\text{transfer}}$  is the readout end of the one principle** ([verification/v303\_ftransfer\_dynamics.py], FR.DYNAMICS.01). The typing above is sharpened by a *dynamical* reading. The main-branch mechanism is a gapped, positivity-preserving relaxation to a *unique* attractor (the local-averaging update  $m \leftarrow (A+2I)/4m \rightarrow$  the  $E_8$  marks, gap  $6 \ln(3/2)$ ). All four  $F_{\text{transfer}}$  instances share that *same shape*: [E]  **$F_{\text{pole}}$**  (Koide) is the seam dynamics — the source $\rightarrow$ pole/branch relaxation is the Möbius map with fixed-point multiplier  $F'=(2/3)^6=\lambda_2$ , the seam-clock subleading eigenvalue, so  $|F'| < 1 \Rightarrow$  a unique attractor at the *seam* rate (v82); [C]  **$F_{\text{Boltzmann}}$**  ( $\eta_B$ ) is a gapped positive contraction (washout  $\kappa_f \in (0, 1)$  toward the unique balance, an H-theorem); [C]  **$F_{\text{relic}}$**  (axion) freezes at an adiabatic fixed point (the comoving



number relaxes to a constant); [E] **F\_QCD** ( $m_p/m_e$ ) is the 1-loop RG flow with the carrier  $b_3 = -7$ , whose Gaussian UV fixed point is the attractor (asymptotic freedom,  $\Lambda_{\text{QCD}}$  generated). *Honest boundary*: only **F\_pole** has the seam rate  $(2/3)^6$  exactly; the other three share the *shape* with external rates (thermal, cosmological, RG). So **F\_transfer** is not a bolt-on but the downstream manifestation of the one discrete→dynamic principle — the compiler fixes the discrete seed + integer kernel, a gapped flow carries it to the observable — and the theory’s simplicity is preserved precisely because the external rates are honestly fenced (**v187**), not because they reduce to the seam. (Even **F\_pole**’s flow time is non-integer, **v93**: the relaxation is the *continuous* semigroup  $e^{t \log F}$  generated by the discrete operator — exactly the diffusion-semigroup reading of the main branch.)

**The single-flow reduction** (**DYN.TRANSFER.UNIVERSAL.01**). The dynamic lens is sharpened to a precise reduction ([**verification/v425\_dyn\_transfer\_universal.py**]): the shared “shape” is literally the *one* native flow of the theory — the seam modular/recovery semigroup (**v238/v240**, gap  $6 \ln(3/2)$ ) — restricted to each sector, extending the static universal-gap principle (**v383**) to a dynamic one. [E] the **v99** Koide generator  $\dot{q} = (\Delta/N_{\text{fam}})(q-2)(q-5)$  linearised at the attractor  $q=2$  has rate  $-\Delta$ , so the time-1 contraction is  $(2/3)^6$  (the recovery eigenvalue,  $q=5$  unstable); [E] the Boltzmann washout is that same contraction, with the CP phase  $4\pi/3$  native (**v320**); [E] the QCD rate is  $b_3 = -7$ . So the transfer *dynamics* is native; only the *anchors* stay external — the IR pole scale and  $M_R$  are  $v_{\text{geo}}$ -class,  $C_p$  is the lattice  $O(1)$ , and  $F_{\text{relic}}$ ’s cosmological initial condition  $\theta_i$  is the one genuine wall. A reduction, *not* internalisation: no new premise, and the firewall (**v187**) holds — the observable stays [C]. So the visible residual is one geometric postulate (**QGeo.SYM.01**), one unit gauge ( $v_{\text{geo}}$ ), and this one typed transfer functor — everything else is out-of-sample data.

## External works built upon

The TFPT discrete core (the two axioms, the  $E_8$  glue, the integer skeleton) is self-contained and machine-checked. Its *bridges* to continuum/AQFT physics, to gravity and to the lattice/singularity picture build on established mathematics, listed here by topic; the body refers to these by author name.

### Modular theory and algebraic quantum field theory.

- M. Takesaki, *Tomita’s Theory of Modular Hilbert Algebras and its Applications*, Lecture Notes in Math. **128**, Springer (1970).
- J. J. Bisognano, E. H. Wichmann, *On the duality condition for a Hermitian scalar field / for quantum fields*, J. Math. Phys. **16** (1975) 985–1007; **17** (1976) 303–321.
- K. Osterwalder, R. Schrader, *Axioms for Euclidean Green’s functions* I, II, Comm. Math. Phys. **31** (1973) 83–112; **42** (1975) 281–305.
- S. Doplicher, R. Haag, J. E. Roberts, *Local observables and particle statistics* I, II, Comm. Math. Phys. **23** (1971) 199–230; **35** (1974) 49–85.
- R. Haag, *Quantum field theories with composite particles and asymptotic conditions*, Phys. Rev. **112** (1958) 669–673; D. Ruelle, *On the asymptotic condition in quantum field theory*, Helv. Phys. Acta **35** (1962) 147–163.
- R. Longo, *Index of subfactors and statistics of quantum fields* I, II, Comm. Math. Phys. **126** (1989) 217–247; **130** (1990) 285–309.
- Y. Kawahigashi, R. Longo, M. Müger, *Multi-interval subfactors and modularity of representations in conformal field theory*, Comm. Math. Phys. **219** (2001) 631–669.

### Conformal nets, vertex operator algebras and lattice constructions.

- I. B. Frenkel, V. G. Kac, *Basic representations of affine Lie algebras and dual resonance models*, Invent. Math. **62** (1980) 23–66; G. Segal, *Unitary representations of some infinite-dimensional groups*, Comm. Math. Phys. **80** (1981) 301–342.

- T. J. Osborne, A. Stottmeister, *Conformal field theory from lattice fermions*, Comm. Math. Phys. **398** (2023) 219–289, [arXiv:2107.13834](#), [doi:10.1007/s00220-022-04521-8](#). (This is the seam scaling-limit theorem the TFPT continuum leg cites — the Koo–Saleur  $\rightarrow$  Virasoro convergence,  $c = \frac{1}{2}$ , and the smeared WZW-current Theorem 5.1. The internal TFPT code-token “MMST” and the legacy claim-ids SEAM.MMST.\* / filenames v\*\_mmst\_\* denote *this* Osborne–Stottmeister theorem.)
- V. Morinelli, G. Morsella, A. Stottmeister, Y. Tanimoto, *Scaling limits of lattice quantum fields by wavelets*, Comm. Math. Phys. **387** (2021) 299–360, [arXiv:2010.11121](#), [doi:10.1007/s00220-021-04152-5](#); and *Operator-algebraic renormalization and wavelets*, Phys. Rev. Lett. **127** (2021) 230601. (The operator-algebraic (wavelet/Wilson–Kadanoff) renormalization *framework* the Osborne–Stottmeister theorem above builds on.)
- M. S. Adamo, Y. Moriwaki, Y. Tanimoto, *Osterwalder–Schrader axioms for unitary full vertex operator algebras*, [arXiv:2407.18222](#) (2024).
- M. S. Adamo, L. Giorgetti, Y. Tanimoto, *Rational and non-rational two-dimensional conformal field theories arising from lattices*, [arXiv:2506.01008](#) (2025); *Representations of conformal nets associated with infinite-dimensional groups*, [arXiv:2508.07109](#) (2025).
- P. Naaijken, D. Penneys, B. Wallick, *(commuting-projector reflection positivity / LTO-RP)*, [arXiv:2605.10693](#).

### Lattices and quadratic forms.

- J. H. Conway, N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Grundlehren der math. Wiss. **290**, Springer (1999).
- C. L. Siegel, *Über die analytische Theorie der quadratischen Formen* (the Minkowski–Siegel mass formula), Ann. of Math. **36** (1935) 527–606.

### Singularities and geometry.

- E. Brieskorn, *Beispiele zur Differentialtopologie von Singularitäten*, Invent. Math. **2** (1966) 1–14.
- J. Milnor, *Singular Points of Complex Hypersurfaces*, Ann. of Math. Studies **61**, Princeton Univ. Press (1968).

### Spectral geometry, gravity and thermal time.

- A. H. Chamseddine, A. Connes, *The spectral action principle*, Comm. Math. Phys. **186** (1997) 731–750.
- A. Connes, C. Rovelli, *Von Neumann algebra automorphisms and time–thermodynamics relation in general covariant quantum theories*, Class. Quantum Grav. **11** (1994) 2899–2918.
- R. M. Wald, *Black hole entropy is the Noether charge*, Phys. Rev. D **48** (1993) R3427–R3431.
- H. Casini, M. Huerta, R. C. Myers, *Towards a derivation of holographic entanglement entropy*, JHEP **05** (2011) 036, [arXiv:1102.0440](#).

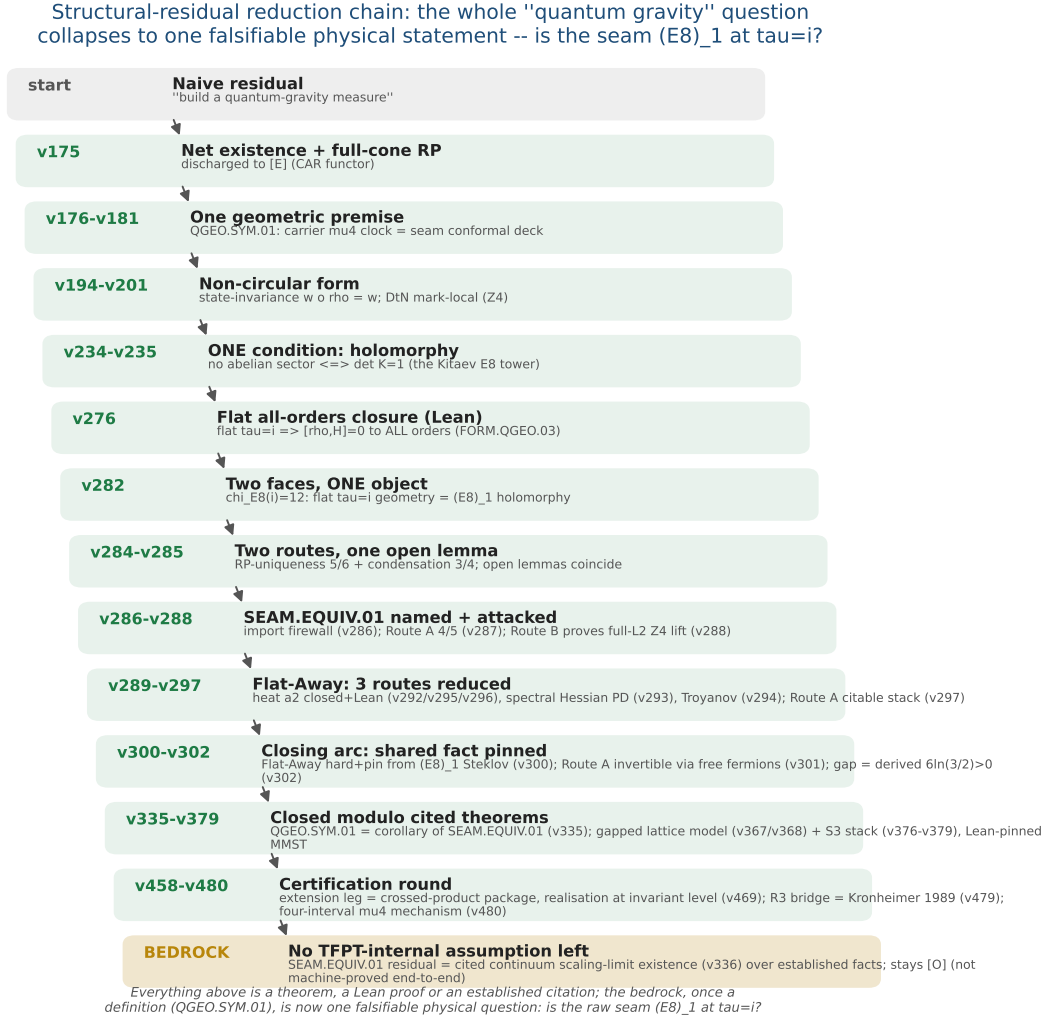


Figure 3: The structural-residual reduction chain ( $v175 \rightarrow$  the certification round). The naive “build a quantum-gravity measure” shrinks, one machine-checked step at a time, from a vague measure to one falsifiable physical statement — *is the raw seam  $(E_8)_1$  at  $\tau = i$ ?* (SEAM.EQUIV.01; its conformal-deck face QGEO.SYM.01 is a corollary, v335). The closing arc ( $v276$ – $v302$ ) pins the shared fact and derives the gap  $6\ln(3/2) > 0$ ; the certification round discharges the extension leg to the crossed-product package (v469), the R3 graph→geometry bridge to Kronheimer 1989 (v479) and exhibits the four-interval  $\mu_4$  mechanism (v480). Every step above the bedrock is a theorem, a Lean proof or an established citation; the residual — the cited continuum scaling-limit existence (v336) — stays honestly [O].

$F_{\text{transfer}}$ : one shape, four rates — a gapped relaxation to a **UNIQUE** attractor

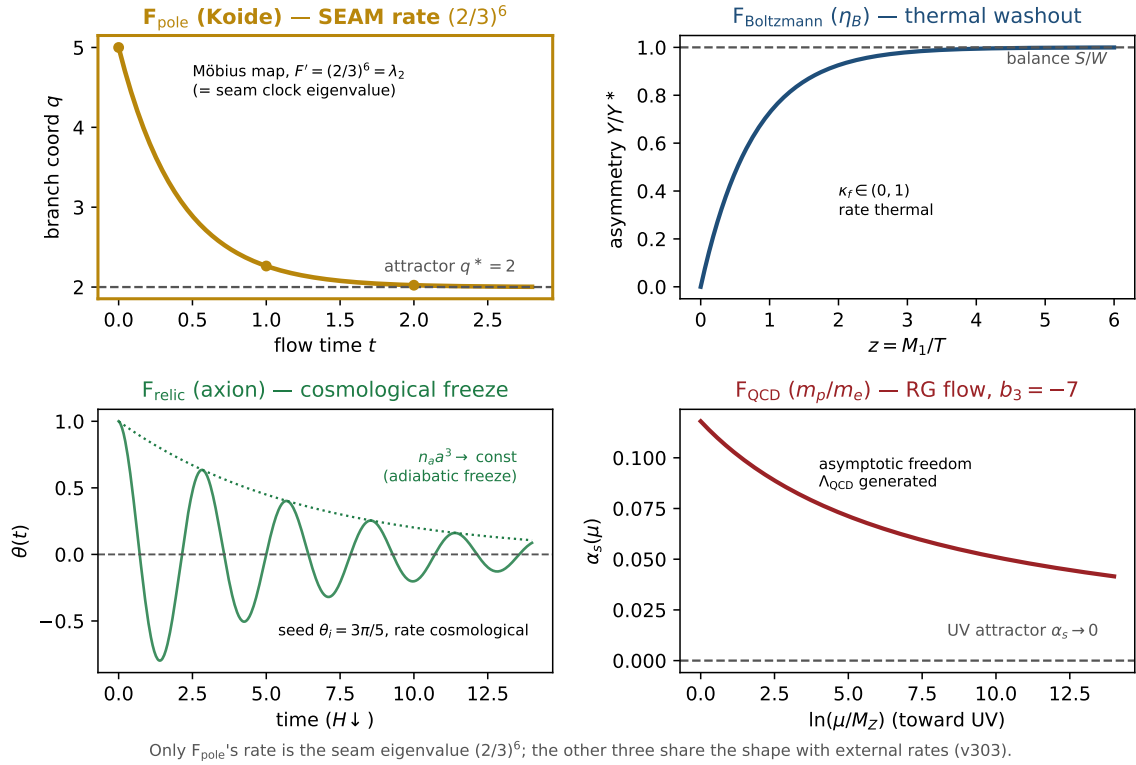


Figure 4: **The discrete→dynamic lens on  $F_{\text{transfer}}$**  ([verification/v303\_ftransfer\_dynamics.py]). All four transfer instances share *one* shape — a gapped, positivity-preserving relaxation to a *unique* attractor, the same Perron-Frobenius / H-theorem / RG-fixed-point structure as the main-branch update  $m \leftarrow (A+2I)/4m \rightarrow$  the  $E_8$  marks. *Only*  $F_{\text{pole}}$  (Koide, gold) runs it at the *seam* rate: its Möbius multiplier is  $F'=(2/3)^6=\lambda_2$ , the seam-clock subleading eigenvalue.  $F_{\text{Boltzmann}}$  ( $\eta_B$ , thermal washout),  $F_{\text{relic}}$  (axion, cosmological freeze) and  $F_{\text{QCD}}$  ( $m_p/m_e$ , the RG flow to the Gaussian UV fixed point with the carrier  $b_3 = -7$ ) share the *shape* with *external* rates. So  **$F_{\text{transfer}}$**  is the readout end of the one principle, not a bolt-on; the theory's simplicity is preserved because the external rates are honestly fenced (v187), not reduced to the seam.